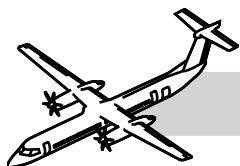


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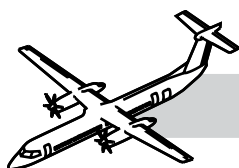
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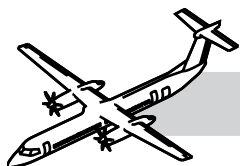
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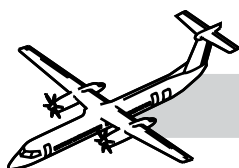
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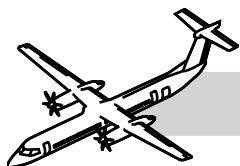


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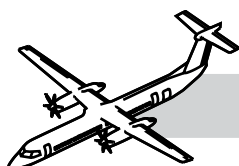


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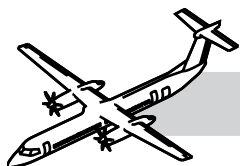
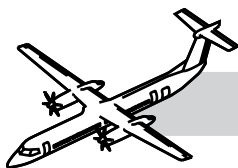


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CHAPTER 29

HYDRAULICS



29-00-00 INTRODUCTION

The main and auxiliary hydraulic systems supply the fluid force necessary to move the hydraulically powered devices.

The Dash 8 Q400 has four hydraulic systems consisting of three main systems designated No.1, No.2, No.3 and an auxiliary system.

All systems use a synthetic phosphate ester-based fluid.

NOTE

Phosphate ester hydraulic fluid causes discomfort when comes in contact with the skin or the eyes. Use protective glove or barrier type hand cream. Wash thoroughly with soap and water if exposed to phosphate ester.

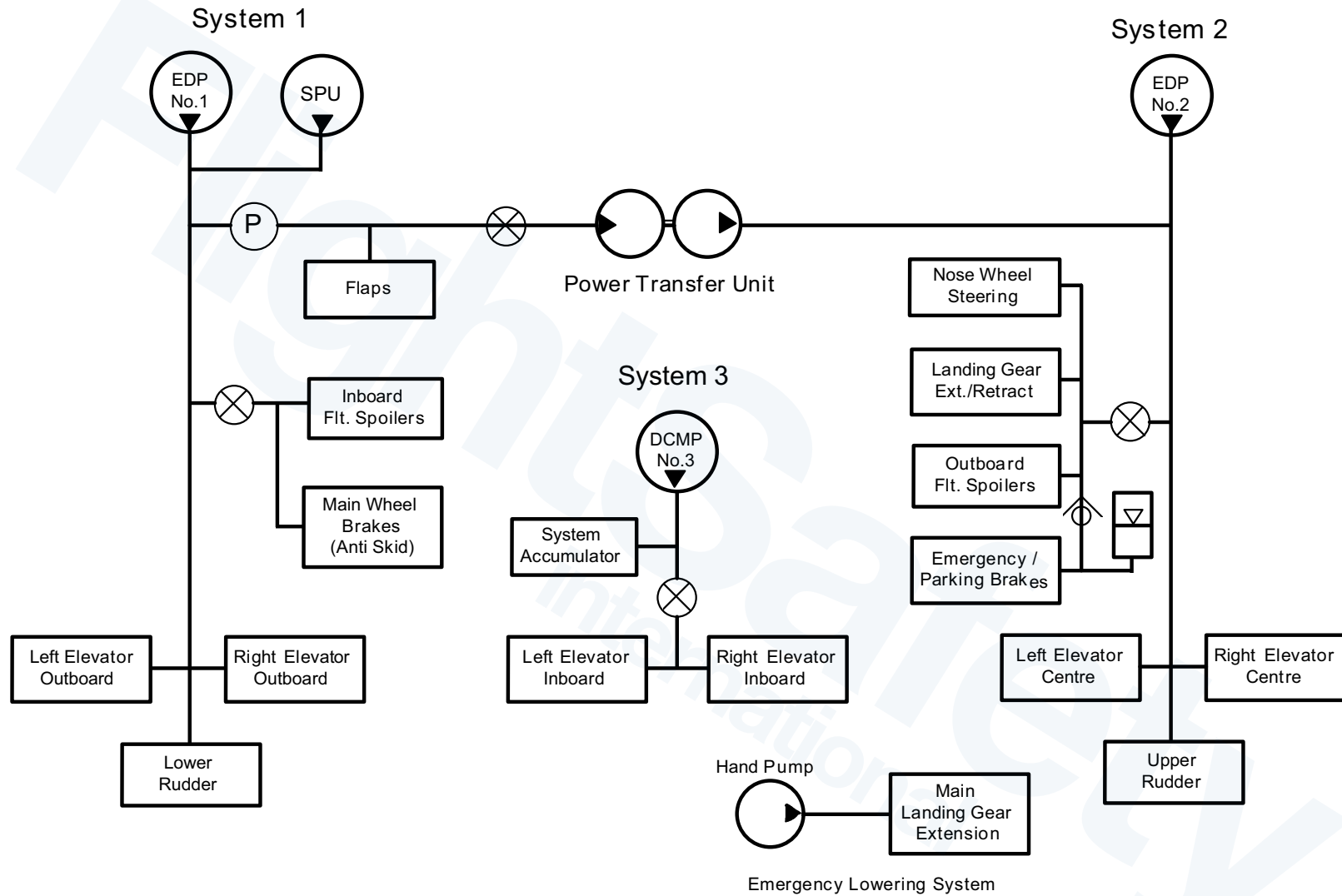
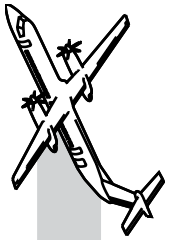
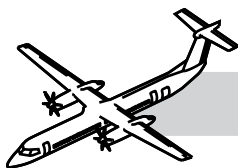


Figure 29-1. Distribution of Hydraulic System





GENERAL

Refer to Figure 29-1. Distribution of Hydraulic System.

The No.1 hydraulic system provides hydraulic power to operate:

- Flaps
- Inboard Roll Spoilers
- Brakes (Normal)
- Elevator (Outboard Pressure Control Units (PCU's)
- Rudder (Lower PCU).

The No.2 hydraulic system provides hydraulic power to operate:

- Gear
- Rudder (Upper PCU)
- Outboard Roll Spoilers
- Park Brake
- Elevator (Center PCU's)
- Steering.

The No.3 hydraulic system supplies hydraulic power to operate the elevators inboard PCU's.

The auxiliary hydraulic system provides power to the alternate main landing gear extension system.

Hydraulic system indications are shown on the:

- Multi-Function Displays (MFDs)
- Caution and Warning Panel (CAWP)
- HYDRAULIC CONTROL Panel
- Remote Indicators
- Fire Protection Panel (FPP).

The indicating system shows:

- Hydraulic pressures
- Hydraulic quantities
- System faults
- System operating status
- Accumulator pressures.

HYDRAULIC SAFETY PRECAUTIONS

General

- Tag the applicable electrical and hydraulic controls when you do maintenance on hydraulic systems or flight controls
- Use approved gloves and goggles that are resistant to hydraulic fluid when you do work on hydraulic components or systems
- If you get hydraulic fluid in your eyes, do as follows:
 - Flush the eyes with cool, clean water for 15 minutes and get medical aid
- If you get hydraulic fluid in your mouth or stomach, get medical aid immediately.

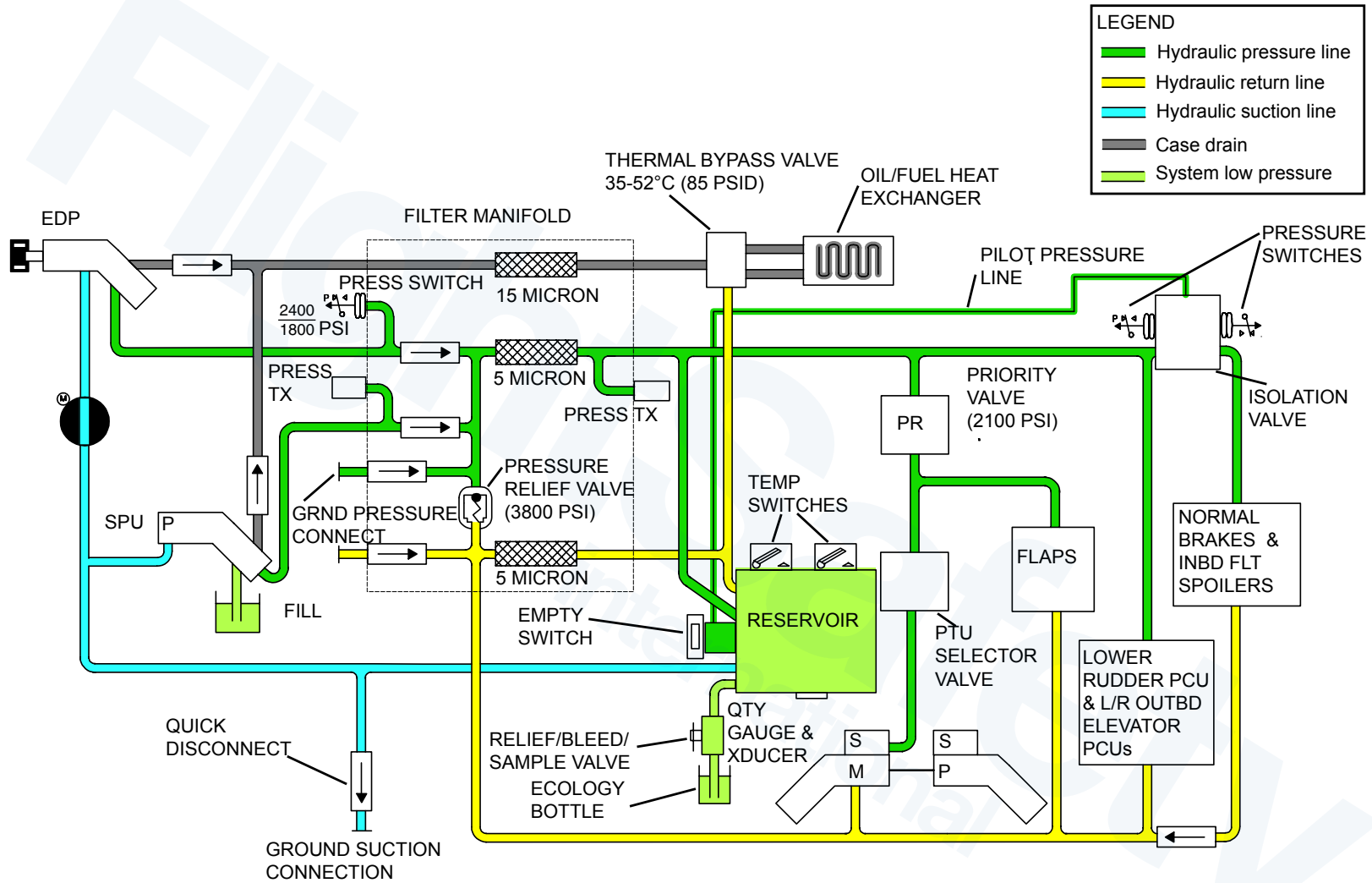
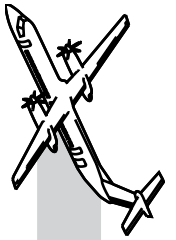
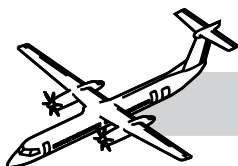


Figure 29-2. Hydraulic Synoptic No.1





29-11-00 MAIN HYDRAULIC SYSTEMS NO.1 AND NO.2

GENERAL

Refer to:

- Figure 29-2. Hydraulic Synoptic No.1.
- Figure 29-3. Hydraulic Synoptic No.2.

Engine Driven Pump (EDP) output pressure is monitored by a pressure switch and then delivered through a check valve to the filter manifold where it is filtered. A transducer measures the pressure and sends data to the Flight Data Processing System (FDPS) for display.

Backup Standby Pressure Unit (SPU) pressure is measured and sent to the FDPS for indication. This flow joins the No.1 system to supply pressure when selected.

A thermal/bypass valve controls the flow of case drain fluid from the EDP and SPU through heat exchangers in the fuel collector bays.

The 3000 psi output fluid is split into two flows. One flow is sent to the reservoir differential piston to keep the reservoir pressurized. This ensures a positive inlet pressure to the SPU and the EDP (through a firewall shut-off valve). The flow is also sent through a pilot pressure line to hold the isolation valve open. The second flow is sent to the hydraulic components that are operated by the No.1 hydraulic system.

The bootstrap reservoirs contain a level sensing valve and switches for control and indication. Provisions for bleed/relief/samples are provided. Any overflow from the system is collected by the ecology bottles.

Reservoir quantity transducers send information to the FDPS for display on the MFDs. The reservoirs also contain direct reading quantity gauges.

No.1 system contains a priority valve that closes if pressure becomes low. This action will remove pressure to the flaps and Power Transfer Unit (PTU). The valve will automatically reset when fluid pressure returns to normal.

A PTU selector valve controls No.1 system pressure to the PTU (motor side).

The No.2 hydraulic system uses a PTU as a backup. The system also contains an accumulator to provide park brake pressure when primary pressure is not available. A transducer supplies park brake pressure to the FDPS. A handpump is provided to pressurize the park brake system if primary pressure is not available.

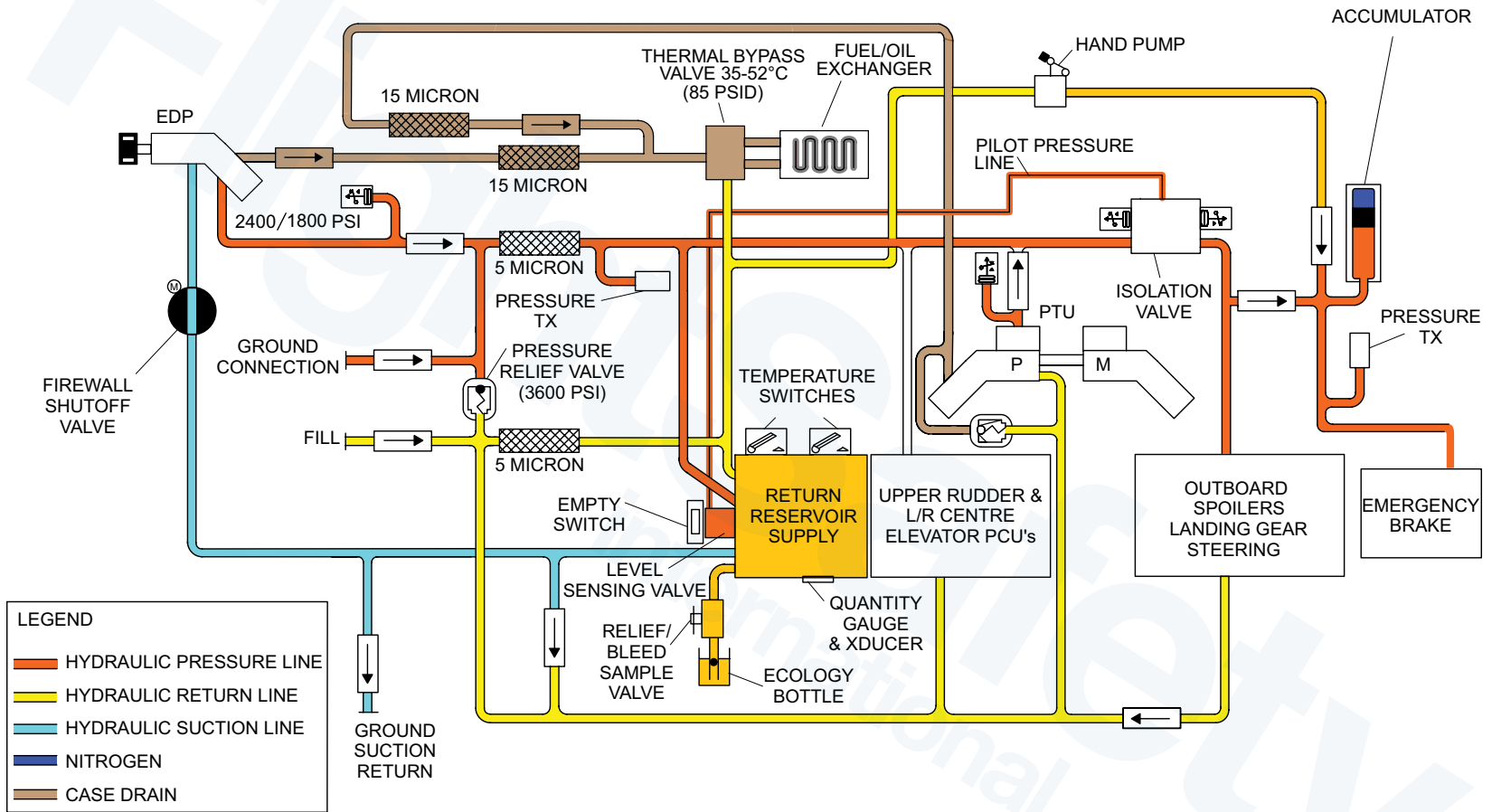
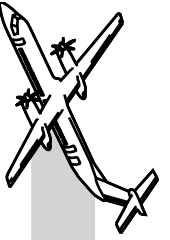
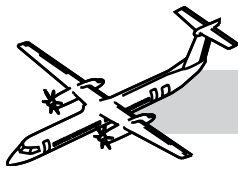


Figure 29-3. Hydraulic Synoptic No.2





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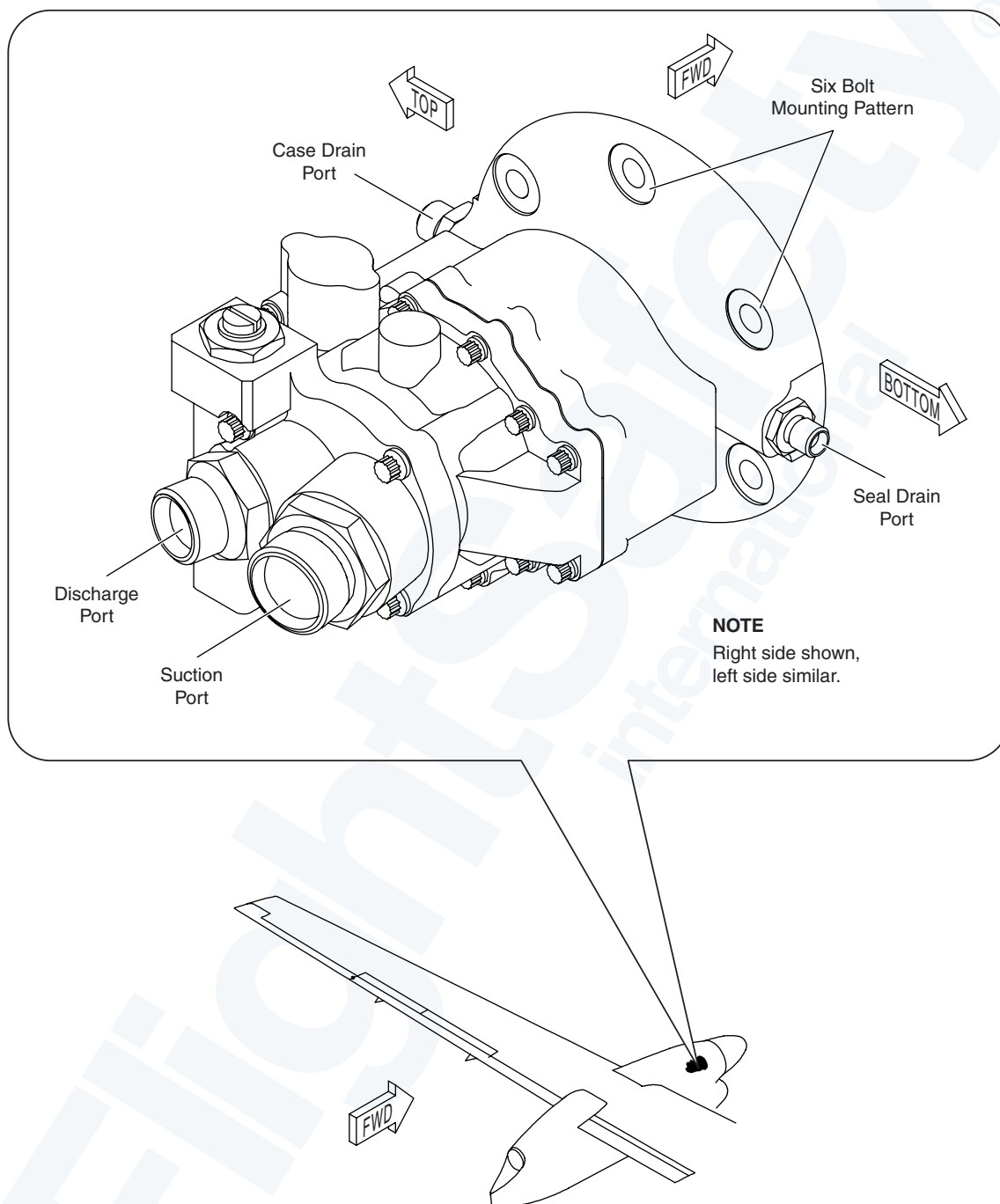
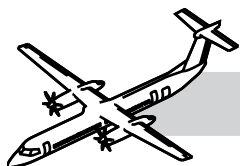
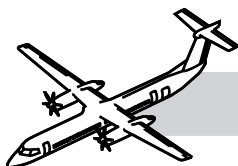


Figure 29-4. Engine Driven Pump



COMPONENT DESCRIPTION

Engine Driven Pumps (EDPs)

Refer to:

- Figure 29-4. Engine Driven Pump.
- Figure 29-5. Engine Driven Pump - Removed.

The EDP is bolted to the reduction gearbox and is driven through a splined input shaft. The discharge, suction, and case drain ports of the pump are connected directly to the aircraft hydraulic system through flexible hoses.

The pump is a variable displacement axial piston type unit, wherein a cylinder barrel containing nine pistons is driven by an external power source (RGB).

The EDPs for the No.1 and No.2 hydraulic systems supply primary power.

REMOVAL/INSTALLATION OF THE VARIABLE DELIVERY HYDRAULIC PUMP

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

NOTE

Do not lubricate the preformed packing with hydraulic fluid.

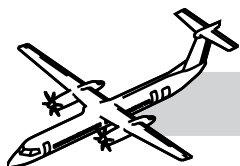
NOTE

Make sure that the case drain port points up and the seal drain port points down.

If the EDP is removed for a mechanical failure, do a flushing of the applicable hydraulic system before the installation of a new EDP, then replace the applicable case-drain filter element for the EDP that was removed.



Figure 29-5. Engine Driven Pump - Removed



Firewall Shut-Off Valves (FSOVs)

Refer to Figure 29-6. Firewall Shut-Off Valve (FSOV).

The hydraulic FSOVs are installed in the No.1 and No.2 engine nacelles (they are located in the wheel well on the forward-right sidewall), in the EDP suction line.

The valve assembly consists of 28 VDC electric motor operated by an actuator mounted on a ball valve (valve body assembly). It has two positions, OPEN and CLOSED with an attached position indicator.

Internal switches provide position status to the FPP.

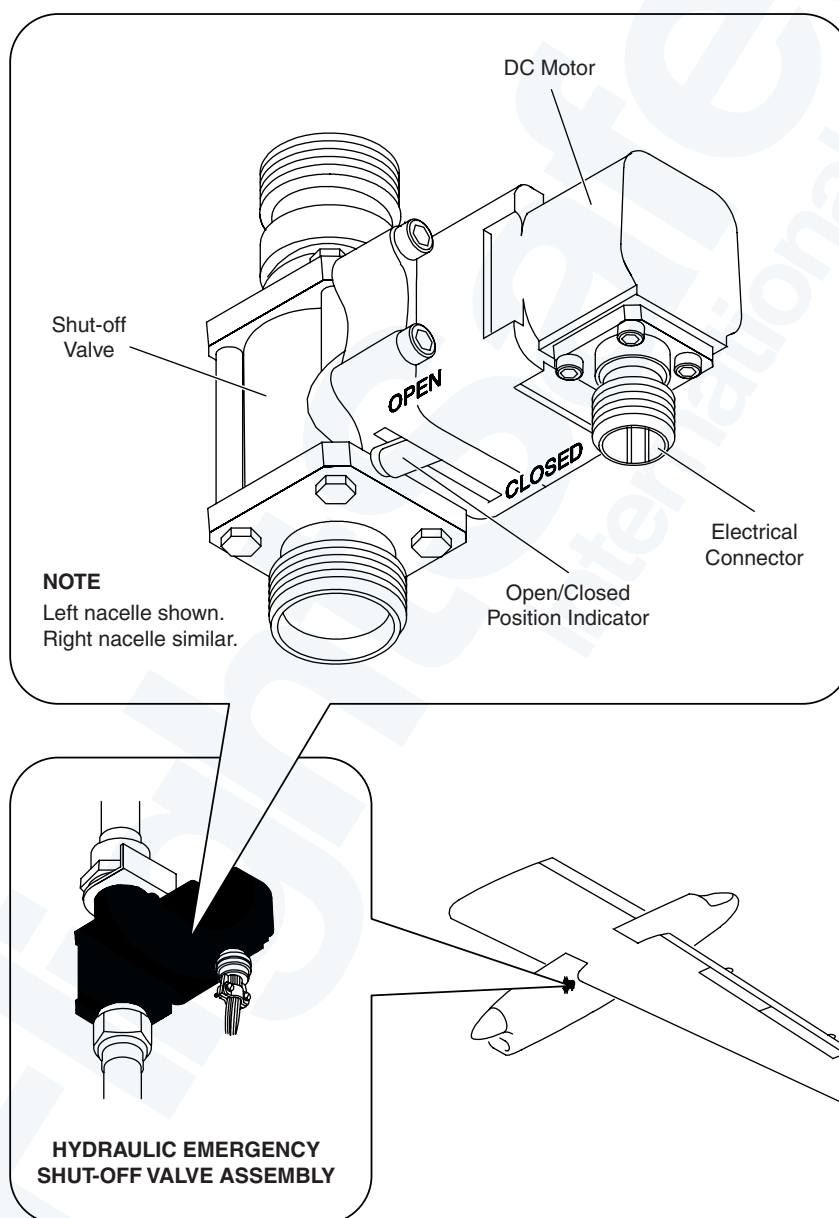
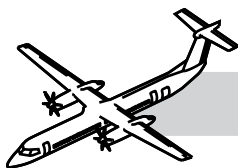


Figure 29-6. Firewall Shut-Off Valve (FSOV)



Isolation Valves

Refer to Figure 29-7. Isolation Valve (No.1 and No.2 Hydraulic Systems).

The No.1 Hydraulic system valve is contained within the LH wing to fuselage fairing. System 2 valve is located on the RH side.

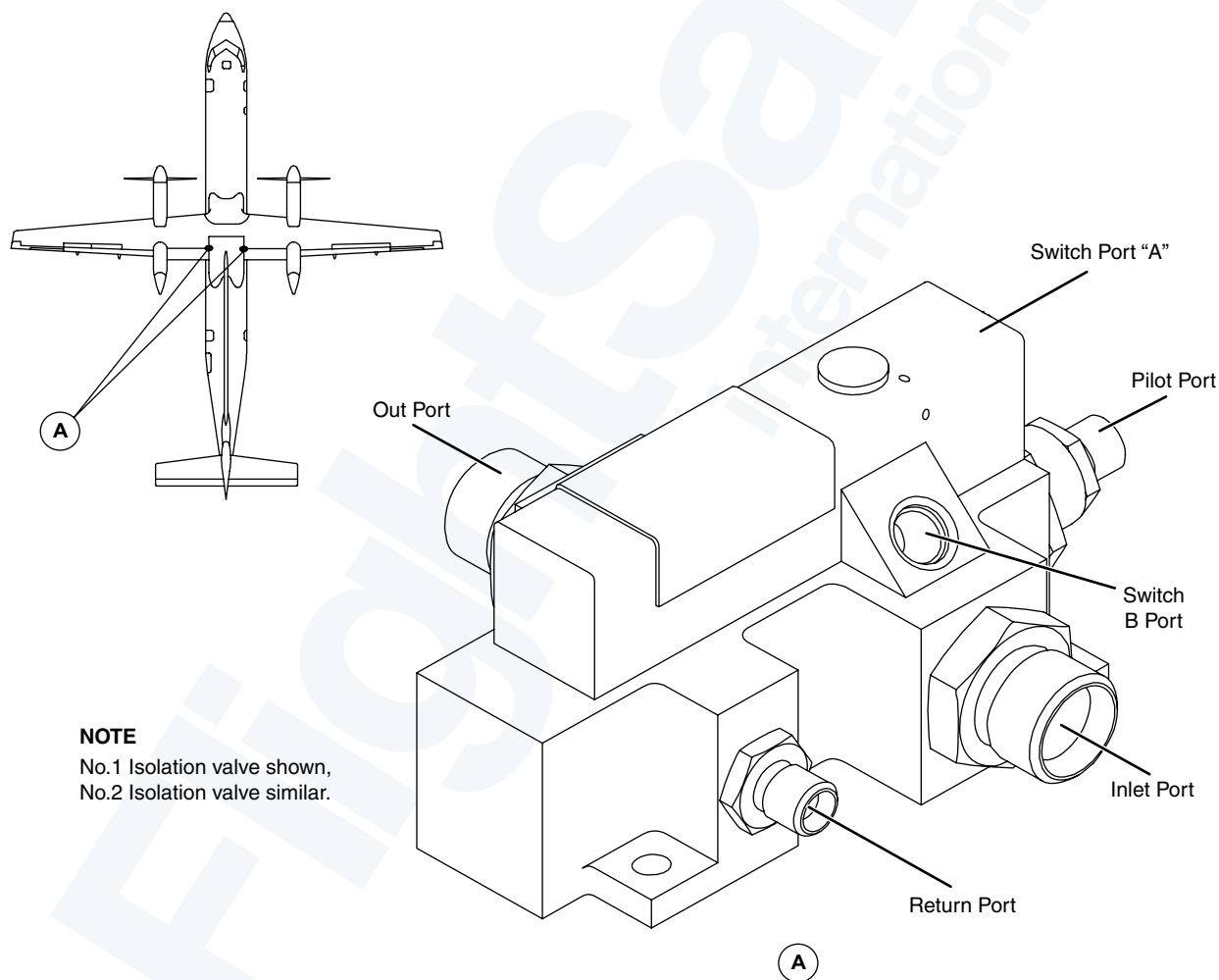
The isolation valves stop the flow of hydraulic fluid to parts of the hydraulic systems, when the reservoir contents drop below 1.3 liters.

The valve has two ports for pressure switches. The inlet switch senses system pressure; the outlet port senses the valve position (open/closed).

A pilot port supplies 3000 psi to hold the valve open. A loss of this pressure causes the valve to close. It is closed by spring pressure.

When closed, the No.1 hydraulic system isolation valve stops fluid flow. As a result, fluid will no longer flow to the normal brakes and the inboard roll spoiler systems.

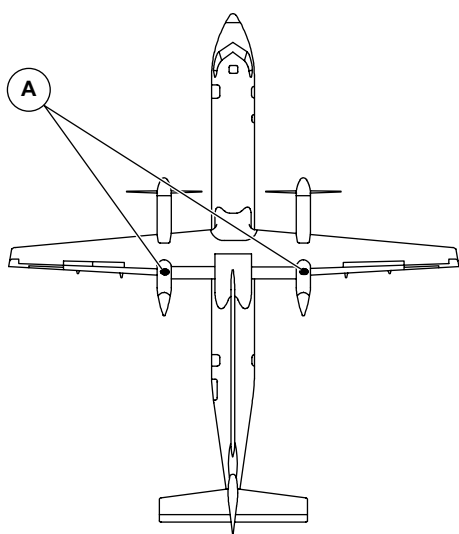
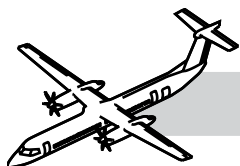
When closed, the No.2 hydraulic system isolation valve stops fluid flow to the landing gear, outboard roll spoilers and nose wheel steering systems.



NOTE

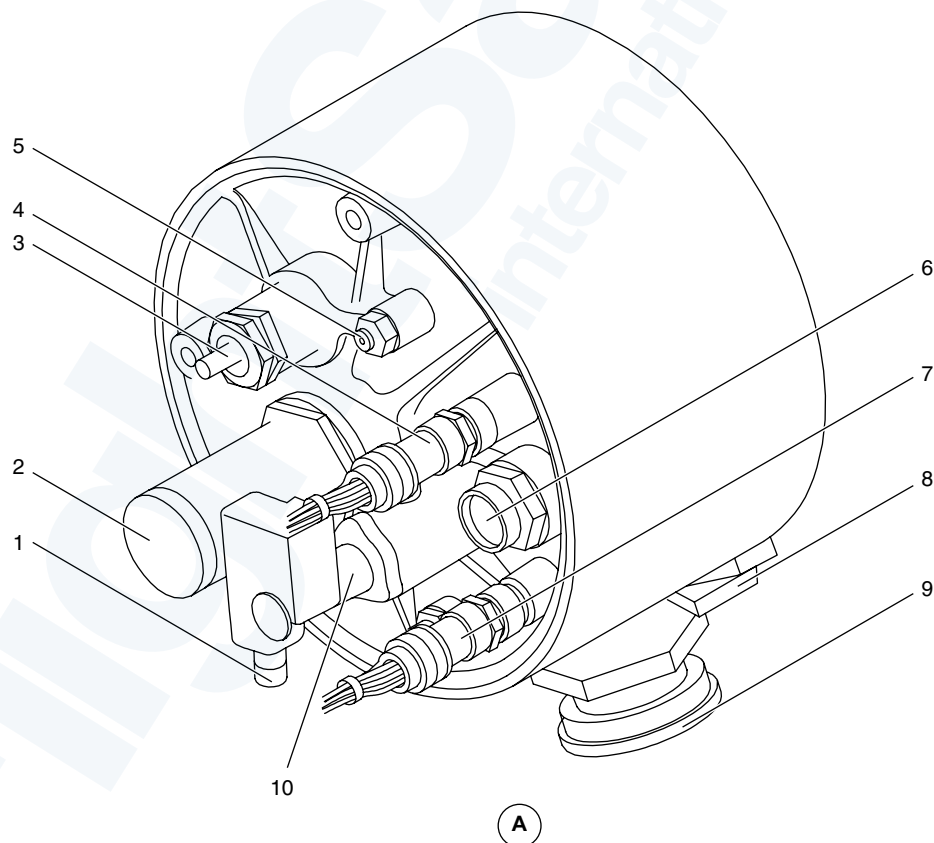
No.1 Isolation valve shown,
No.2 Isolation valve similar.

Figure 29-7. Isolation Valve (No.1 and No.2 Hydraulic Systems)



LEGEND

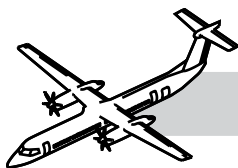
1. Reservoir Empty Switch
2. Differential Piston
3. Relief Valve
4. 107°C (225°F) Temperature Switch
5. Bleed Port
6. Return Port
7. 135°C (275°F) Temperature Switch
8. Quantity Indicator
9. Transducer
10. Reservoir Level Sensing Valve.



NOTE

Left side shown,
right side similar.

Figure 29-8. Hydraulic Reservoir



No.1 and 2 Hydraulic System Reservoirs

Refer to Figure 29-8. Hydraulic Reservoir.

The No.1 hydraulic system reservoir is installed in the No.1 nacelle (wheel well mid-right sidewall). The No.2 hydraulic system reservoir is installed in the No.2 nacelle (wheel well mid-right sidewall).

No.1 and 2 Hydraulic System Reservoir Components

1. Reservoir Empty Switch

The normally open switch closes when the fluid level inside the reservoir is empty - Close Hydraulic FSOV.

2. Differential Piston

The small-area side of the piston is exposed to system pressure (3000 psi) and transfers the force to pressurize the reservoir to 55 psi.

3. Relief Valve

The valve opens at 80 psig which ports fluid to the ecology bottle.

4. 107°C Temperature Switch

The normally open switch closes when the fluid inside the reservoir is $>107^{\circ}\text{C}$. This causes the No.1 or No.2 HYD FLUID HOT caution lights to illuminate.

5. Bleed Port

Provides manual bleed capabilities.

6. Return Port

The port allows hydraulic fluid into the reservoir.

7. 135°C Temperature Switch

The normally open switch closes when the fluid inside the reservoir is $>135^{\circ}\text{C}$ - Close Hydraulic FSOV.

8. Transducer

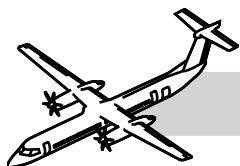
A pulley rotates a 1000 Ohm potentiometer to provide an electrical signal to the IFC (IOM) for the hydraulic system quantity display.

9. Quantity Indicator

A wire cable attached to the low pressure piston drives a spring loaded pulley that is attached to a pointer on the dial gauge.

10. Reservoir Level Sensing Valve

A spool and sleeve linear valve is operated by the low pressure piston. The valve controls the pressure to the isolation valve pilot pressure line and actuates the reservoir empty switch.



Ecology Bottles

Refer to Figure 29-9. Ecology Bottle.

The No.1, No.2 and SPU hydraulic system ecology bottles are installed in their related nacelles.

The No.1 system ecology bottle collects leakage from the No.1 hydraulic system reservoir bleed/relief valve and the balanced relief (priority) valve.

The SPU ecology bottle collects leakage from the SPU shaft seal.

The No.2 hydraulic system reservoir collects leakage from the No.2 reservoir bleed/relief valve.

Each bottle has a fluid capacity of 0.793 US qt (750 milliliters).

NOTE

A 3/8" hole is drilled through the intercostal and the ecology bottle lid to prevent the bottle from collapsing due to differential pressure.

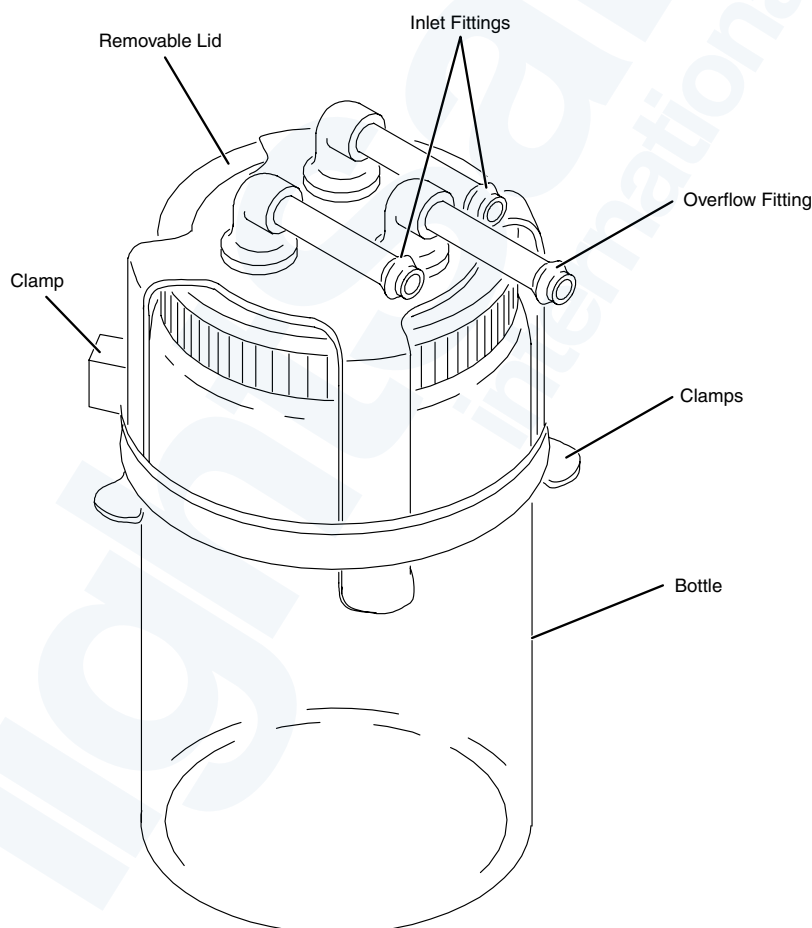
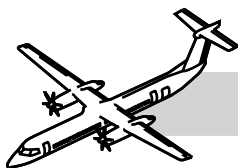


Figure 29-9. Ecology Bottle



Hydraulic Filter Manifold

Refer to Figure 29-10. Hydraulic Filter Manifold.

The No.1 and No.2 hydraulic system filter manifolds are installed in their related nacelle wheel wells (mid-right sidewall).

The filter manifolds contain:

- Filters to remove contaminants from the pressure, return, case drain, filling and ground power fluid circuits
- Check valves, switches and transducers for system monitoring
- Differential Pressure Indicators (DPI) that “pop out” when a differential pressure across the filter exceeds 70 PSI.

NOTE

No.1 Manifold shown,
No.2 Manifold shown.

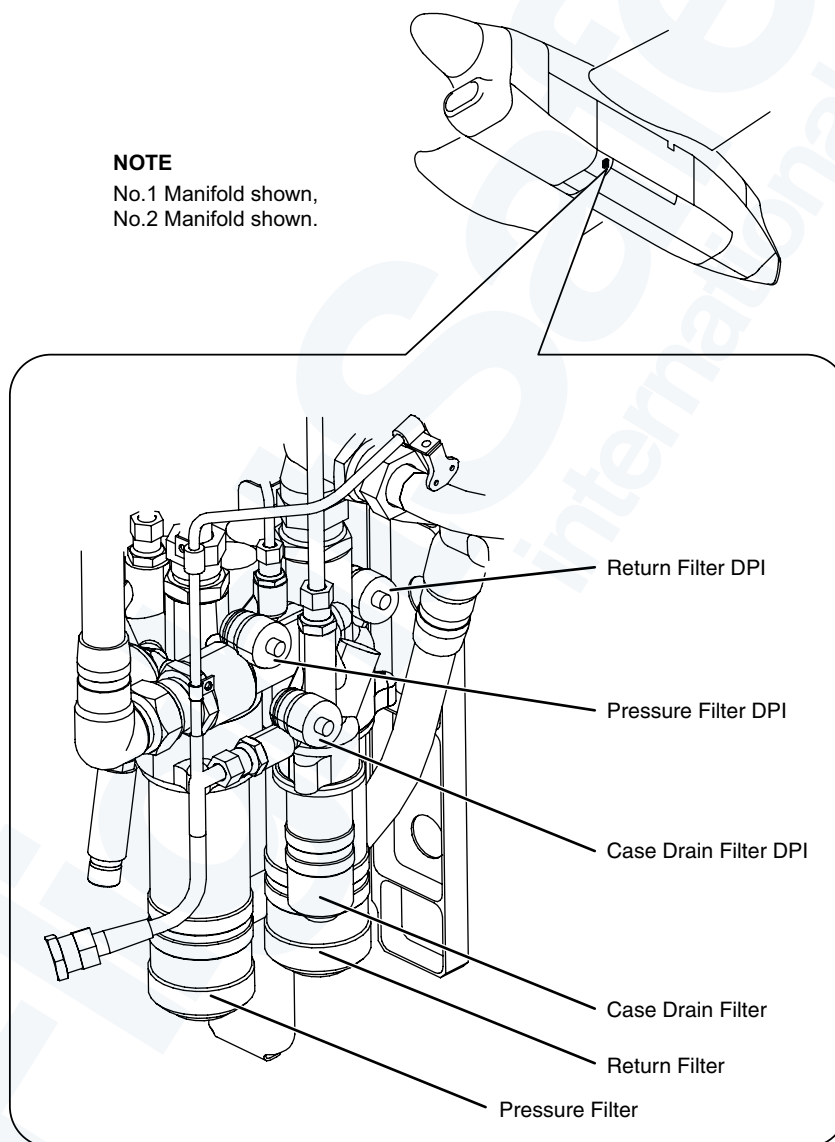
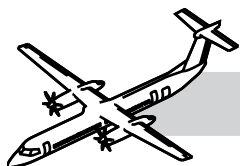


Figure 29-10. Hydraulic Filter Manifold



Pressure Filter

Refer to Figure 29-11. Pressure Filter (No.1 and No.2 Hydraulic Systems).

One Non-Bypass disposable pressure filter is installed in each filter manifold element. The filter removes 100% of all contamination particles larger than 5 microns.

It consists of a filter element with a bowl, a differential pressure indicator, and an automatic shut-off device.

Return Filter

One Bypass return filter is installed in each filter manifold. The filter removes 100% of all contamination particles larger than 5 microns.

It consists of a filter element with a bowl, a differential pressure indicator, an automatic shut-off device, and a bypass relief valve.

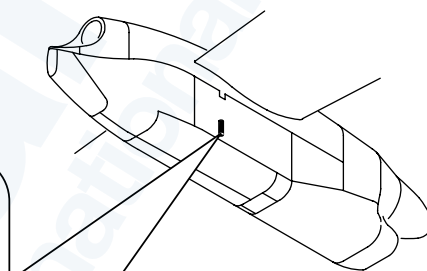
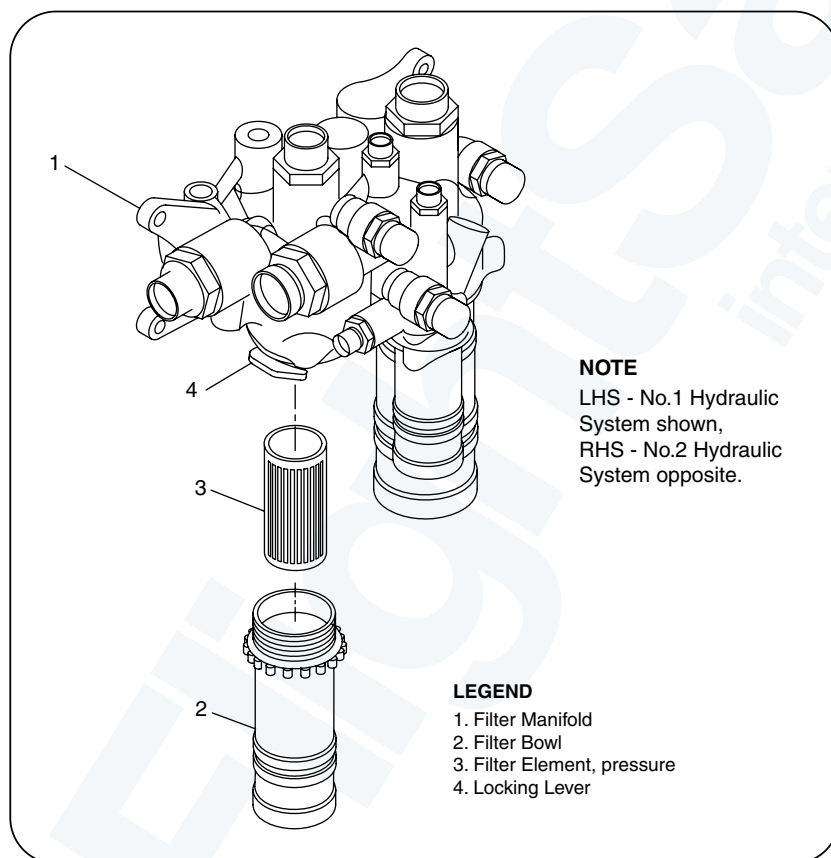
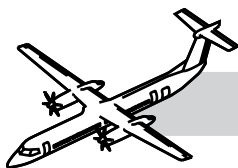


Figure 29-11. Pressure Filter (No.1 and No.2 Hydraulic Systems)



Case Filter (Pre Mod SB)

Refer to Figure 29-12. Case Filter Pre Mod SB (No.2 Hydraulic Systems).

One Bypass disposable case-drain filter installed in each filter manifold removes 100% of all contamination particles larger than 15 microns.

It is made up of a filter element and bowl, a differential pressure indicator, an automatic shut-off device, and a bypass relief valve.

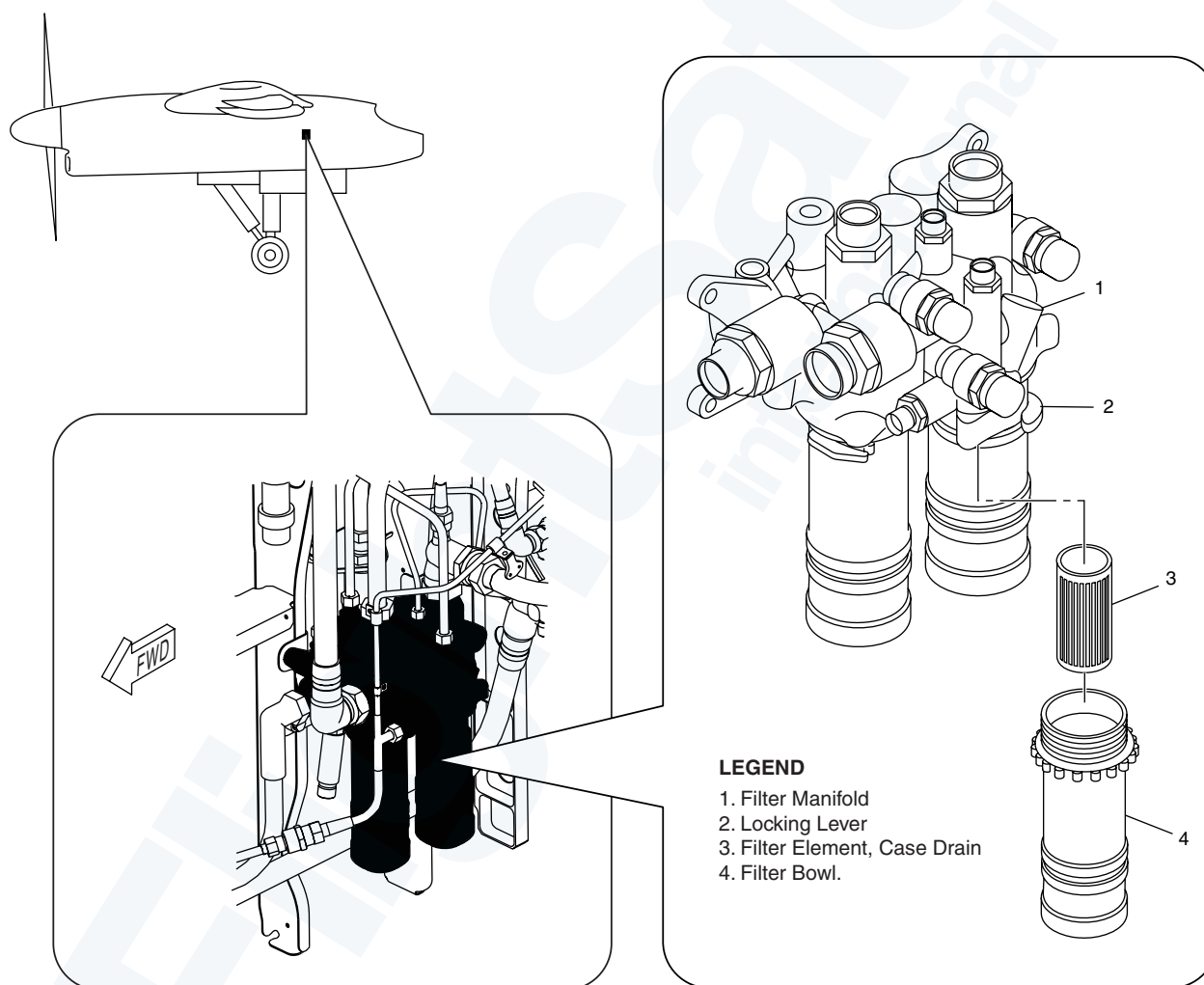
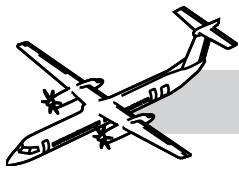


Figure 29-12. Case Filter Pre Mod SB (No.2 Hydraulic Systems)



Case Filter (Post Mod SB)

Refer to Figure 29-13. Case Filter Post Mod SB (No.2 Hydraulic Systems).

A second case drain filter has been added for the PTU. This modification is designed to improve reliability of system No.2 EDP by separating EDP and PTU case drain lines.

The second filter reduces back pressure, increases cooling flow and provides independent filtering of the pump case drain lines.

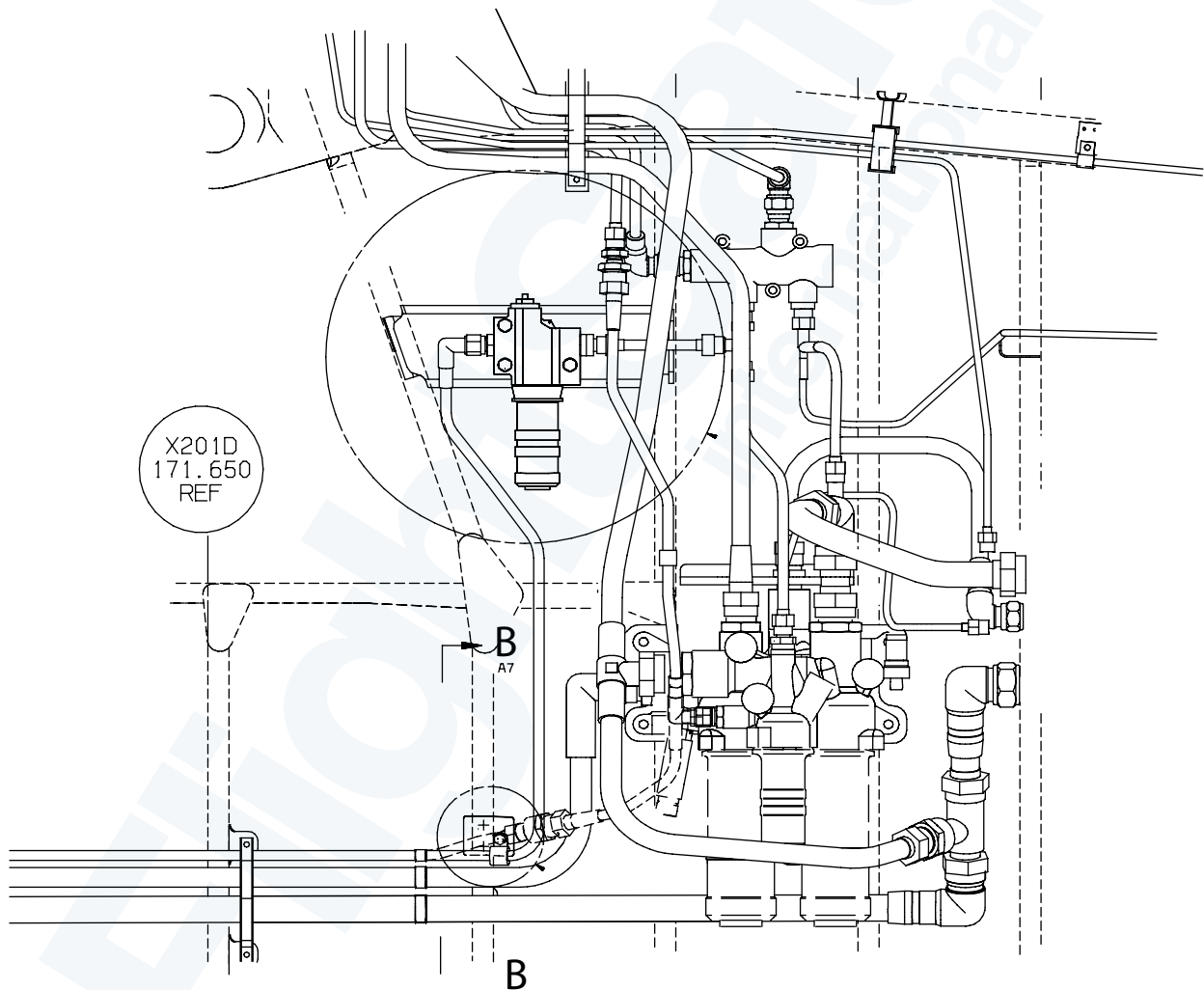
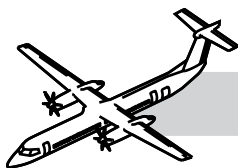


Figure 29-13. Case Filter Post Mod SB (No.2 Hydraulic Systems)



No.1 Hydraulic System Standby Power Unit (SPU)

Refer to Figure 29-14. Standby Power Unit.

The SPU installed in the No.1 nacelle (upper-mid wheel well) provides a backup hydraulic source for the No.1 system.

The SPU control power is from (right) main bus (CB Q4) STBY HYD PUMP No.1. The SPU motor power 115V AC from the right contactor box 60 A circuit breaker (CB 18).

An internal temperature switch monitors the motor windings. See Figure 29-29 No.1 Hydraulic System Electrical Schematic - SPU Windings Hot in this chapter.

The SPU is commanded on for normal takeoff and landing configurations. It can also be selected on by the crew.

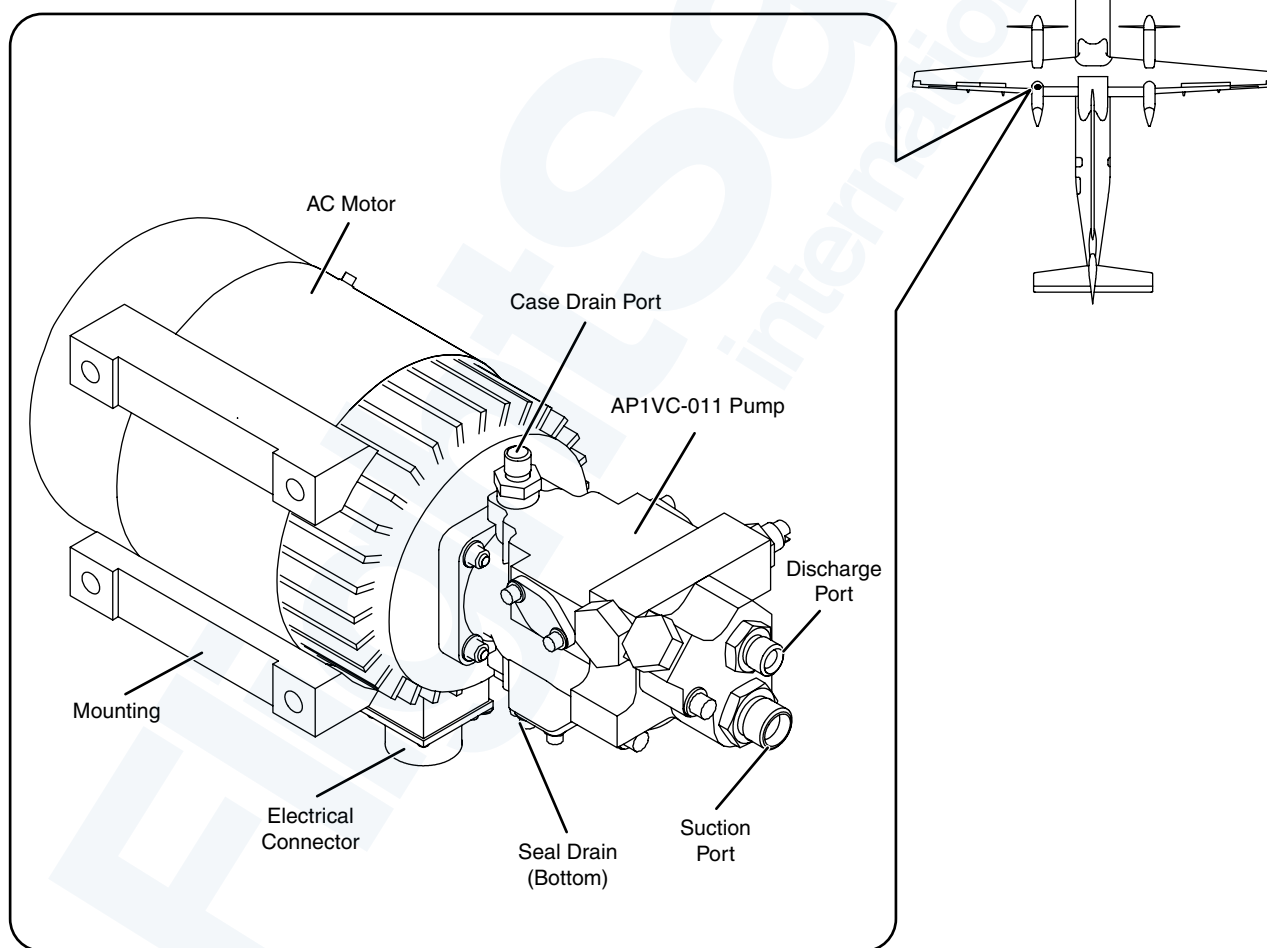
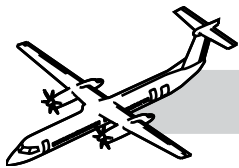


Figure 29-14. Standby Power Unit



Heat Exchanger

Refer to Figure 29-15. Heat Exchanger.

The No.1 and No.2 hydraulic systems have heat exchangers located in the fuel tank to cool the hydraulic fluid.

REMOVAL/INSTALLATION OF THE HYDRAULIC FLUID TO FUEL HEAT EXCHANGER

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

Cut the inlet and return hydraulic tubes of the heat exchanger at the swage attachment.

NOTE

It may be necessary to remove the sealant from the heat exchanger flange.

NOTE

Make sure that the surfaces are clean with no corrosion, paint, grease or preservation fluids.

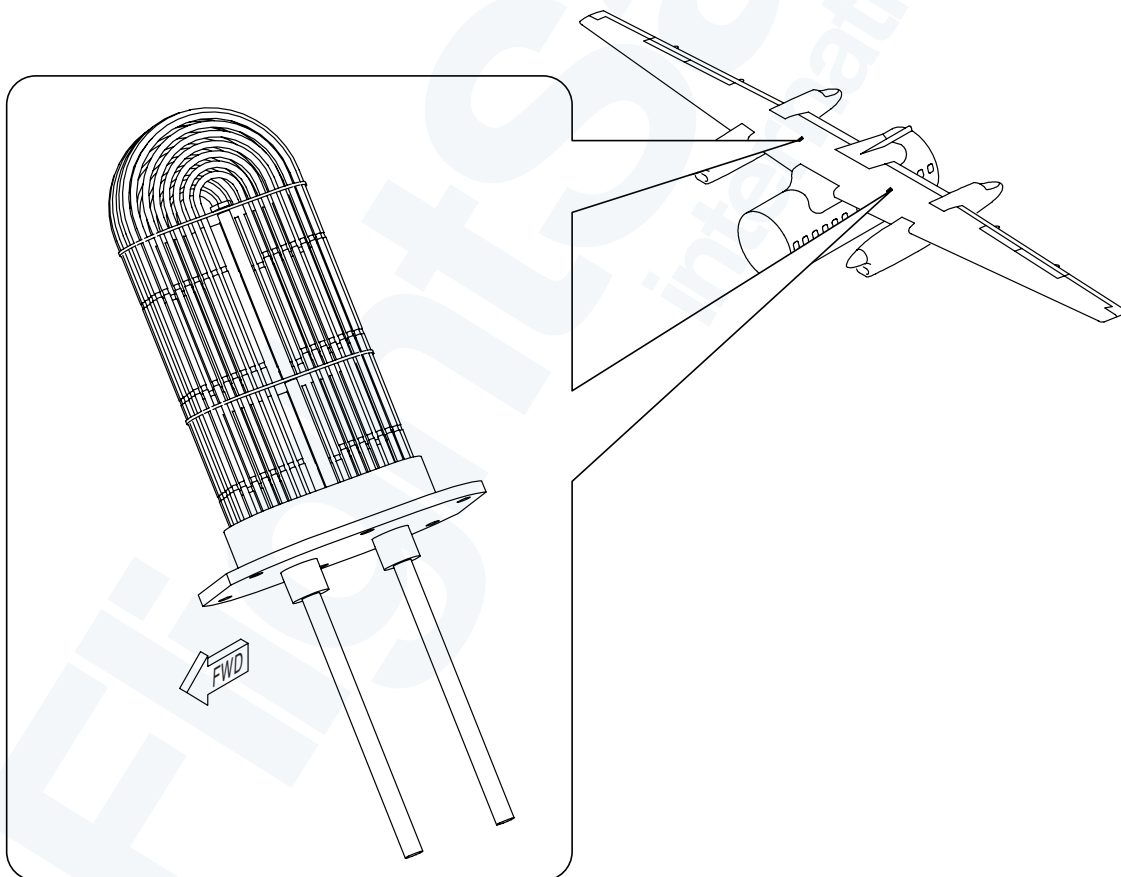
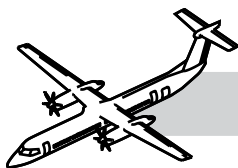


Figure 29-15. Heat Exchanger



Heat Exchanger Bypass Valve

Refer to Figure 29-16. Heat Exchanger Bypass Valve.

Heat exchanger bypass valves, installed between the heat exchanger inlet and outlet lines of the No.1 and No.2 systems, control the flow of hydraulic case drain fluid from the pumps to the heat exchangers. The valve starts to open at 35°C. It is fully open at 52°C. The valve relieves at pressure greater than 85 psid.

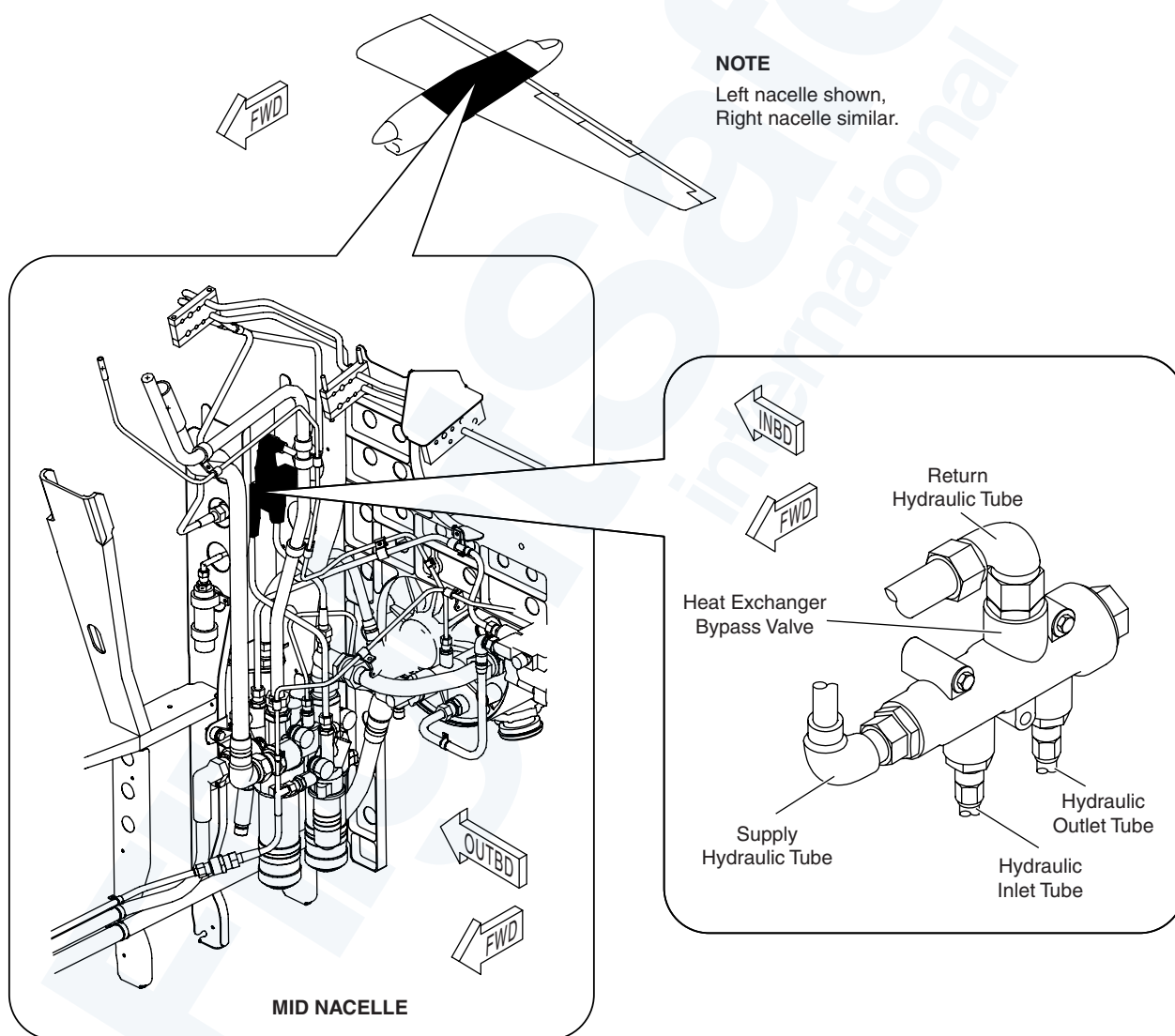


Figure 29-16. Heat Exchanger Bypass Valve

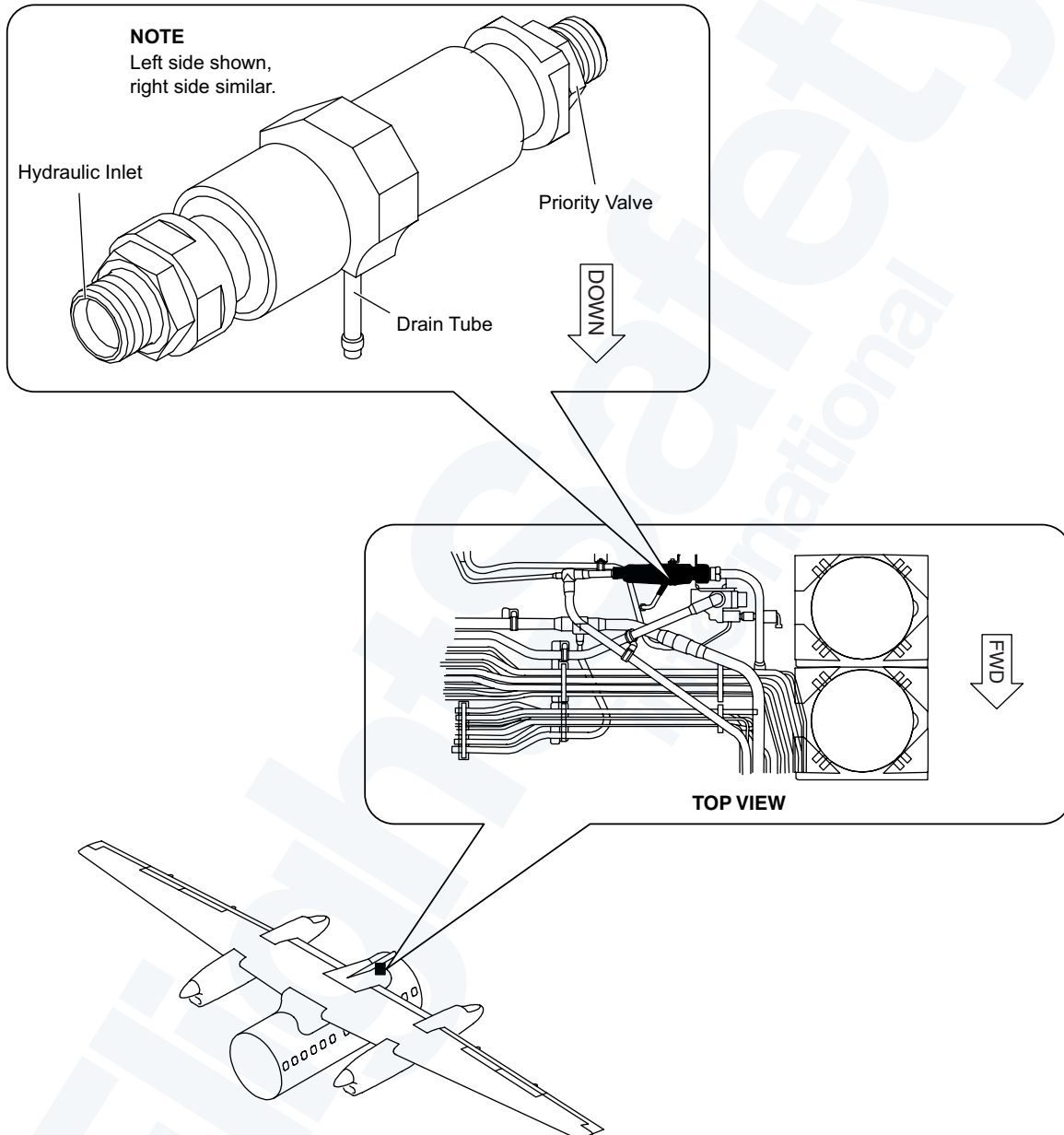
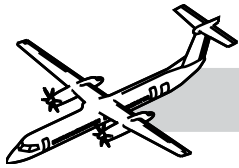
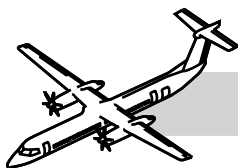


Figure 29-17. No.1 Hydraulic System Balanced Relief Valve



No.1 Hydraulic System Balanced Relief Valve

NOTES

Refer to Figure 29-17. No.1 Hydraulic System Balanced Relief Valve.

The No.1 hydraulic system balanced relief valve is located in the rear left wing-to-fuselage fairing. If No.1 hydraulic system pressure drops to less than 2100 psi, then the valve removes hydraulic power to the flaps and PTU.

The valve is spring-loaded to the closed position.

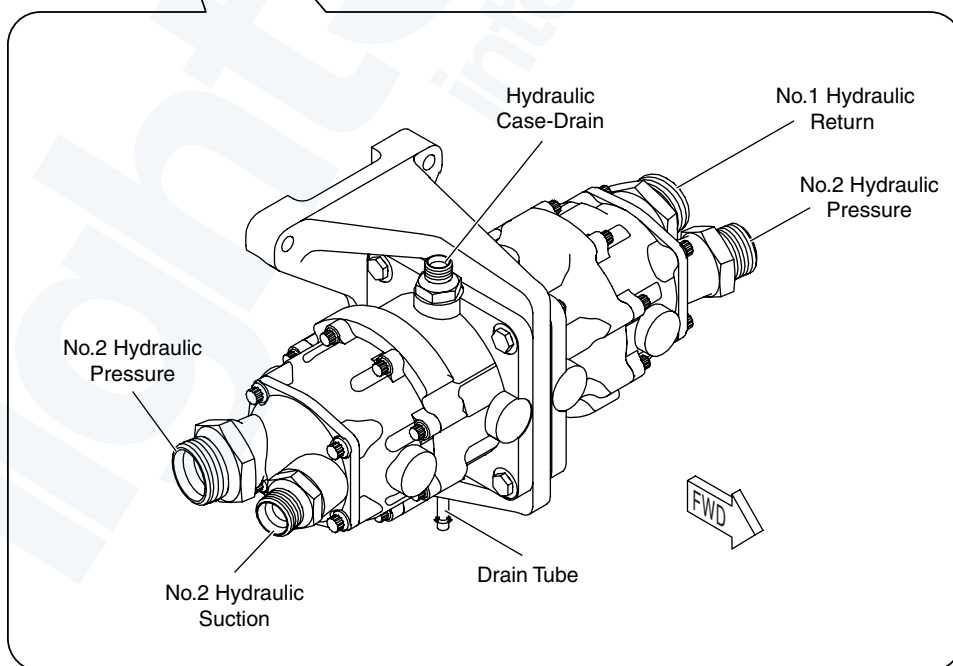
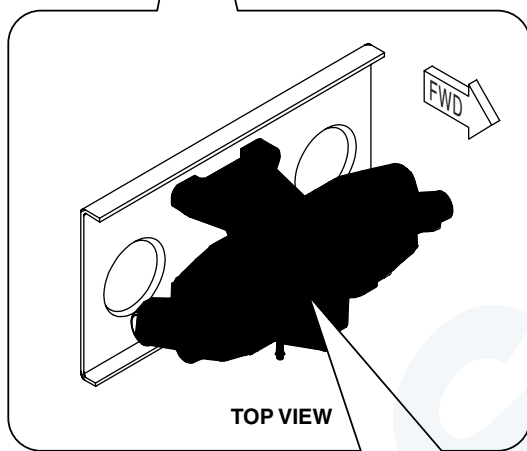
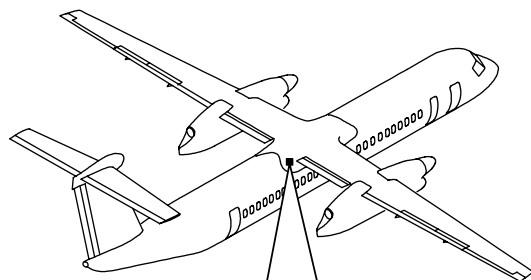
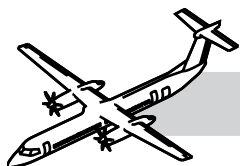
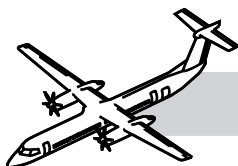


Figure 29-18. Power Transfer Unit (PTU)



Power Transfer Unit (PTU)

Refer to:

- Figure 29-18. Power Transfer Unit (PTU).
- Figure 29-19. Power Transfer Unit (PTU).

The PTU is a fixed displacement component installed in the fuselage-to-wing fairing (rear-right). It supplies backup hydraulic pressure to the No.2 hydraulic system.

Hydraulic No.1 system pressure drives the motor-side and the pressure pumps No.2 hydraulic system fluid.

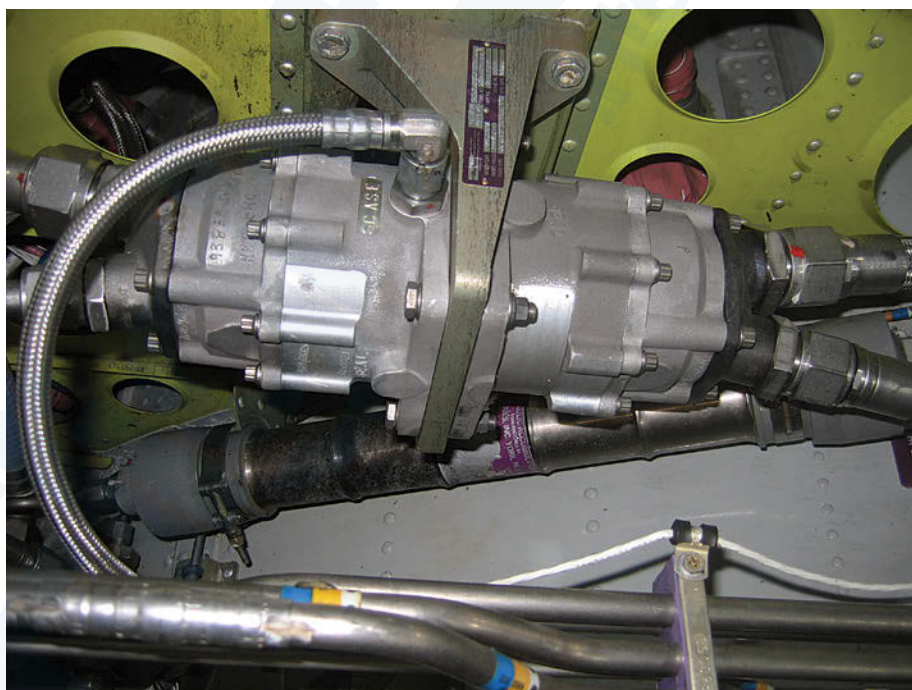


Figure 29-19. Power Transfer Unit (PTU)

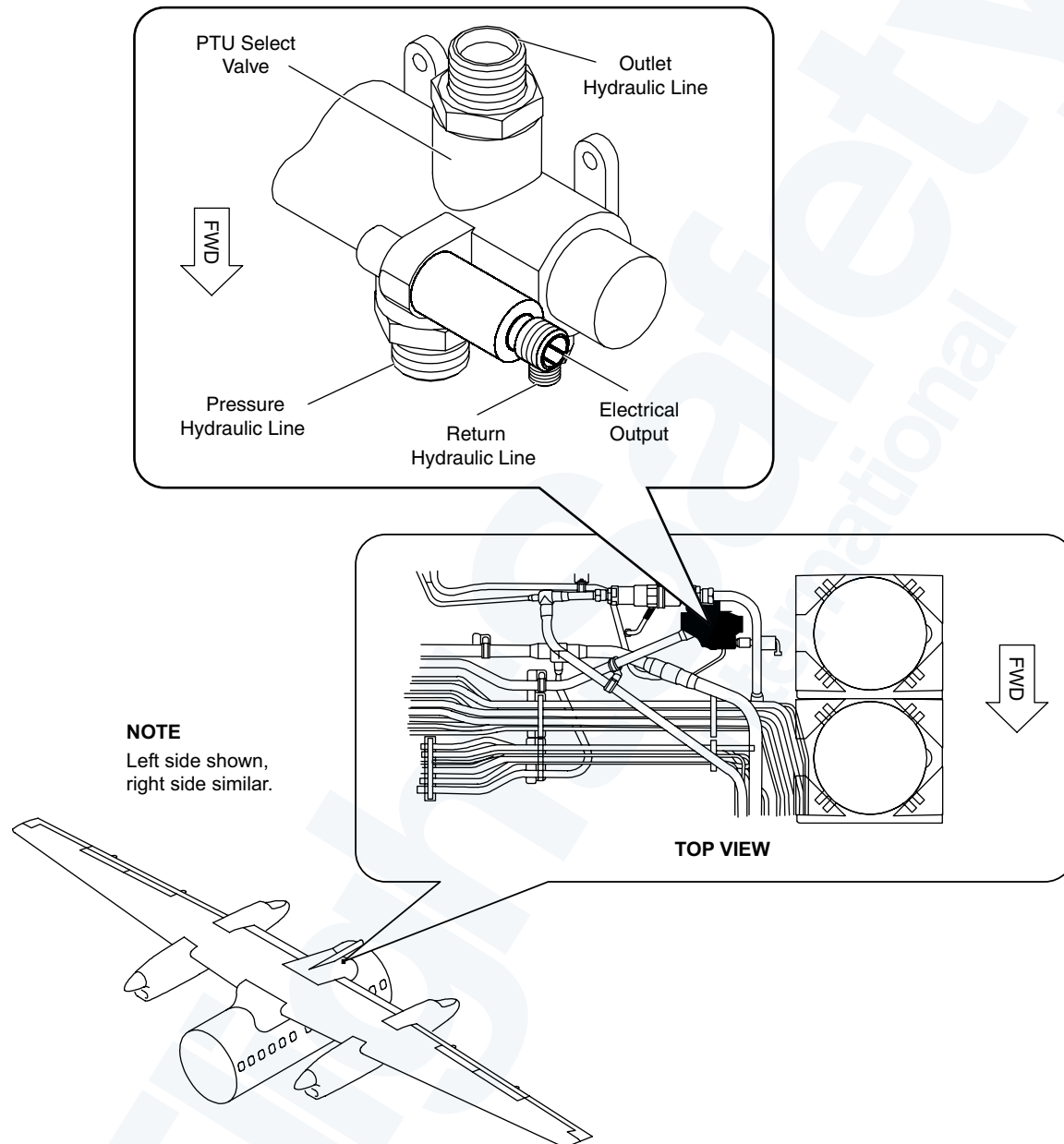
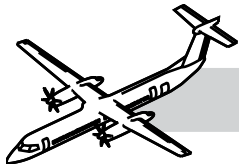
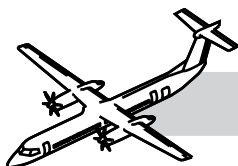


Figure 29-20. PTU Select Valve



PTU Select Valve

Refer to Figure 29-20. PTU Select Valve.

The PTU select valve is installed in the wing-to-fuselage fairing (rear left). The valve controls the flow of hydraulic pressure to the PTU motor.

CONTROLS AND INDICATIONS

Hydraulic Control Panel/Switch

Refer to Figure 29-21. Standby Hydraulic Pressure Switchlight.

The switches that follow are alternate action switches.

Standby Hydraulic Pressure Switchlight (Green)

- Energizes SPU on manually
- The ON (green) segment illuminates when the switch is manually selected.

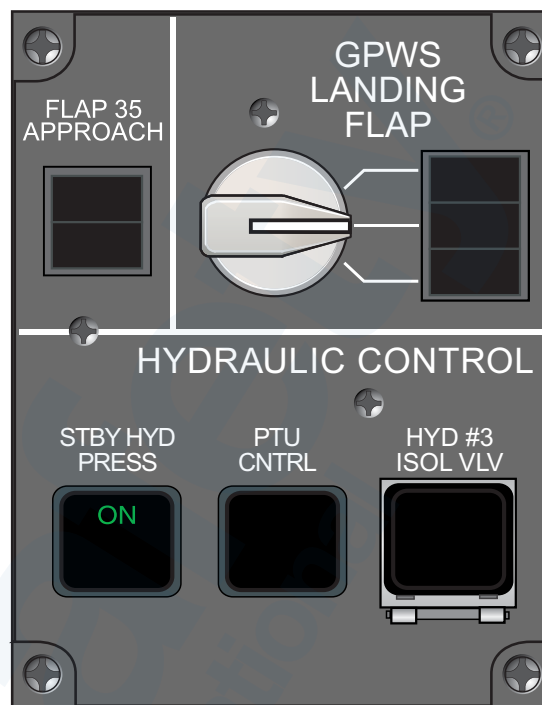


Figure 29-21. Standby Hydraulic Pressure Switchlight

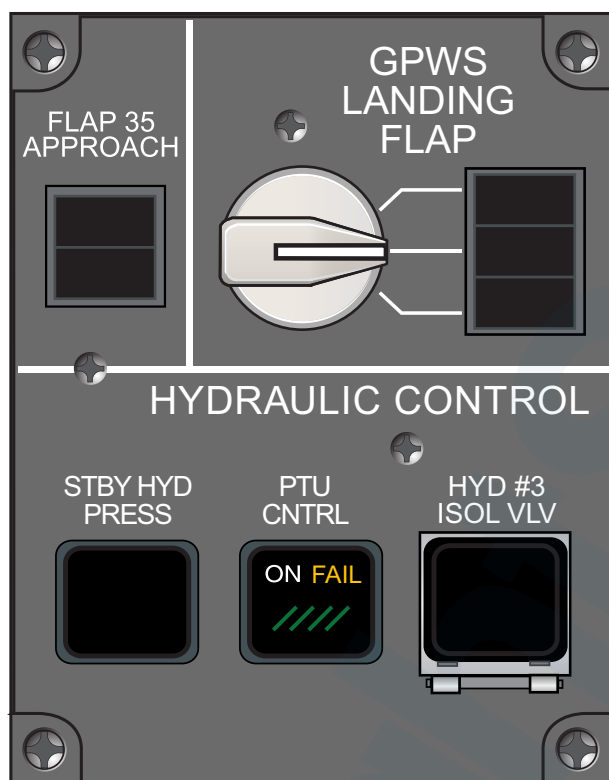
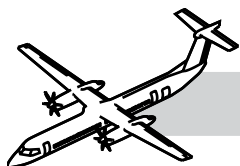
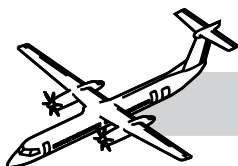


Figure 29-22. Power Transfer Unit Switchlight



Figure 29-23. Ground Service Connections - System Pressure Transducer



Power Transfer Unit Switchlight

NOTES

Refer to:

- Figure 29-22. Power Transfer Unit Switchlight.
- Figure 29-23. Ground Service Connections - System Pressure Transducer.
- The ON (White) segment illuminates when the switch is manually selected on.
- The FAIL (amber) segment will illuminate and latch on whenever the PTU has failed to pressurize five seconds after being commanded on (manual or automatic), or failed during operation. It can only be cancelled / reset by cycling electrical power.
- The PTU CNTRL switch (////) segment illuminates when the PTU is pressurized.

NOTE

ALL OF THE ABOVE ARE DIMMABLE, AND WILL TEST DURING THE ADVISORY LIGHT TEST.

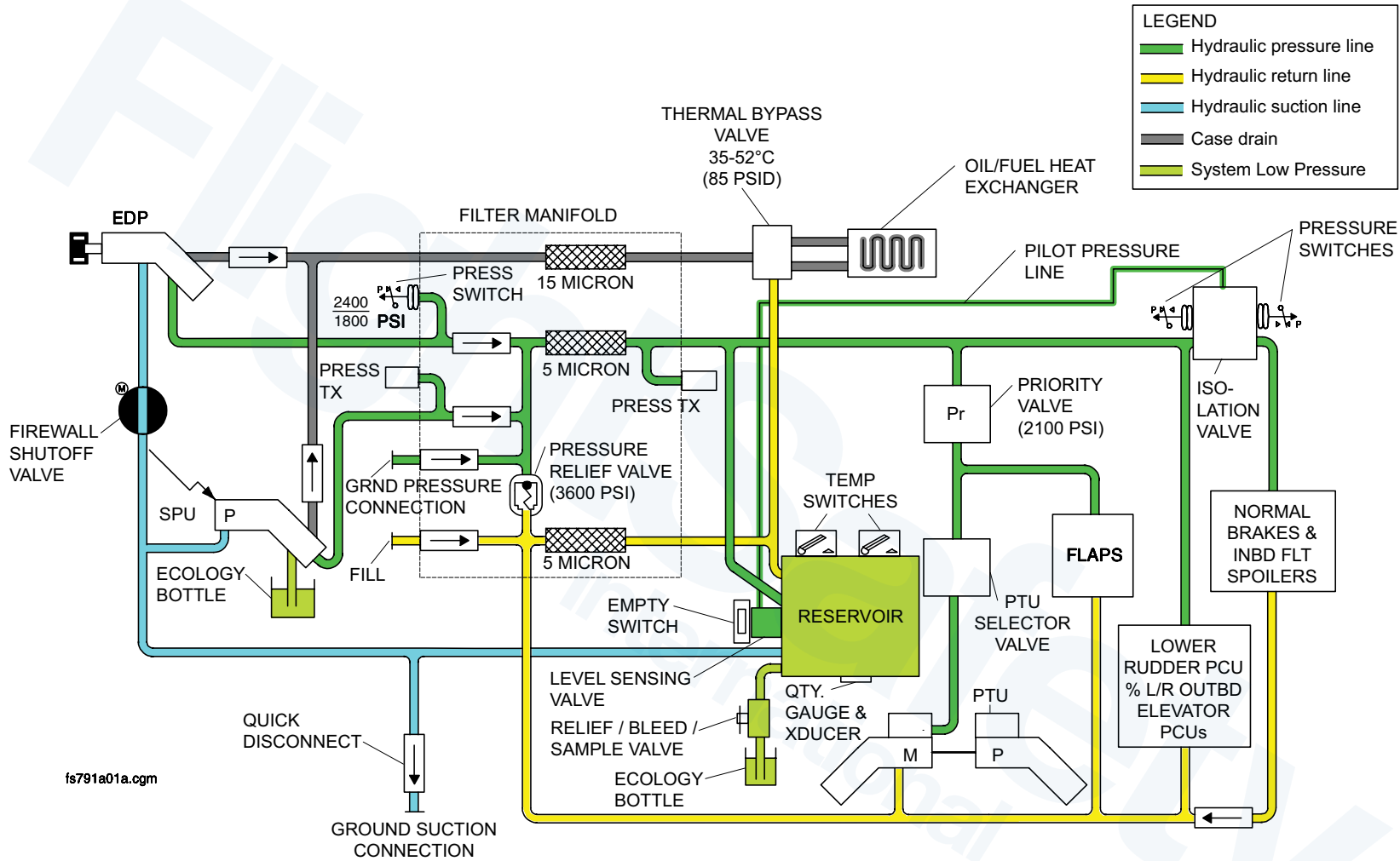
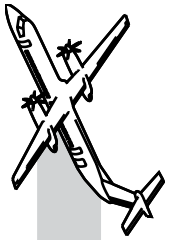
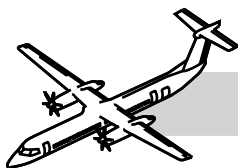


Figure 29-24. No.1 Hydraulic System - Firewall Shut-Off Valve (FSOV) CLOSED Signal





OPERATION

NOTES

No.1 System Operation

Refer to:

- Figure 29-24. No.1 Hydraulic System - Firewall Shut-Off Valve (FSOV) CLOSED Signal.
- Figure 29-25. No.1 Hydraulic System - Firewall Shut-Off Valve (FSOV) CLOSED.

Any of the following three condition signals will provide a ground to energize relay K2, removing power from the valve OPEN contact, and applying the power to the CLOSED contact:

- Pulling the PULL FUEL/HYD handle
- The 107°C temperature switch and the 135°C temperature switch are closed
- The hydraulic reservoir quantity empty switch is closed.

The 107°C temperature switch provides a ground to pin 29 on the CAWP, causing the NO.1 HYD FLUID HOT caution light to illuminate. The switch also provides a ground to the 135°C temperature switch.

Firewall Shut-off Valves (CLOSED)

Refer to Figure 29-26. Three-Position Level Sensing Valve.

The reservoir includes a three-position level sensing valve operated by the low pressure portion of the differential piston.

During normal operation, the isolation valve is held open through the pilot pressure line. If the fluid volume falls below a pre-set level, then the piston makes contact with the level sensing valve, positioning it to the mid-position. This closes the pilot pressure line, thus removing pressure to the isolation valve pilot port. The valve spring-loads close.

If the fluid volume in the reservoir continues to fall, the level sensing valve contacts the empty switch. The FSOV will be commanded to close. This is the normal position when operating from a hydraulic ground cart.

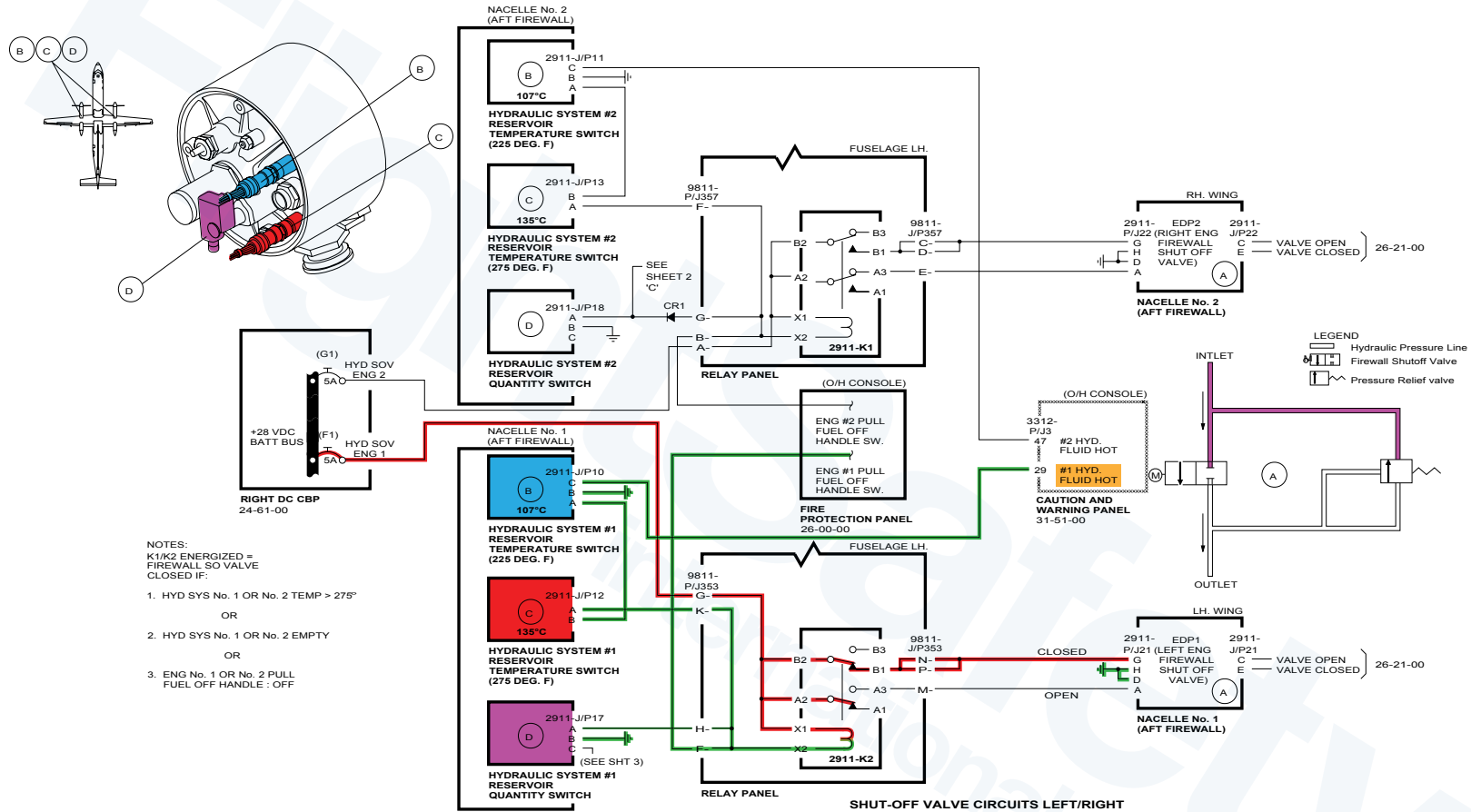
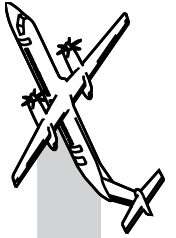


Figure 29-25. No.1 Hydraulic System - Firewall Shut-Off Valve (FSOV) CLOSED



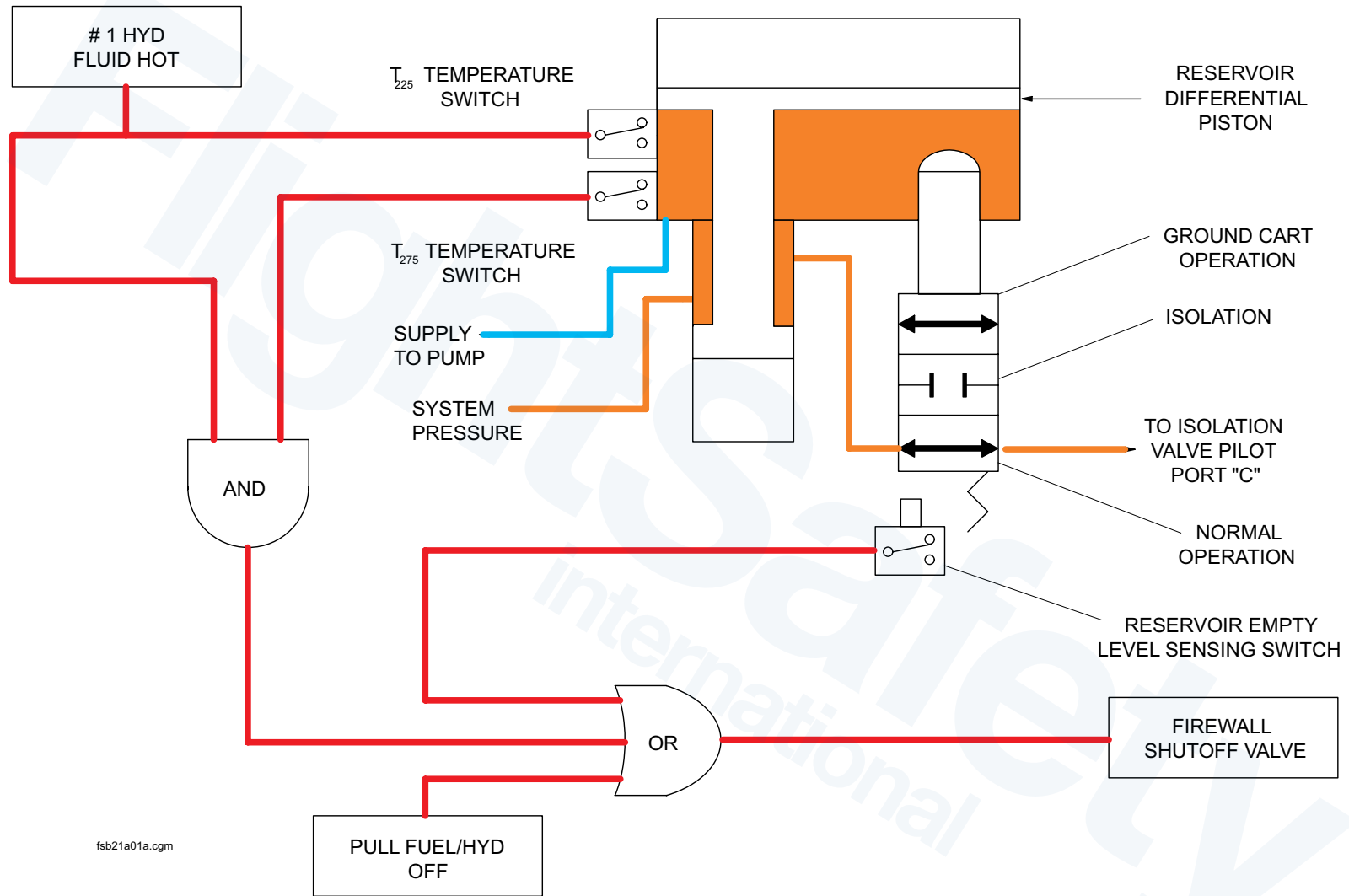
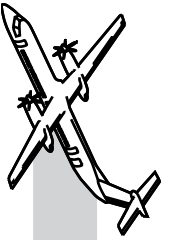


Figure 29-26. Three-Position Level Sensing Valve

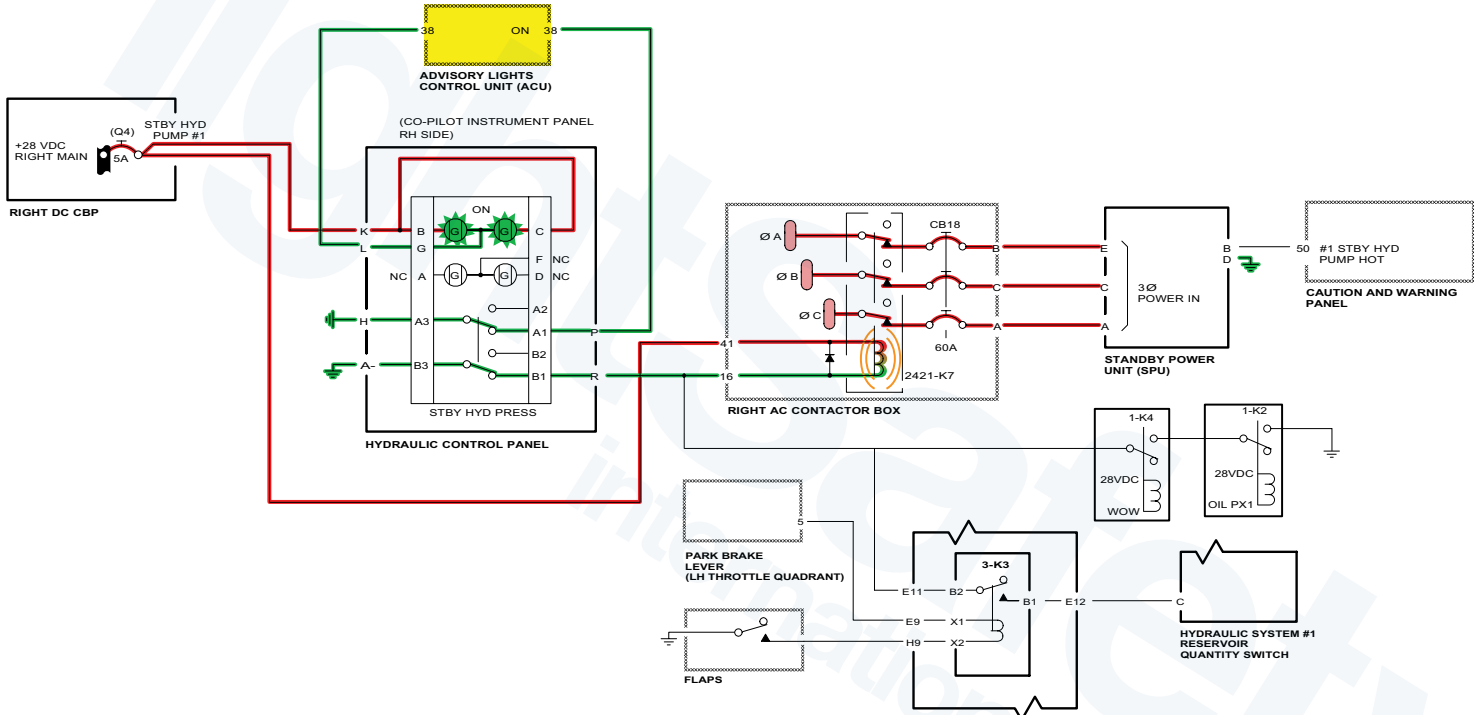
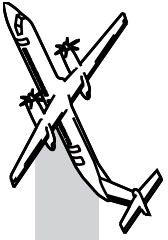
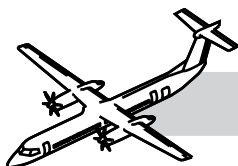


Figure 29-27. SPU Operation - Manual ON



SPU Operation

The SPU will operate:

- Manually if selected by “STBY HYD PRESS” switchlight
- Automatically if the flaps are set > 0, parking brake not set and the No.1 reservoir is not empty
- Automatically if the engine No.1 oil pressure is low and the aircraft is WOFFW.

SPU Operation (Manual)

Refer to Figure 29-27. SPU Operation - Manual ON.

The SPU system operates manually if selected on with the switchlight and if 115 VAC power is available.

Power from the right 28 VDC Main Bus (CB Q4) to:

- Pin K on Hydraulic Control Panel
- Pin 41 on right AC contactor box to relay K-7.

Pushing the STBY HYD PRESS switchlight:

- Provides a ground through the ACU to cause the switchlight ON indication to illuminate
- Provides a ground to energize relay K-7 that will close the three contacts supplying 115 VAC to energize the SPU.

SPU Operation (Automatic)

Refer to:

- Figure 29-27. SPU Operation - Manual ON.
- Figure 29-28. SPU Operation - Automatic - Park Brake Off - Flaps Greater than 0.
- Figure 29-29. SPU Operation - Automatic W Off W - No.1 ENG LOW OIL PX.

The SPU operates automatically if flaps are greater than 0° and the parking brake is selected OFF and 115VAC power is available and No.1 hydraulic reservoir is not empty.

Park brake OFF supplies power from pin 5 to pin E9 on relay 3-K-3.

Flap > 0° provides ground to energize relay 3-K3.

Relay 3-K3 energized:

- Contacts B2-B1 provides ground for relay K7 supplying 115 VAC to energize the SPU (Reservoir not empty).

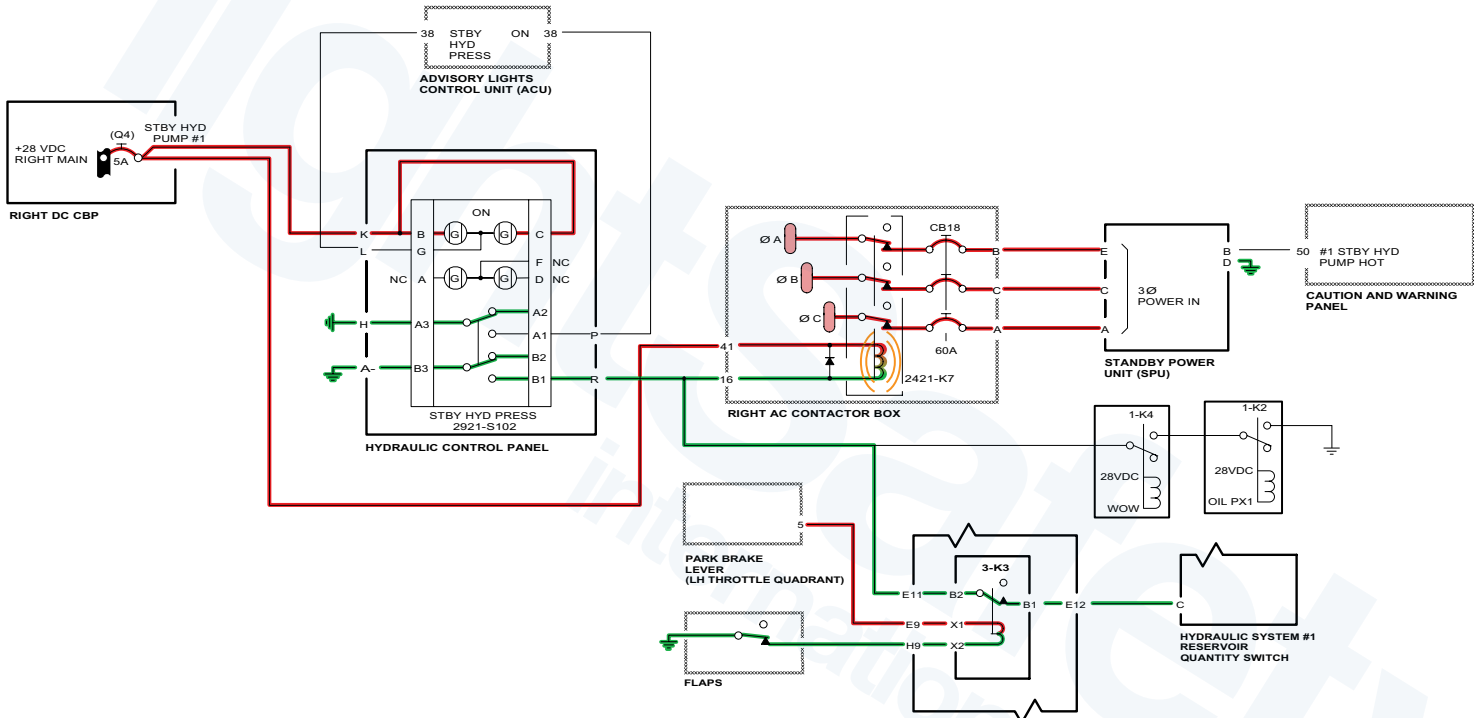
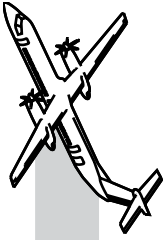


Figure 29-28. SPU Operation - Automatic - Park Brake Off - Flaps Greater than 0

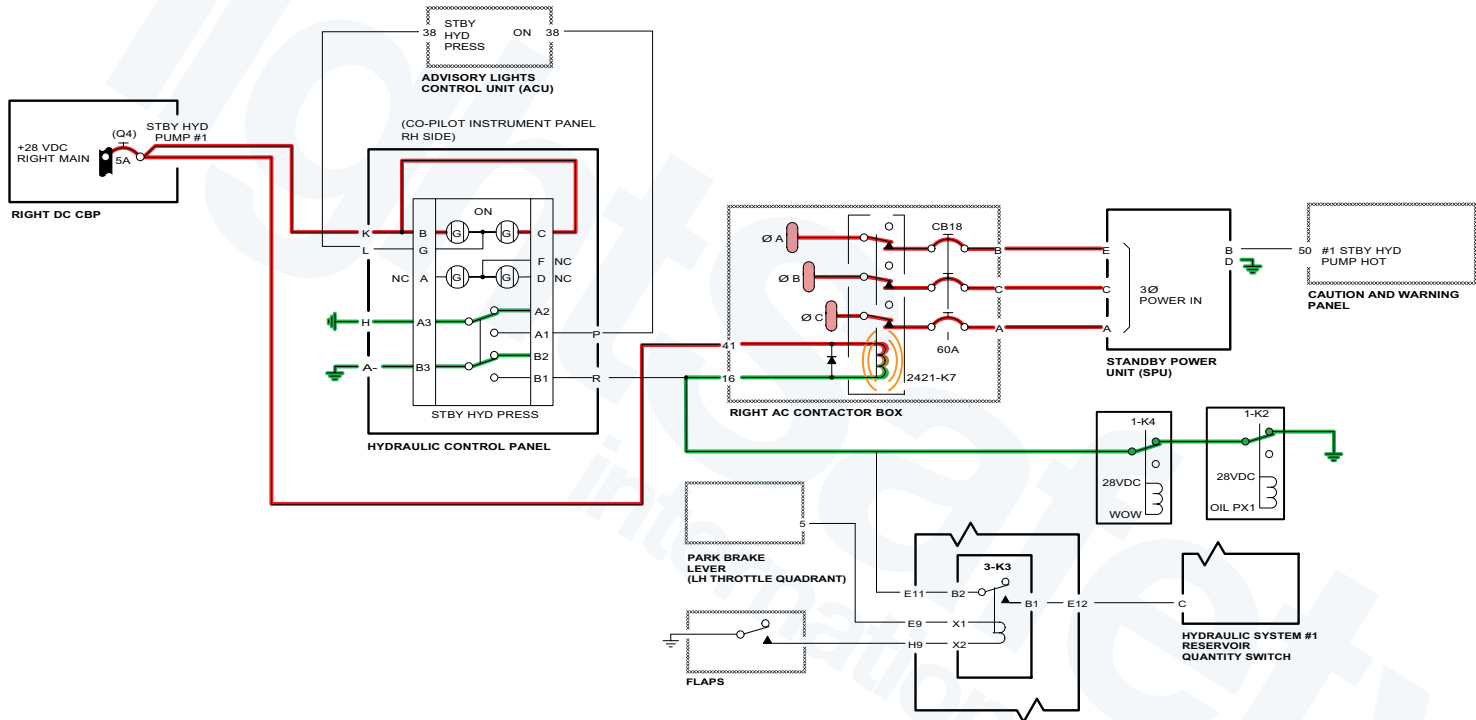
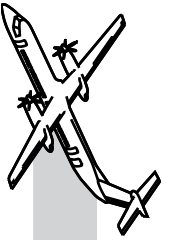
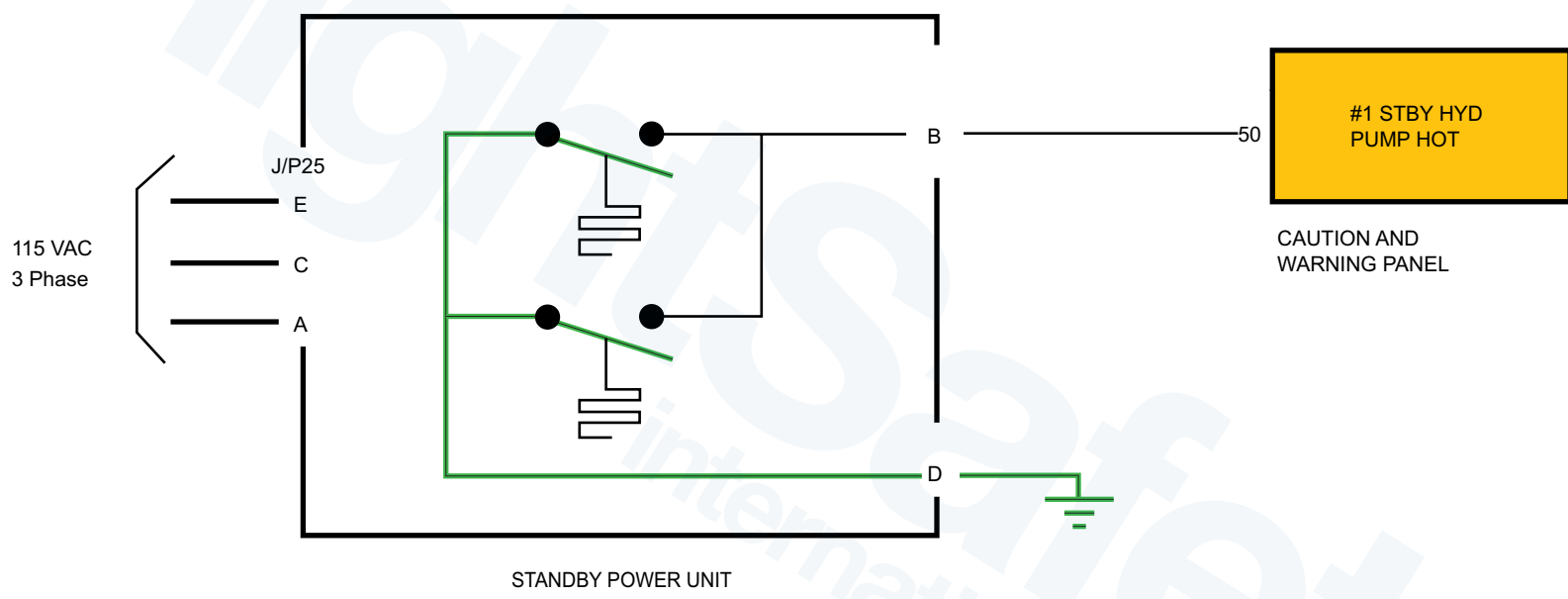
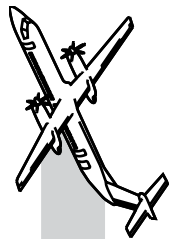


Figure 29-29. SPU Operation - Automatic W Off W - No.1 ENG LOW OIL PX

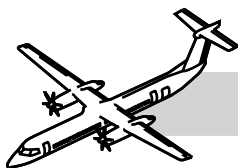


29 HYDRAULICS

29-38

FOR TRAINING PURPOSES ONLY

Figure 29-30. No.1 Hydraulic System Electrical Schematic - SPU Windings Hot



SPU Operation (Motor Hot)

NOTES

Refer to Figure 29-30. No.1 Hydraulic System Electrical Schematic - SPU Windings Hot.

Internal temperature switches monitor the motor windings. A closed switch provides a ground to pin 50 on the CAWP, causing the #1 STBY HYD PUMP HOT light to illuminate. This does not cause the pump to shut-off.

WARNING

BE VERY CAREFUL WHEN YOU APPLY AC ELECTRICAL POWER TO THE AIRCRAFT WHILE IT IS ON JACKS OR IN A WEIGHT-OFF-WHEELS CONDITION (FAR TARGETS). IF YOU DO THIS, THE STANDBY POWER UNIT (SPU) WILL START AUTOMATICALLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO THE AIRCRAFT.

BE VERY CAREFUL BECAUSE THE STANDBY POWER UNIT (SPU) WILL START AUTOMATICALLY IF YOU HAVE THESE CONDITIONS AT THE SAME TIME:

- YOU APPLY EXTERNAL AC ELECTRICAL POWER TO THE AIRCRAFT
- THE PARKING BRAKE IS NOT ON
- THE FLAPS ARE NOT AT ZERO DEGREES.

WARNING

IF YOU ARE NOT CAREFUL, YOU CAN CAUSE INJURY TO PERSONS AND DAMAGE TO THE EQUIPMENT.

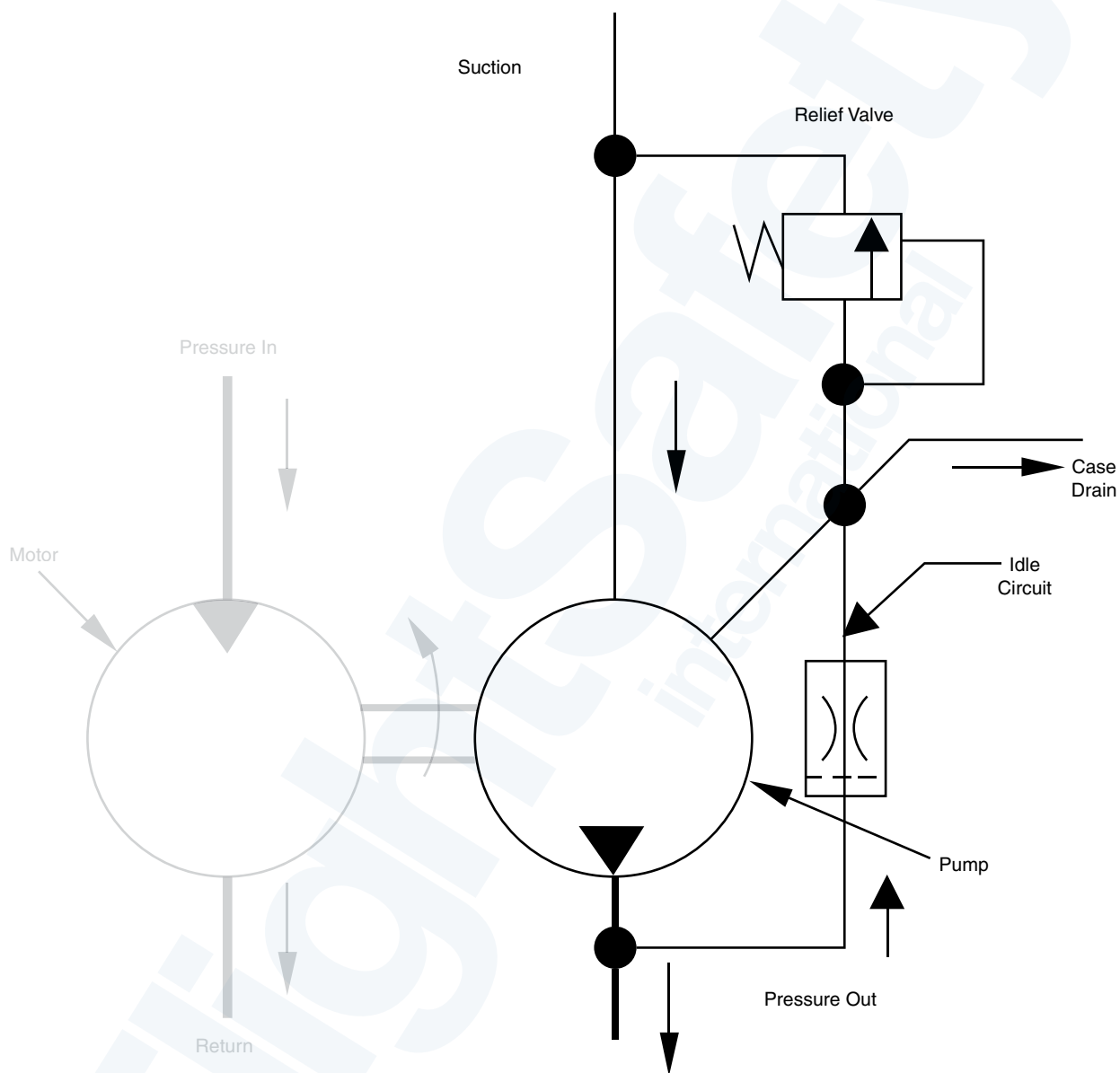
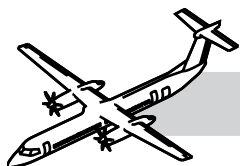
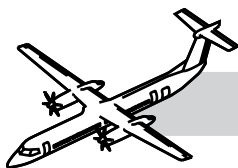


Figure 29-31. PTU Hydraulic System Schematic



No.2 Hydraulic System Operation

NOTES

The PTU selector valve provides the control by starting and stopping the flow of pressurized hydraulic fluid from the No.1 hydraulic system to the PTU motor.

The PTU pump takes fluid from the reservoir through a check valve and the return / suction line to the inlet of the pump. The PTU has a bypass orifice to maintain a steady rotation under no flow demands.

Refer to Figure 29-31. PTU Hydraulic System Schematic.

This bypass flow is ported through the case drain circuit to the system heat exchanger. This action offsets the additional heat created by the PTU operation. The PTU displacement ratio is balanced (flow / pressure) to achieve delivery pressure similar to the EDP throughout the required range. Output flow rate is dependant upon input pressure and flow. (i.e. EDP or SPU).

PTU Operation

Refer to Figure 29-31. PTU Hydraulic System Schematic.

The PTU system has three parts:

- Selector valve
- Motor
- Pump.

The PTU operates:

- Manually if selected on with the switchlight
- Automatically if:
 - Parking Brake OFF and
 - Flaps > 0° and
 - No.2 hydraulic reservoir NOT empty and
 - No.1 system pressure is > 2400 psi.

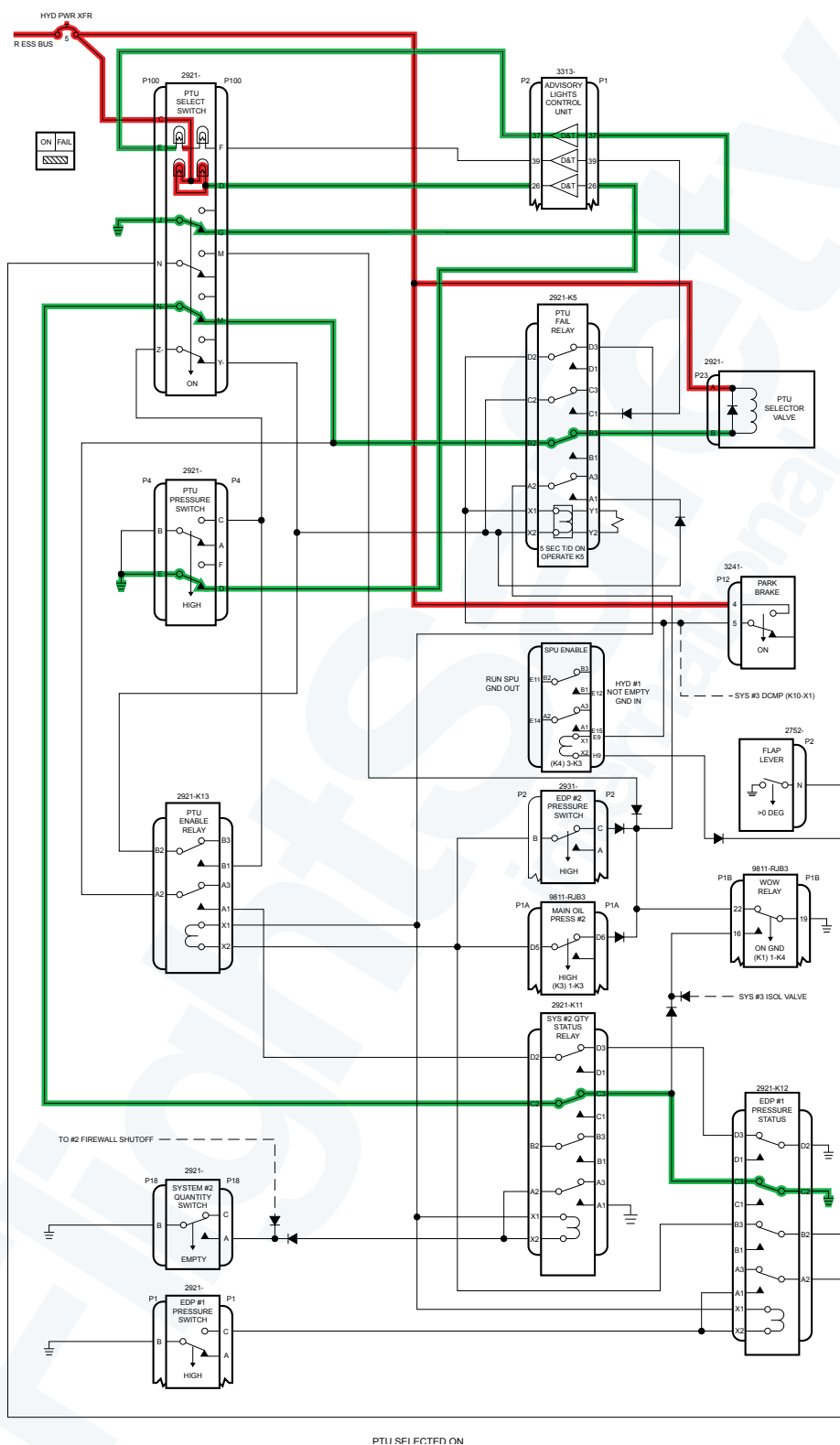
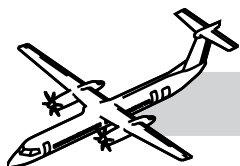
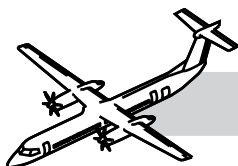


Figure 29-32. No.2 Hydraulic System Electrical Schematic - PTU Manual Operation



PTU Operation with Manual Selection

Refer to Figure 29-32. No.2 Hydraulic System Electrical Schematic - PTU Manual Operation.

Power is supplied to the PTU SELECT SWITCH from the R ESS Bus.

This power is also supplied to the PTU SELECTOR VALVE.

When the PTU SELECT SWITCH is selected ON, a ground is supplied to the ON light of the PTU SELECT SWITCH through the ADVISORY LIGHTS CONTROL UNIT.

A ground is supplied through the PTU PRESSURE SWITCH, through the ADVISORY LIGHTS CONTROL UNIT to the SELECT SWITCH to bring on the green bar.

A ground is supplied through the EDP #1 PRESSURE STATUS de-energized relay K12 to SYS #2 QTY STATUS de-energized RELAY K11, through the SELECT SWITCH, through the PTU FAIL RELAY K5 to the PTU SELECTOR VALVE.

The PTU comes ON.

WARNING

YOU MUST INSTALL THE LOCKPINS ON THE MAIN LANDING GEAR. MAKE SURE YOU ENGAGE THE NOSE GEAR LOCK. IF YOU DO NOT DO THIS, THE LANDING GEAR CAN ACCIDENTALLY RETRACT. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO THE EQUIPMENT.

WARNING

INSTALL THE DOOR LOCKPINS. IF YOU DO NOT DO THIS, YOU CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO THE EQUIPMENT.

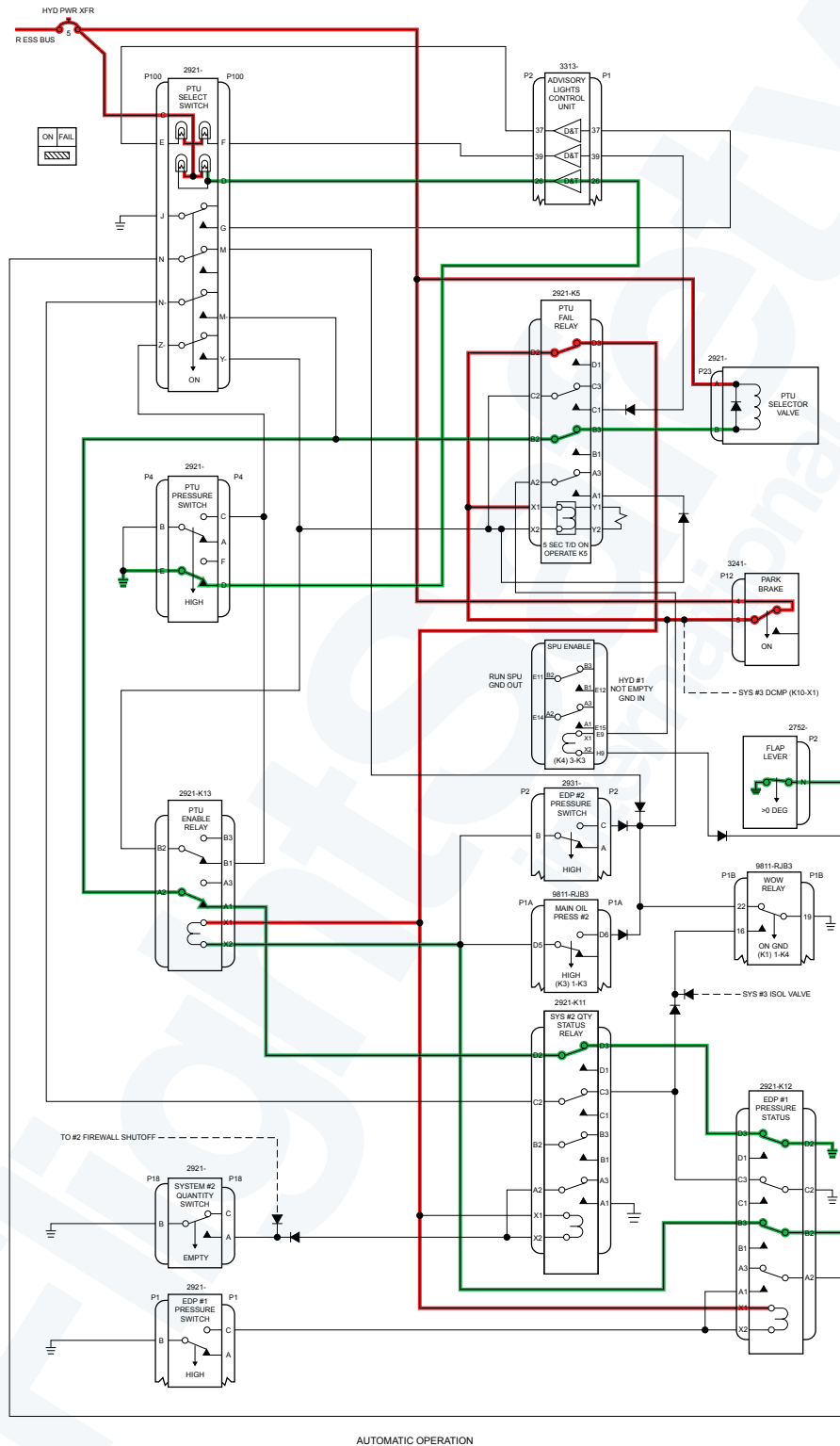
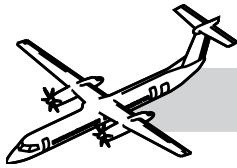
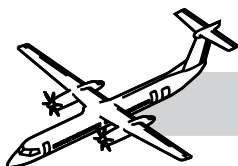


Figure 29-33. PTU Automatic Operation



PTU Automatic Operation

NOTES

Refer to Figure 29-33. PTU Automatic Operation.

With the PARK BRAKE selected OFF ESS power from the ESS BUS is supplied through the de-energized PTU FAIL RELAY K5 to the PTU ENABLE RELAY K13, the ground for K13 is through the de-energized EDP #1 PRESSURE STATUS RELAY K12 and the flap lever in the 0 degree position.

The PTU SELECTOR VALVE receives its ground through the de-energized PTU FAIL RELAY K5, the energized PTU ENABLE RELAY K13, and the de-energized SYS #2 QTY STATUS RELAY K11 and EDP #1 PRESSURE STATUS RELAY K12.

The PTU comes ON.

The green bar has power from the R ESS Bus and finds a ground through ADVISORY LIGHTS CONTROL UNIT and the PTU PRESSURE SWITCH in high.

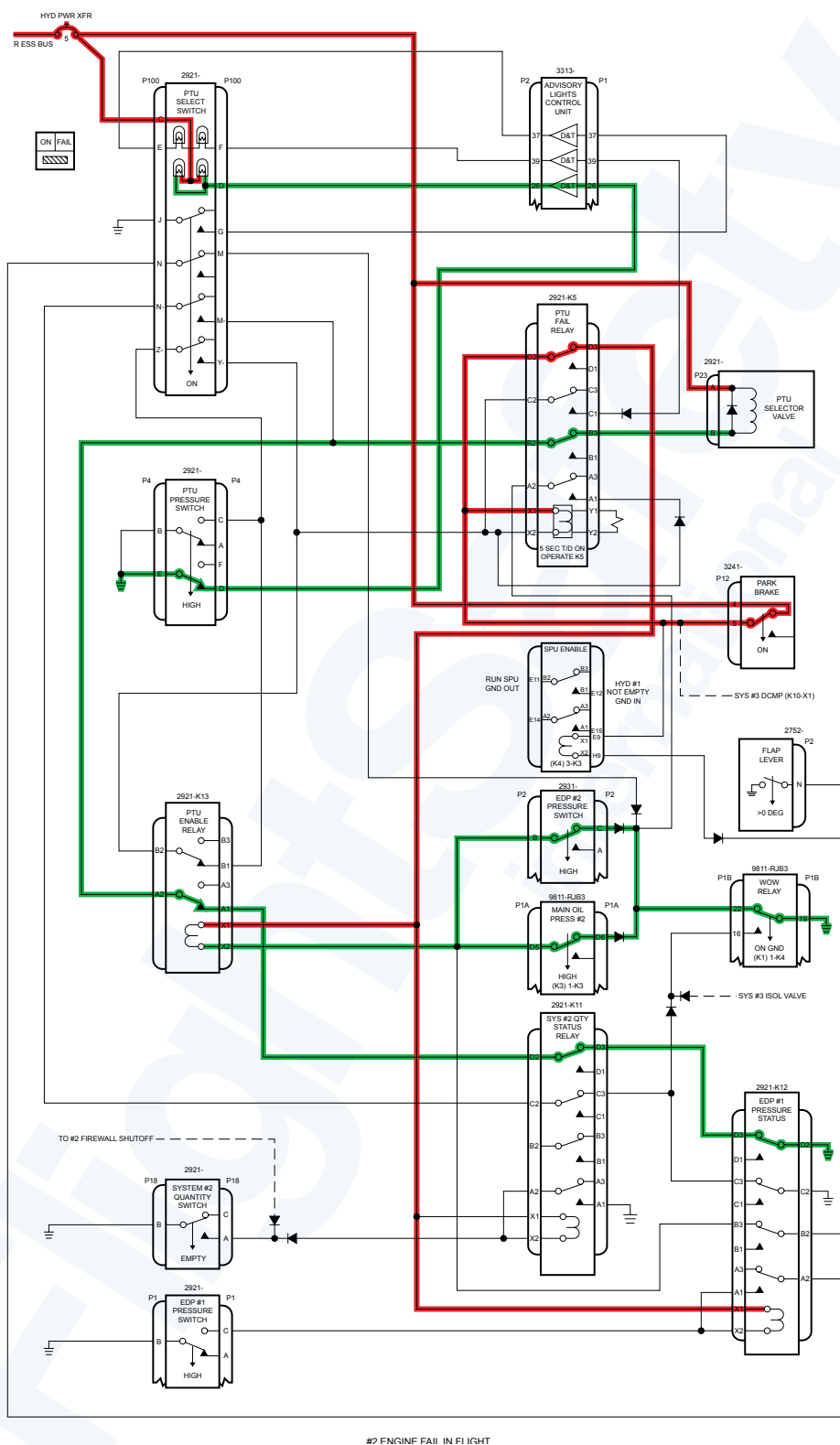
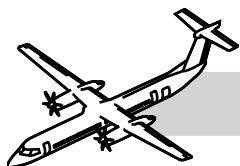
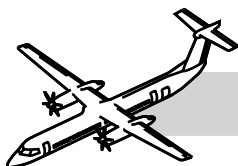


Figure 29-34. No.2 Engine Fail In Flight



PTU Operation for No.2 Engine Failure in Flight and/or #2 EDP Failure in Flight

NOTES

Refer to Figure 29-34. No.2 Engine Fail In Flight.

The power supply remains the same as previously described, but the ground is different.

The ground for the PTU ENABLE RELAY K13 is through the MAIN OIL PRESS #2 SWITCH in low, or the EDP #2 PRESSURE SWITCH in low and then through the WOW RELAY to ground.

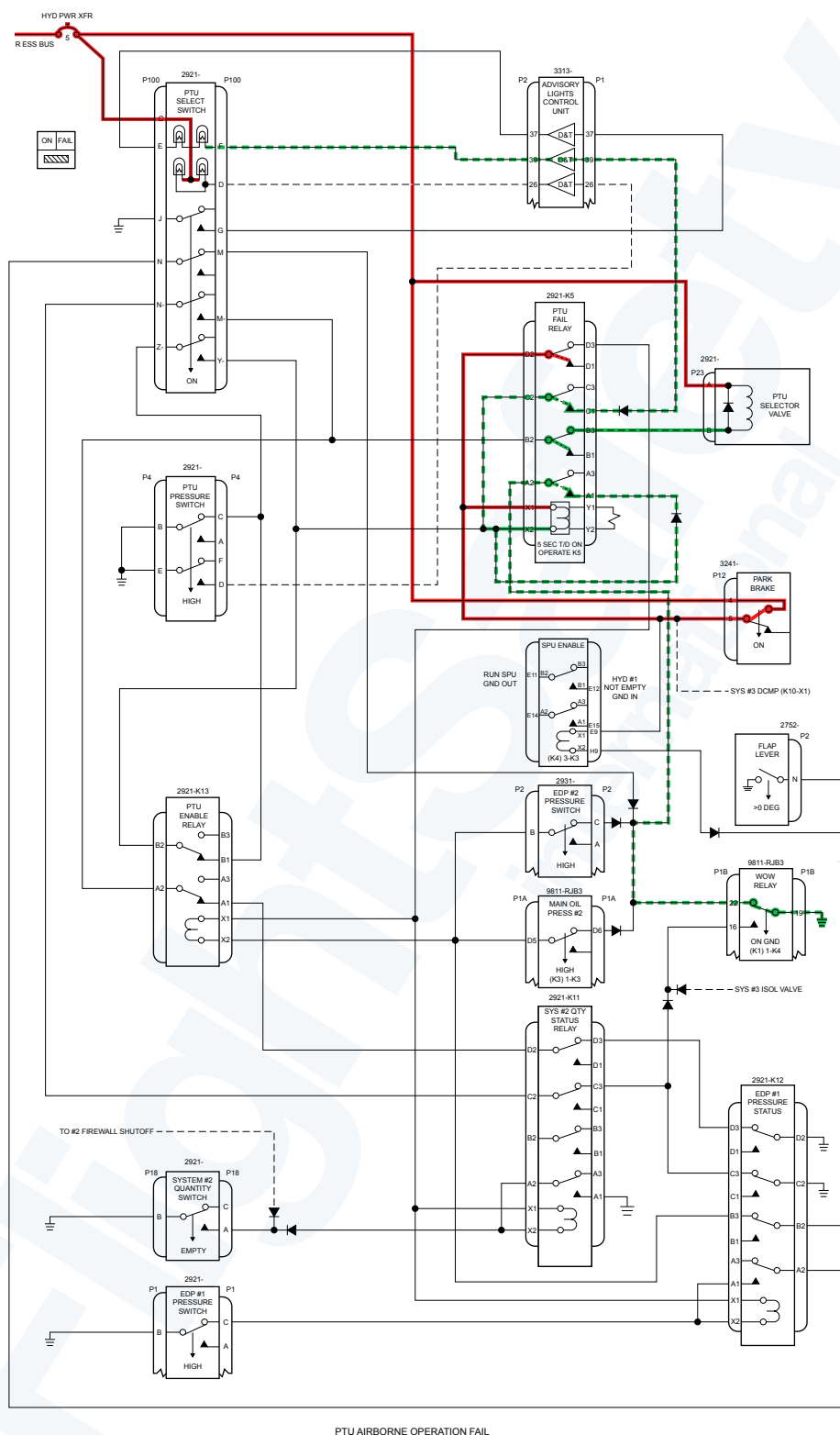
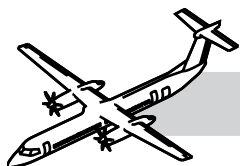
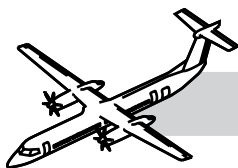


Figure 29-35. PTU Failure



PTU Failure

NOTES

Refer to Figure 29-35. PTU Failure.

Power to the PTU FAIL RELAY K5 is through the PARK BRAKE switch when the park brake is selected OFF.

In normal operation of the PTU there is no ground for the K5 relay because of the position of the PTU PRESSURE SWITCH in the high pressure mode.

If the pressure from the PTU fails, the PTU PRESSURE SWITCH will provide a ground to the K5 RELAY through the PTU ENABLE RELAY K13.

After 5 seconds time delay, K5 relay will energize removing the power from the PTU SELECTOR VALVE to shutdown the PTU, and provide a ground for the FAIL indication on the PTU SELECT SWITCH.

In flight a ground through the WOW 1-K4 will keep relay K5 energized following a loss of PTU pressure, isolating the PTU SELECTOR VALVE.

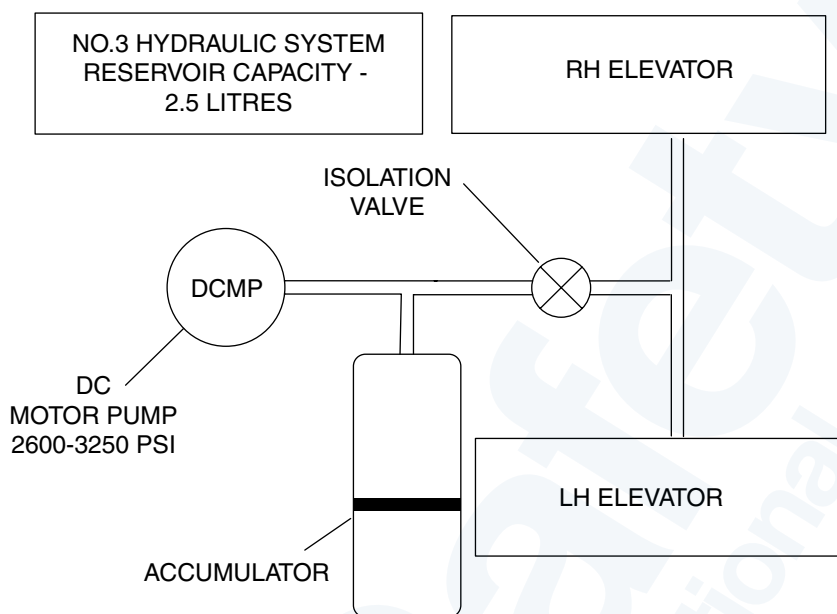
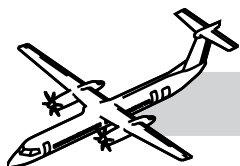


Figure 29-36. No.3 Hydraulic System Block Diagram

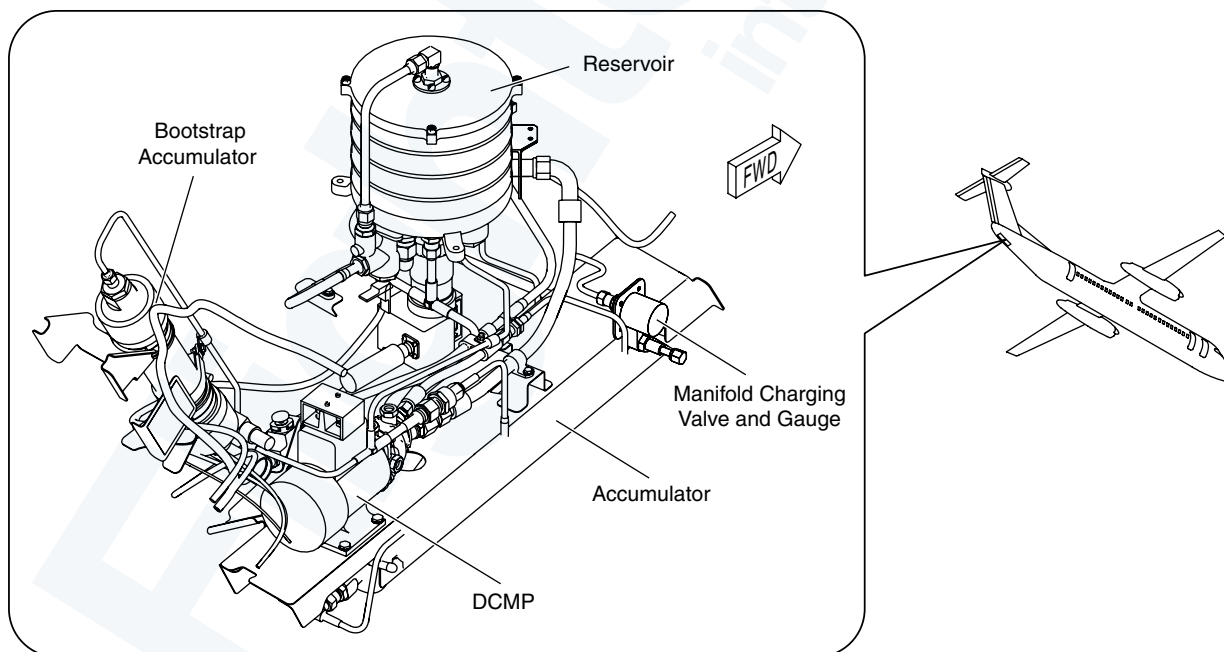
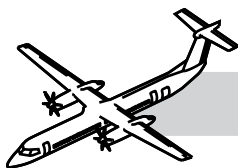


Figure 29-37. No.3 Hydraulic System PreMod



29-12-00 NO.3 MAIN HYDRAULIC SYSTEM

GENERAL

CAUTION

If WOFFW No. 3 Hydraulic System operates automatically when No.1 and / or No.2 Hydraulic System pressure is low, or when both engines oil pressure indicates low oil pressure. No. 3 Hydraulic System can be manually engaged by setting the switch of the isolation valve of No. 3 Hydraulic System to OPEN.

NOTE

No. 3 Hydraulic System will be isolated and automatically switched off when No.1 or No.2 Hydraulic System is re-pressurized.

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF THE MOVEMENT OF THE ELEVATOR SURFACES. MOVEMENT OF THE FLIGHT CONTROL SURFACES CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO THE EQUIPMENT.

The No.3 hydraulic system installed in the aft fuselage supplies backup hydraulic power to the left and right elevator inboard PCU's.

SYSTEM DESCRIPTION

Refer to:

- Figure 29-36. No.3 Hydraulic System Block Diagram.
- Figure 29-37. No.3 Hydraulic System PreMod.

During normal operation, the system is in an active standby mode with the accumulator isolation valve closed. In the event of a dual engine failure, when both EDPs and the SPU are unavailable, or when either system 1 or 2 isolation valve input pressure switches indicate low pressure, the No.3 hydraulic system will power the inboard PCU's.

The system has the main components that follow:

- DCMP
- No.3 Hydraulic system reservoir
- Hydraulic accumulators
- Hydraulic accumulator isolation valve
- Hydraulic pressure filter manifold
- Hydraulic pressure filter element
- Pressure switches
- Manifold charging gauge.

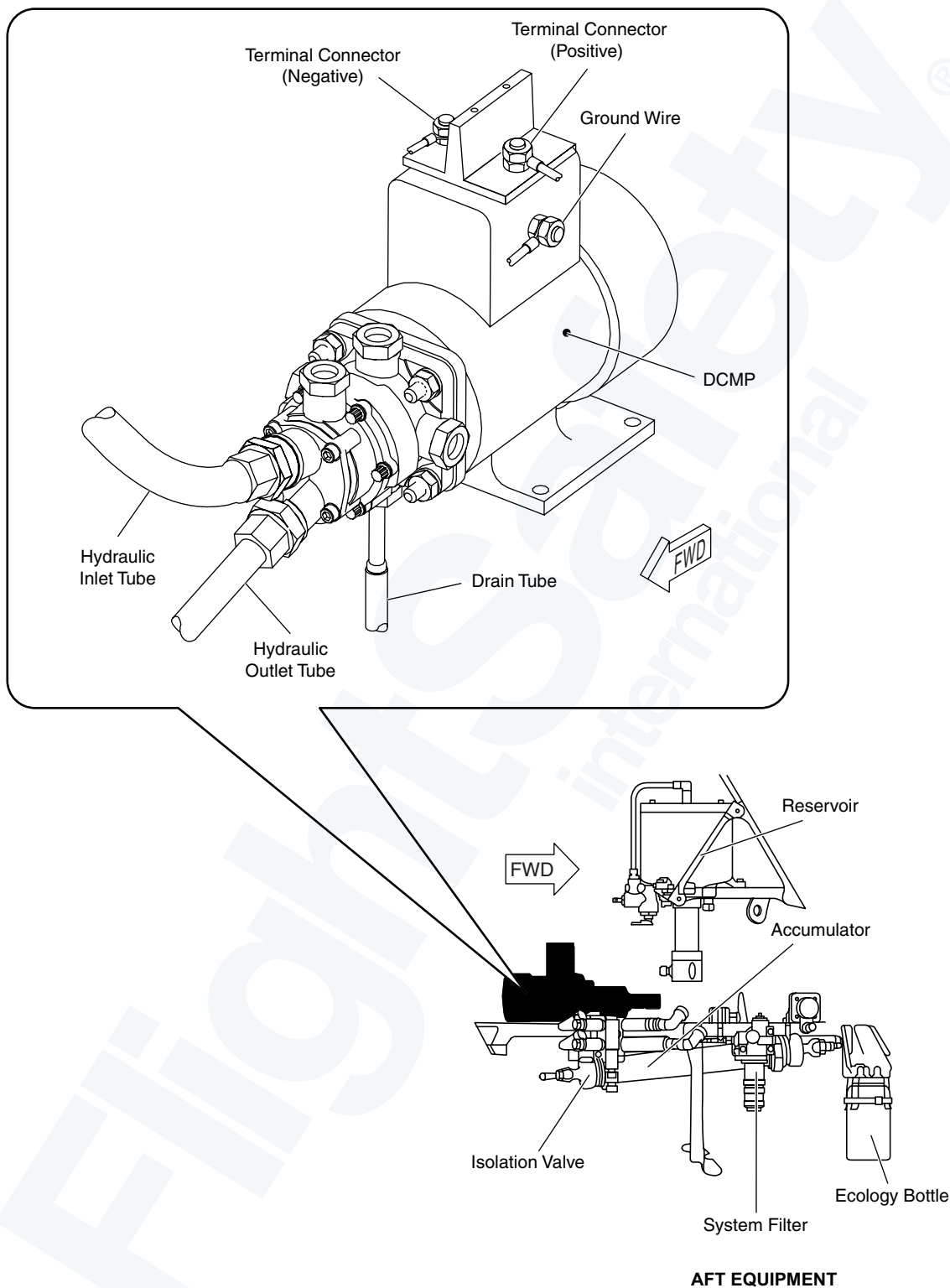
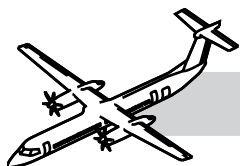
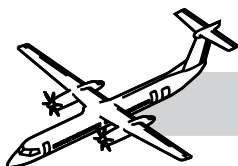


Figure 29-38. DC Motor Driven Pump (DCMP)



COMPONENT DESCRIPTION

DC Motor Driven Pump (DCMP)

Refer to:

- Figure 29-38. DC Motor Driven Pump (DCMP).
- Figure 29-39. View Looking Outboard PostMod.

The No. 3 DC Motor Pump (DCMP) is used to pressurize the accumulator in the No. 3 hydraulic system. Electrical power is supplied to the DCMP by the STBY battery through the dc battery bus (standby battery). The DCMP will operate intermittently during normal flight to maintain a No. 3 system pressure range of 2600 to 3250 psi. DCMP operation is controlled by two pressure switches installed on the accumulator isolation valve.

NOTE

The DCMP will operate intermittently on the ground when the dc battery bus is energized by the batteries or external ground power.

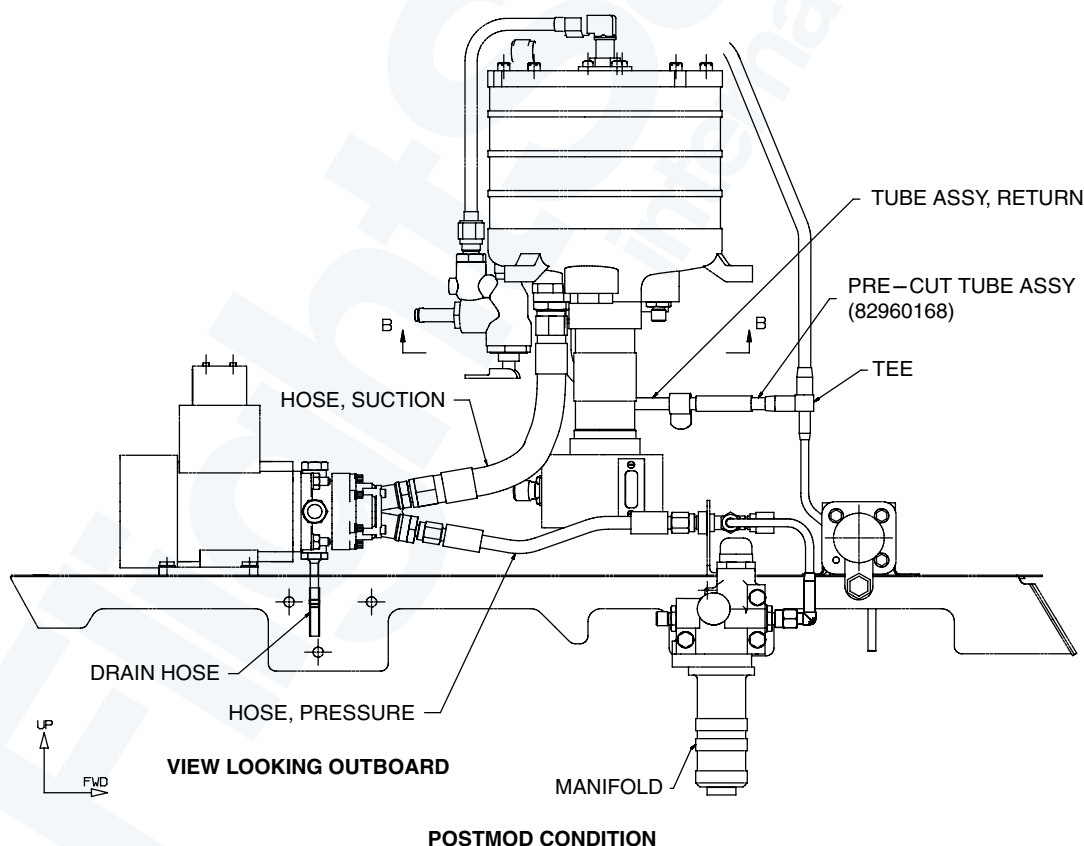
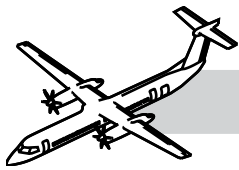


Figure 29-39. View Looking Outboard PostMod



No.3 Hydraulic System Reservoir

Refer to Figure 29-40. No.3 Hydraulic System Reservoir.

A bootstrap reservoir stores hydraulic fluid for the No.3 hydraulic system. The reservoir provides local and remote fluid quantity indication.

A bleed/relief valve installed in the reservoir provides manual bleeding and pressure relief capabilities.

The bootstrap cylinder maintains reservoir pressure at approximately 55 psi.

NOTE

The volume of the No.3 hydraulic system reservoir is 2.5 liters.

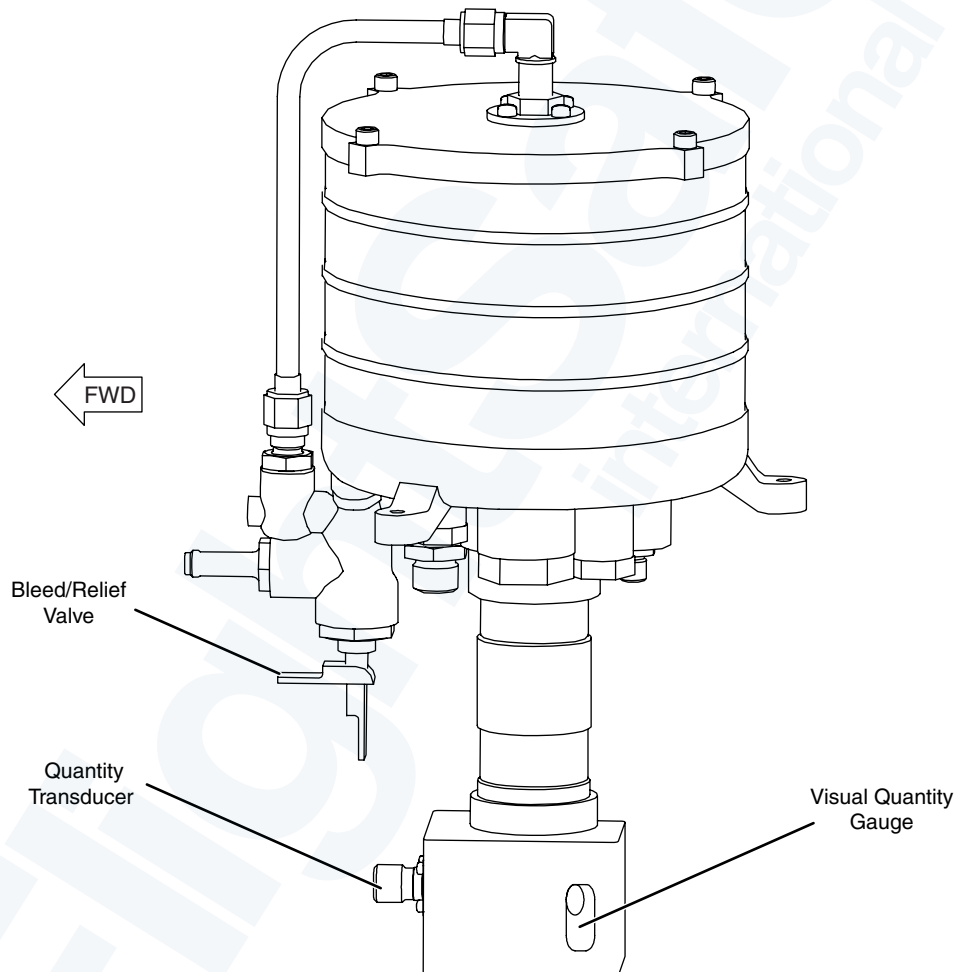
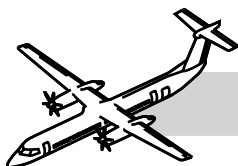


Figure 29-40. No.3 Hydraulic System Reservoir



Ecology Bottle

Refer to Figure 29-41. Hydraulic System Ecology Bottle.

A 0.75 liter plastic ecology bottle collects hydraulic fluid released from the reservoir bleed/relief valve and leakage from the DCMP shaft seal.

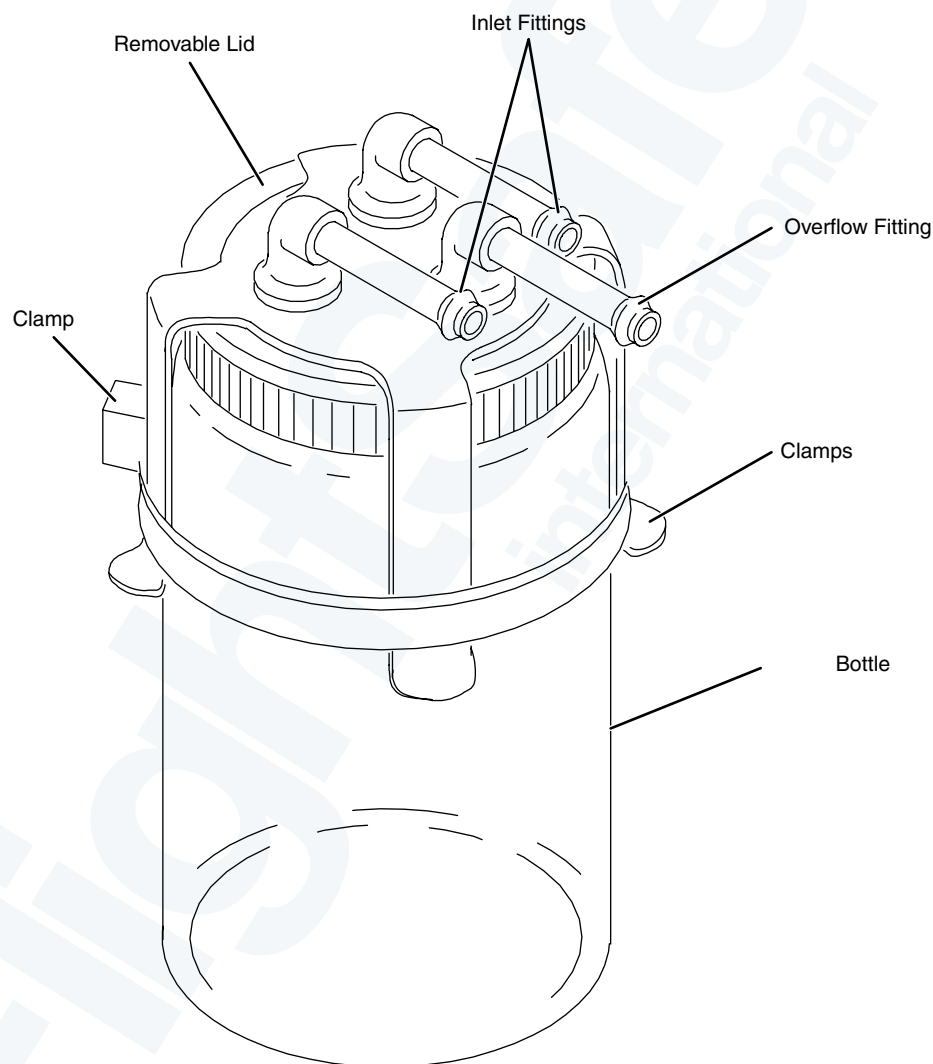
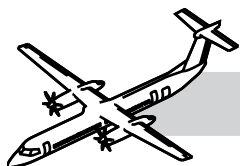


Figure 29-41. Hydraulic System Ecology Bottle



No.3 Hydraulic Accumulator

Refer to Figure 29-42. No. 3 Hydraulic System Accumulator.

The accumulator supplies the hydraulic pressure to the inboard elevator PCUs when required.

The nitrogen pre-charge pressure is dependant upon temperature. When the temperature is between 18 and 37°C (63-99°F), the gas pre-charge pressure will be between 1900 and 2000 psi.

Servicing is through a remote charging valve and indicator.

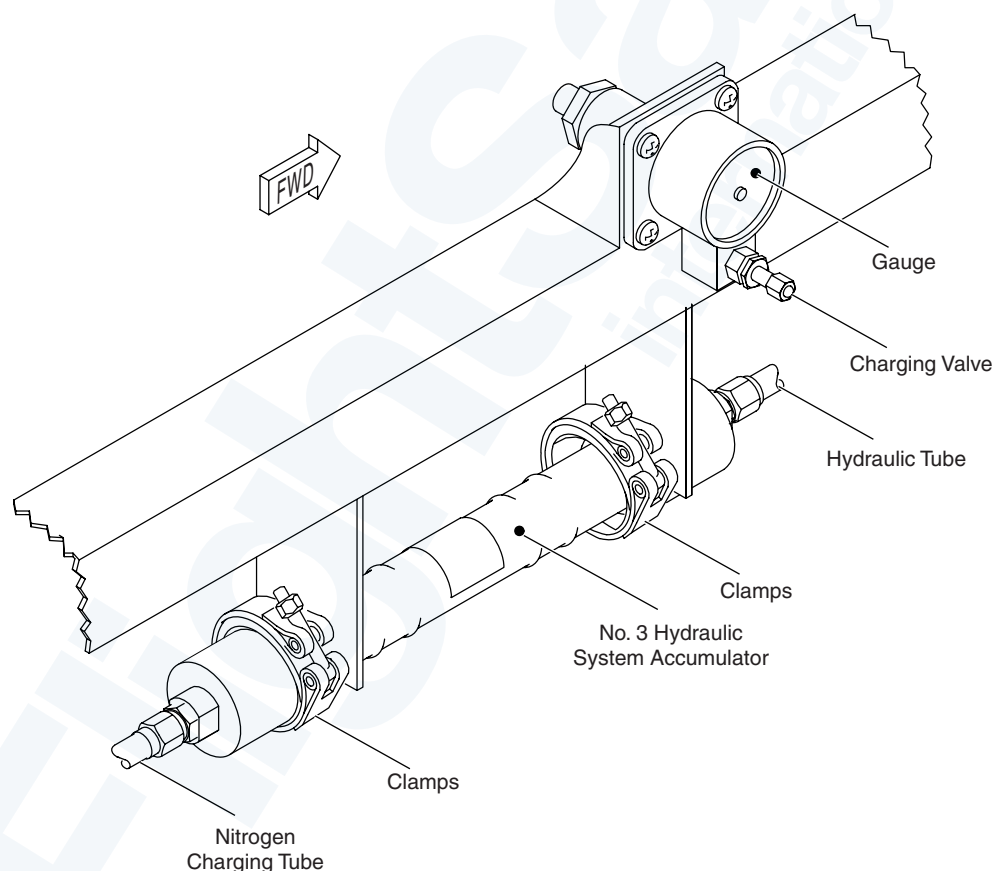
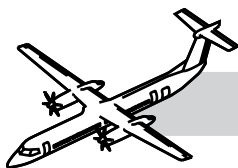


Figure 29-42. No. 3 Hydraulic System Accumulator



No.3 Hydraulic Bootstrap Accumulator

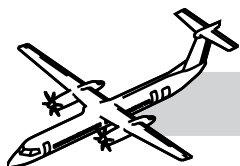
Refer to Figure 29-43. No.3 Hydraulic Bootstrap Accumulator.

A bootstrap accumulator keeps pressure on the reservoir to ensure a positive inlet pressure to the DCMP.

The nitrogen pre-charge pressure is dependant upon temperature. Between 17 and 38°C the gas pre-charge pressure is between 1100 and 1200 psi. The gas inlet end of the accumulator has a gas charging valve and pressure gauge for servicing.



Figure 29-43. No.3 Hydraulic Bootstrap Accumulator



No.3 Hydraulic Accumulator Isolation Valve

Refer to Figure 29-44. No.3 Hydraulic Accumulator Isolation Valve.

During normal flight operations the isolation valve is energized closed to isolate the inboard elevator PCU's from hydraulic pressure. When open, the isolation valve allows hydraulic pressure to power the inboard elevator PCU's.

The isolation valve de-energizes open for the conditions that follow:

- Weight Off Wheels (WOFFW) and
- No.1 and/or No.2 hydraulic system pressure low or
- No.1 and No.2 engine oil pressure low.

The isolation valve can be manually opened from the flight compartment when the HYD No.3 ISOL VLV switch on the HYDRAULIC CONTROL Panel is set to OPEN.

A dump/thermal relief valve is provided for pressure relief capability and to allow the accumulator to be manually de-pressurized for maintenance.

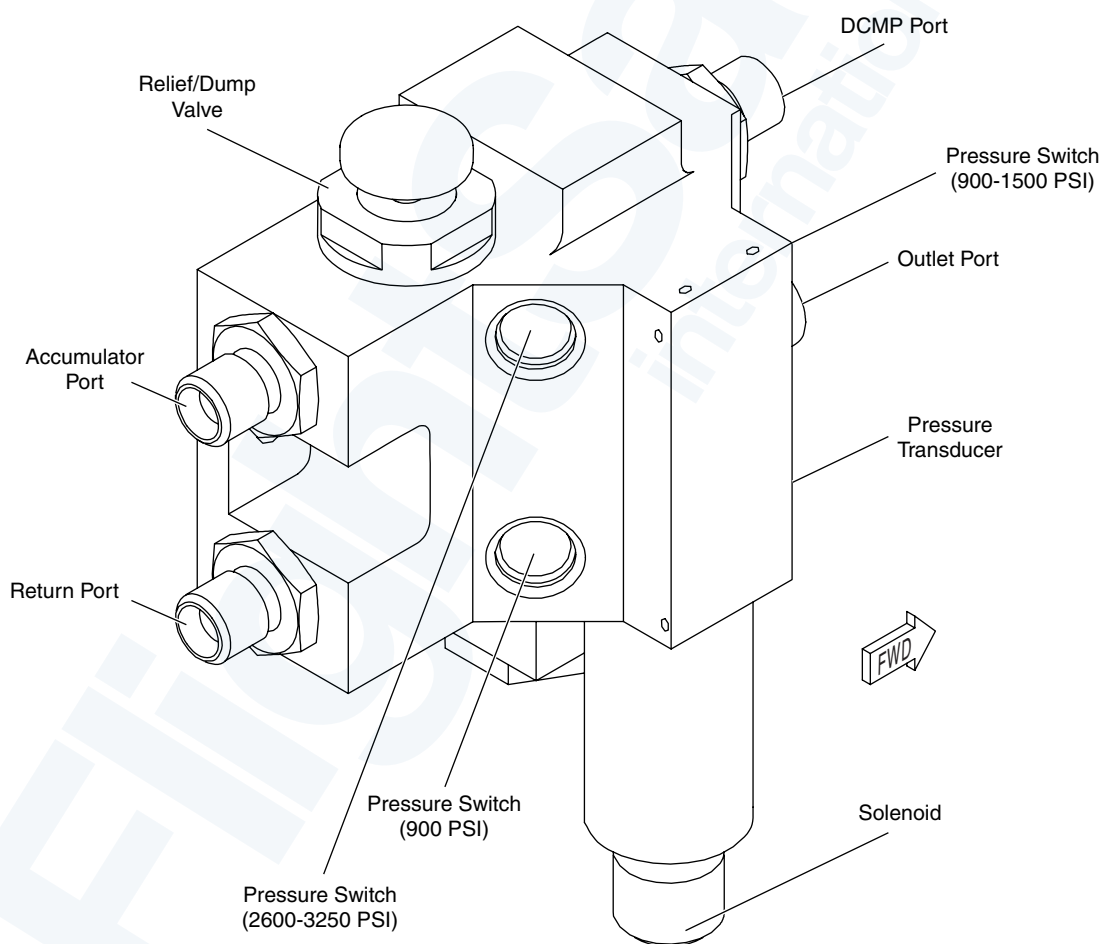
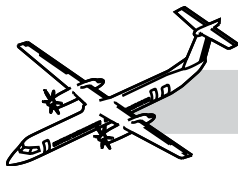


Figure 29-44. No.3 Hydraulic Accumulator Isolation Valve



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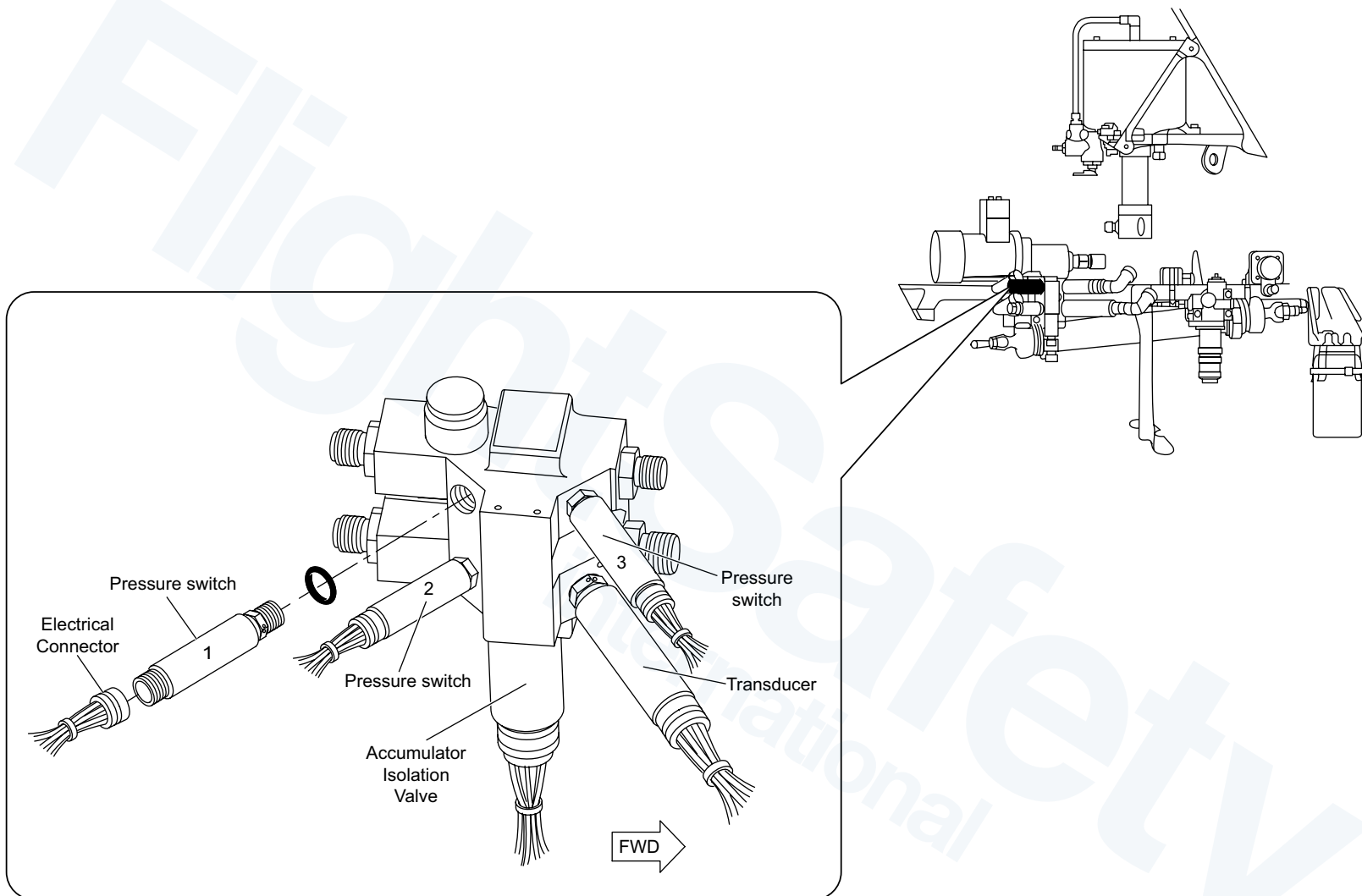
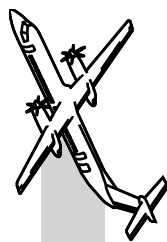
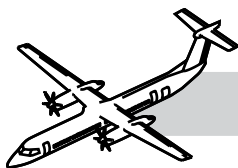


Figure 29-45. Pressure Switches Synoptic





No.3 Hydraulic System Pressure Switches

Refer to:

- Figure 29-45. Pressure Switches Synoptic.
- Figure 29-46. Pressure Switches.

There are three pressure switches and a transducer installed on the accumulator isolation valve.

The No.1 switch cycles the DCMP on/off to keep the accumulator pressure between 2600 psi and 3250 psi.

The No.2 switch causes the #3 HYD PUMP caution light to illuminate when:

- The accumulator pressure is less than 900 psi or
- The pump has been operating for longer than 60 seconds on the ground.

The No.3 switch is installed downstream of the isolation valve. It causes the ELEV PRESS caution light to illuminate if all three systems are supplying power to the elevator PCU's.

The pressure transducer sends signals to the FDPS for pressure display on the MFDs.

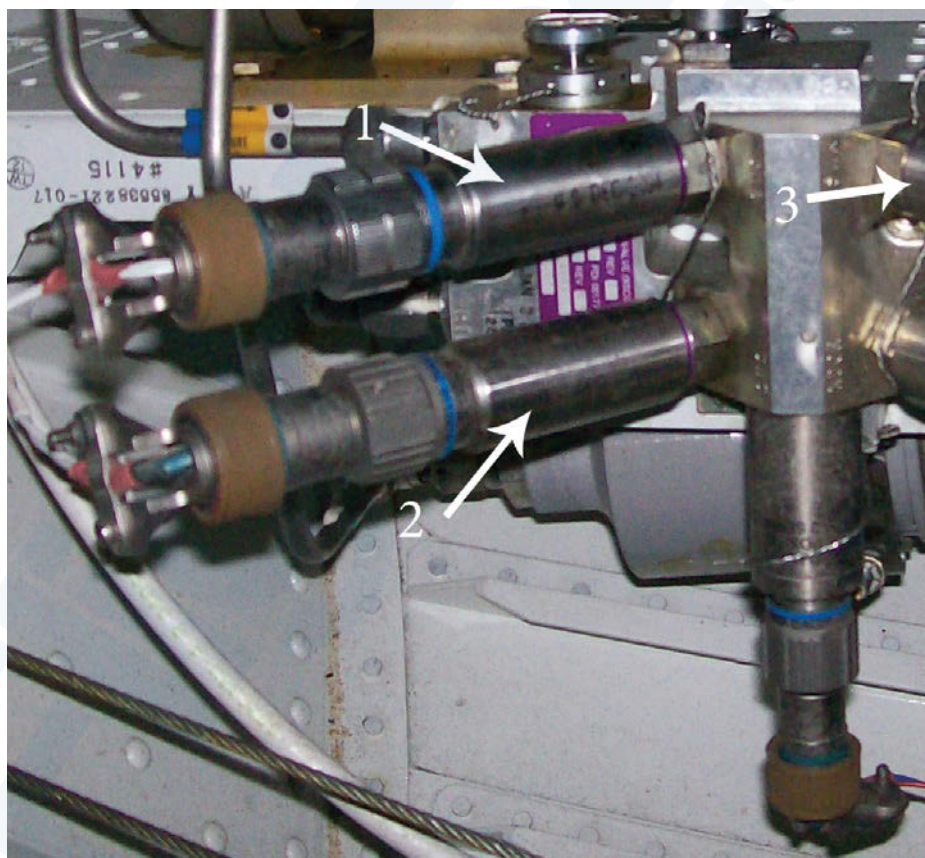


Figure 29-46. Pressure Switches

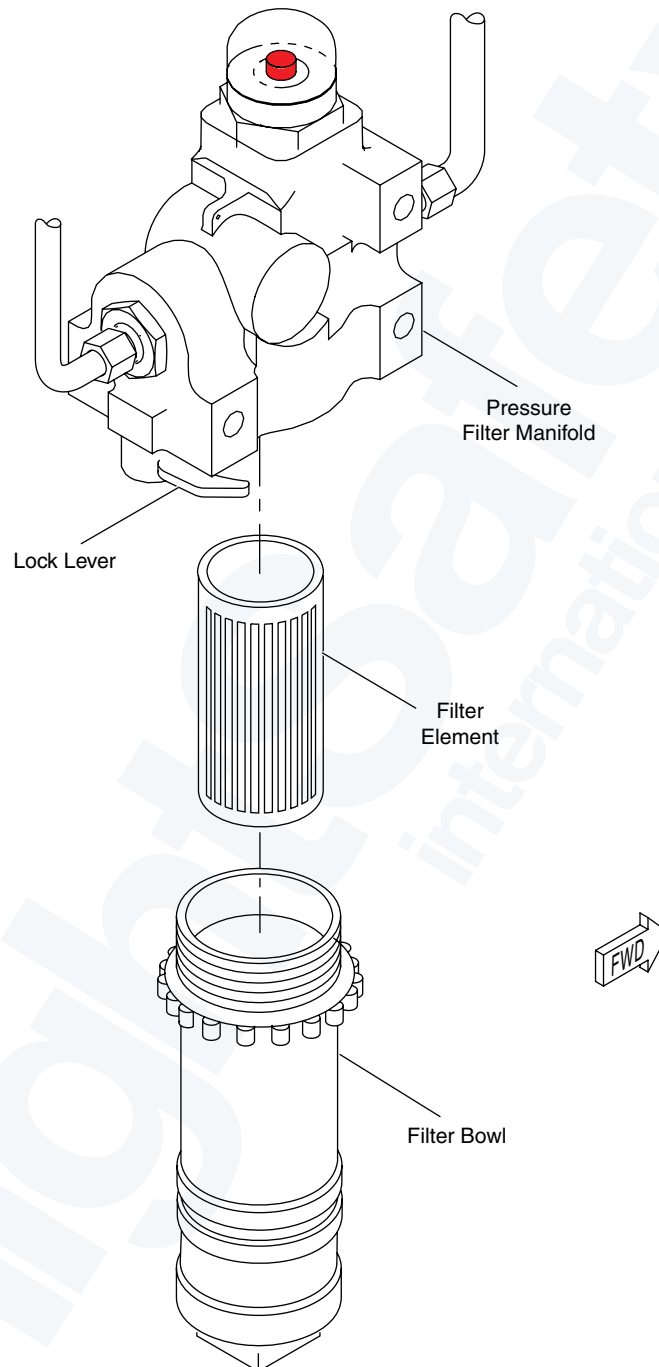
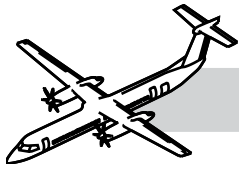
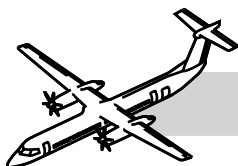


Figure 29-47. Pressure Filter Manifold (System No.3)



No.3 Hydraulic System Pressure Filter Manifold

Refer to:

- Figure 29-47. Pressure Filter Manifold (System No.3).
- Figure 29-48. No.3 Hydraulic Pressure Filter Manifold.

The No.3 hydraulic system contains a non-bypass, 15 micron pressure filter element and bowl and a Differential Pressure Indicator (DPI).

A shut-off device stops fluid flow through the filter when the filter bowl is removed.

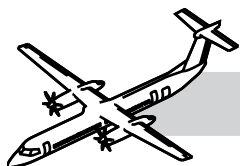
The resettable DPI gives an indication of differential pressure across the filter element at 70 ± 10 psi. The indicator will not actuate with a low fluid temperature.

NOTE

The No.3 hydraulic system pressure filter bowl is fully interchangeable with the case drain filter bowls of the No.1 and No.2 hydraulic system filter manifolds.



Figure 29-48. No.3 Hydraulic Pressure Filter Manifold



Manifold Charging Gauge

Refer to Figure 29-49. Manifold Charging Gauge.

A charging valve and gauge are used to charge the accumulator with dry nitrogen in accordance with chart located nearby.

The fitting is a standard MS size and the gauge reads from 0 to 5000 psi with 1000 psi as major increments.

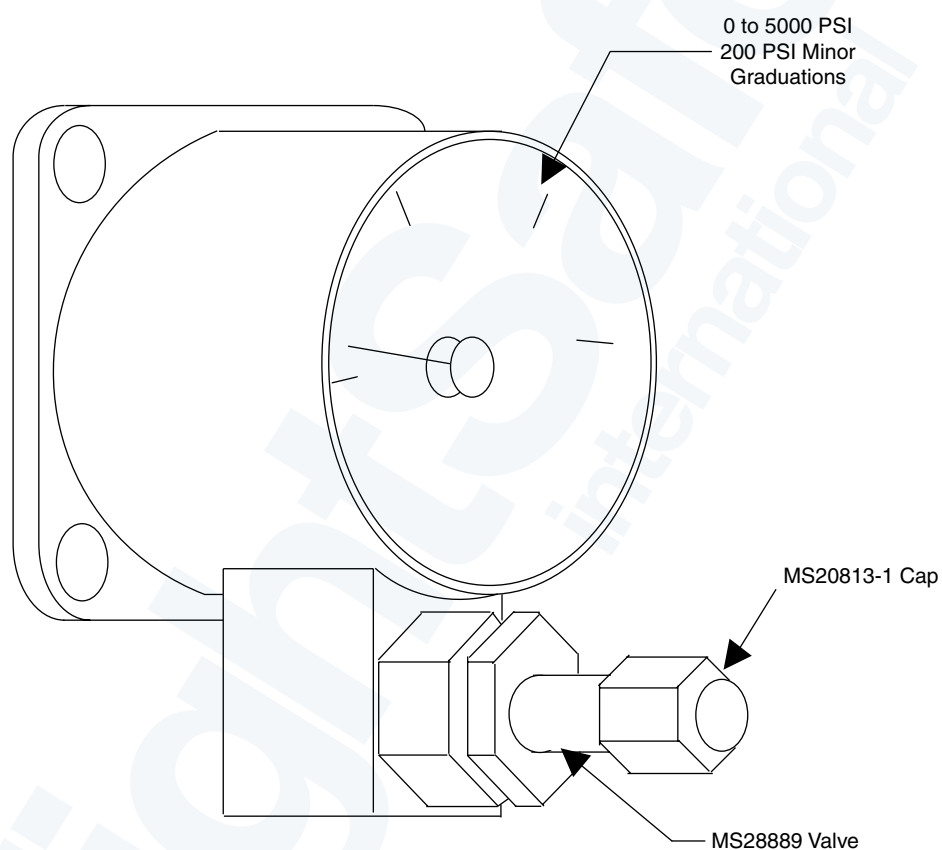
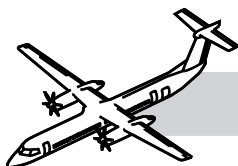


Figure 29-49. Manifold Charging Gauge



CONTROLS AND INDICATIONS

Refer to Figure 29-50. Hydraulic Control Panel - Detail.

The No.3 hydraulic system isolation valve switchlight is blank when the isolation valve is in the NORM (closed) position. Selection of the guarded **HYD No.3 ISOL VLV** switch manually opens the valve and the amber colored **OPEN** segment indication is displayed.



Figure 29-50. Hydraulic Control Panel - Detail

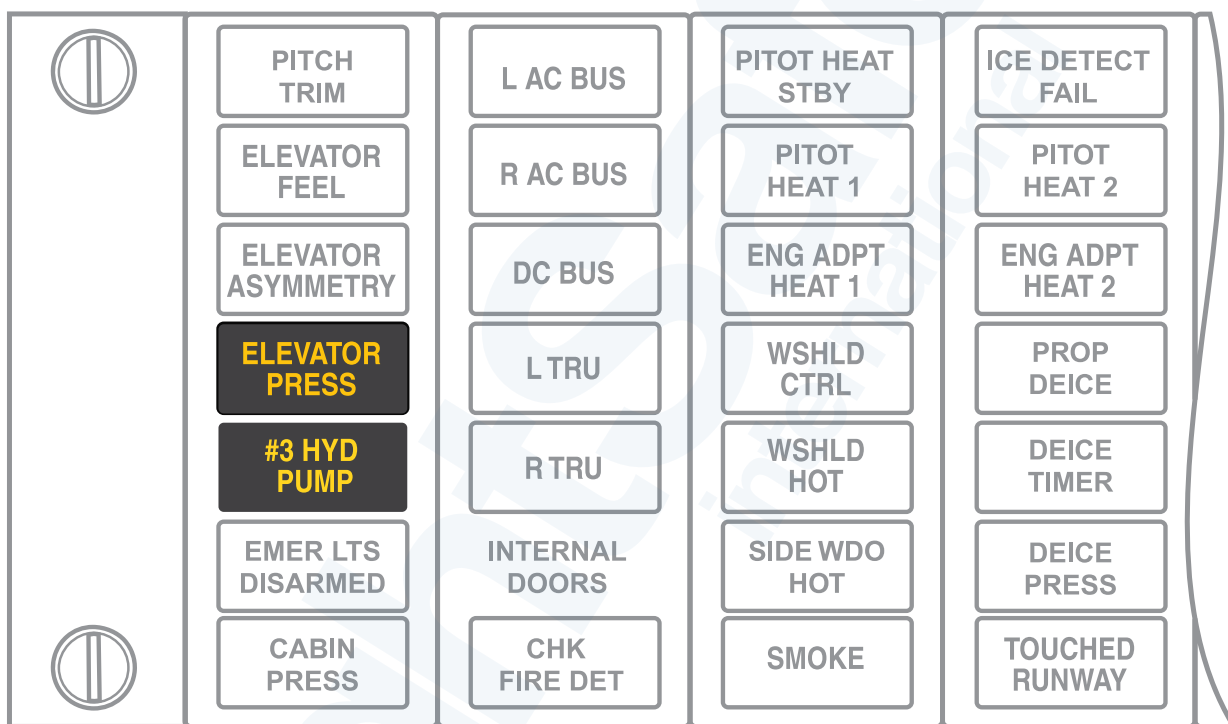
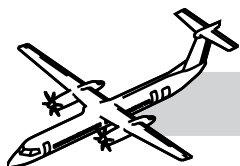
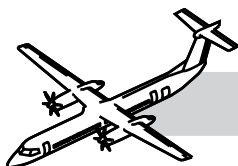


Figure 29-51. Cautions - CAWP



Elevator Press

NOTES

Refer to Figure 29-51. Cautions - CAWP.

A pressure switch downstream of the No.3 isolation valve and the No.1 and No.2 isolation valve inlet pressure switches are effectively connected in series. If all three hydraulic systems are supplying pressure to the elevator PCUs, then the ELEVATOR PRESS caution light illuminates.

No.3 HYD PUMP

Refer to Figure 29-51. Cautions - CAWP.

The No.3 HYD PUMP caution light illuminates when:

- The system pressure falls to 900 psi (6205 kPa) or
- The pump has been operating for longer than 60 seconds on the ground.

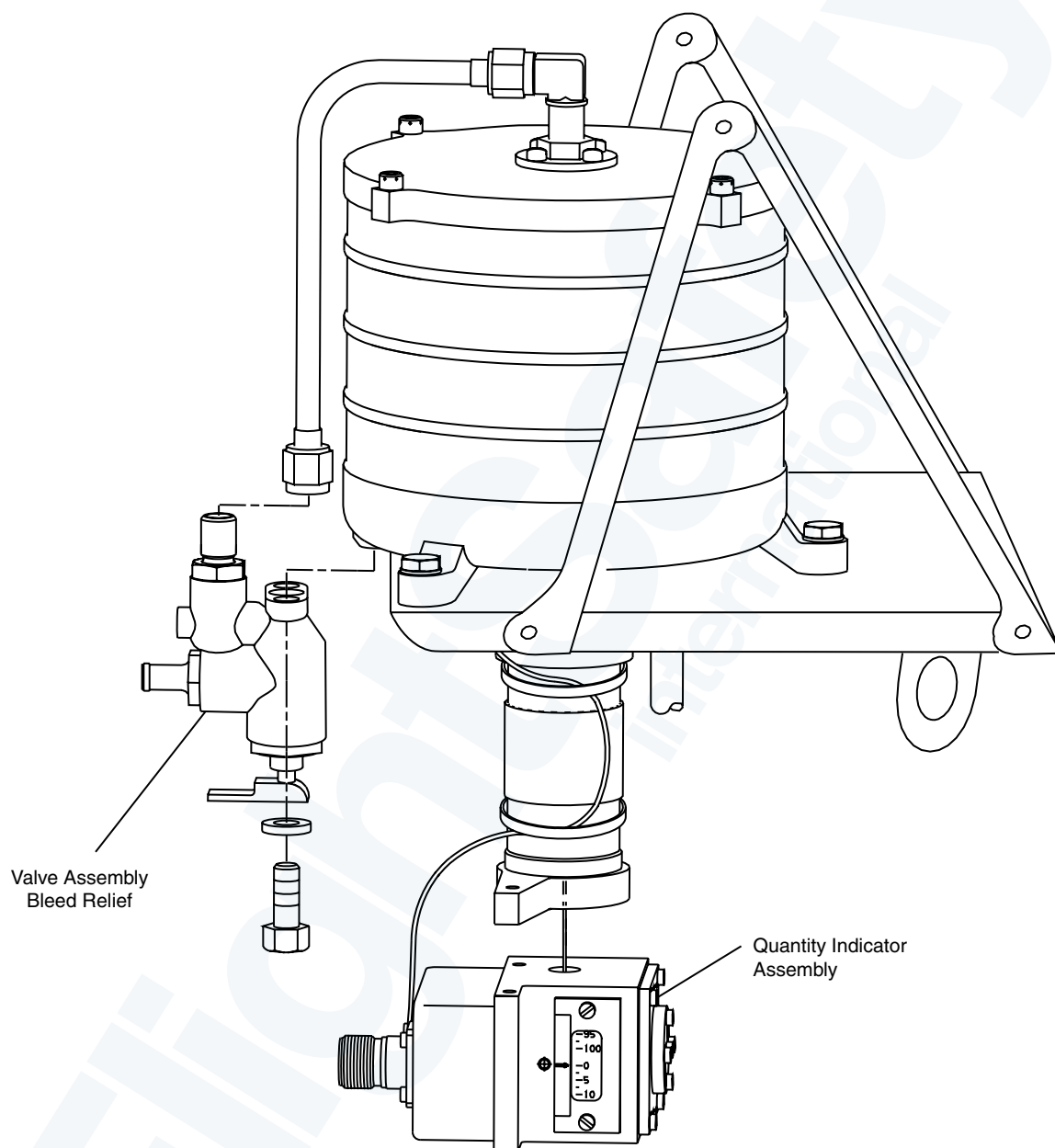
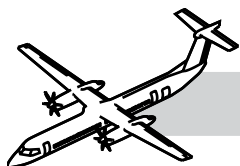
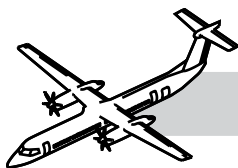


Figure 29-52. Quantity Indicator in No.3 Hydraulic System Reservoir



Quantity Indicator

Refer to:

- Figure 29-52. Quantity Indicator in No.3 Hydraulic System Reservoir.
- Figure 29-53. Quantity Indicator in No.3 Hydraulic System.

A direct reading quantity gauge is operated mechanically by the differential piston in the reservoir. This mechanical movement also rotates a 1000 ohm potentiometer to provide an input to the FDPS for display on the MFDs.

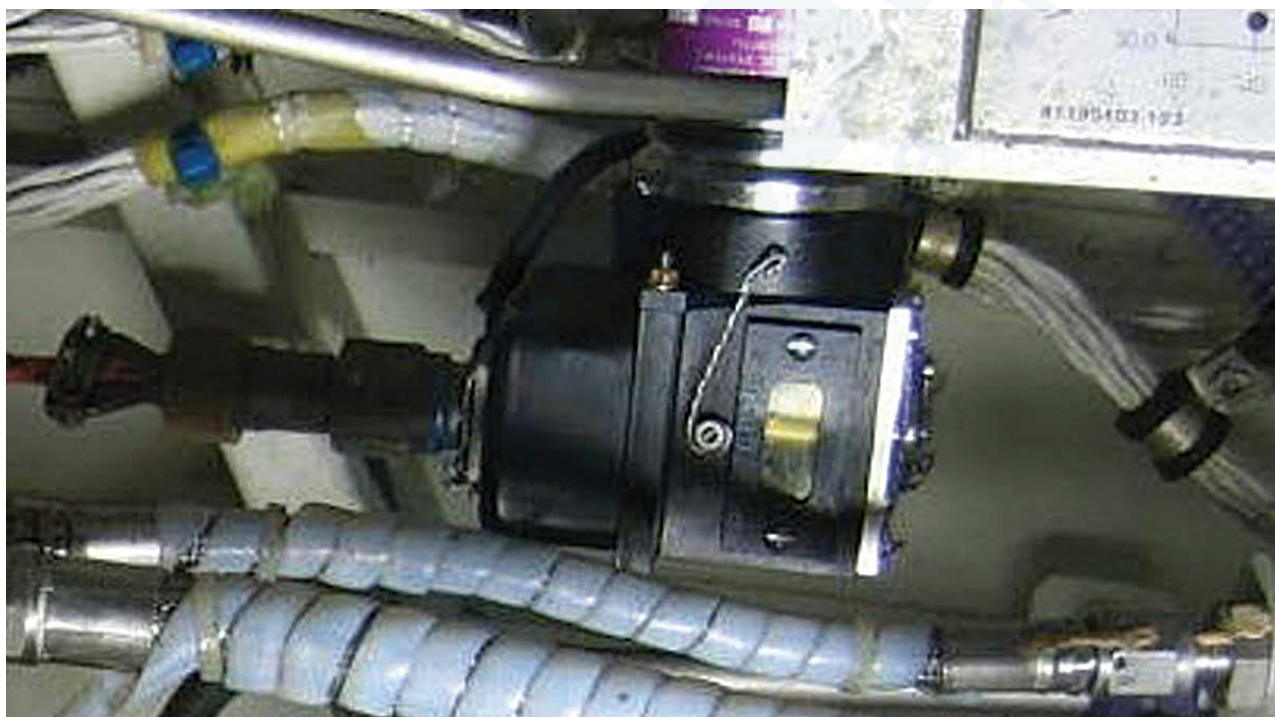


Figure 29-53. Quantity Indicator in No.3 Hydraulic System

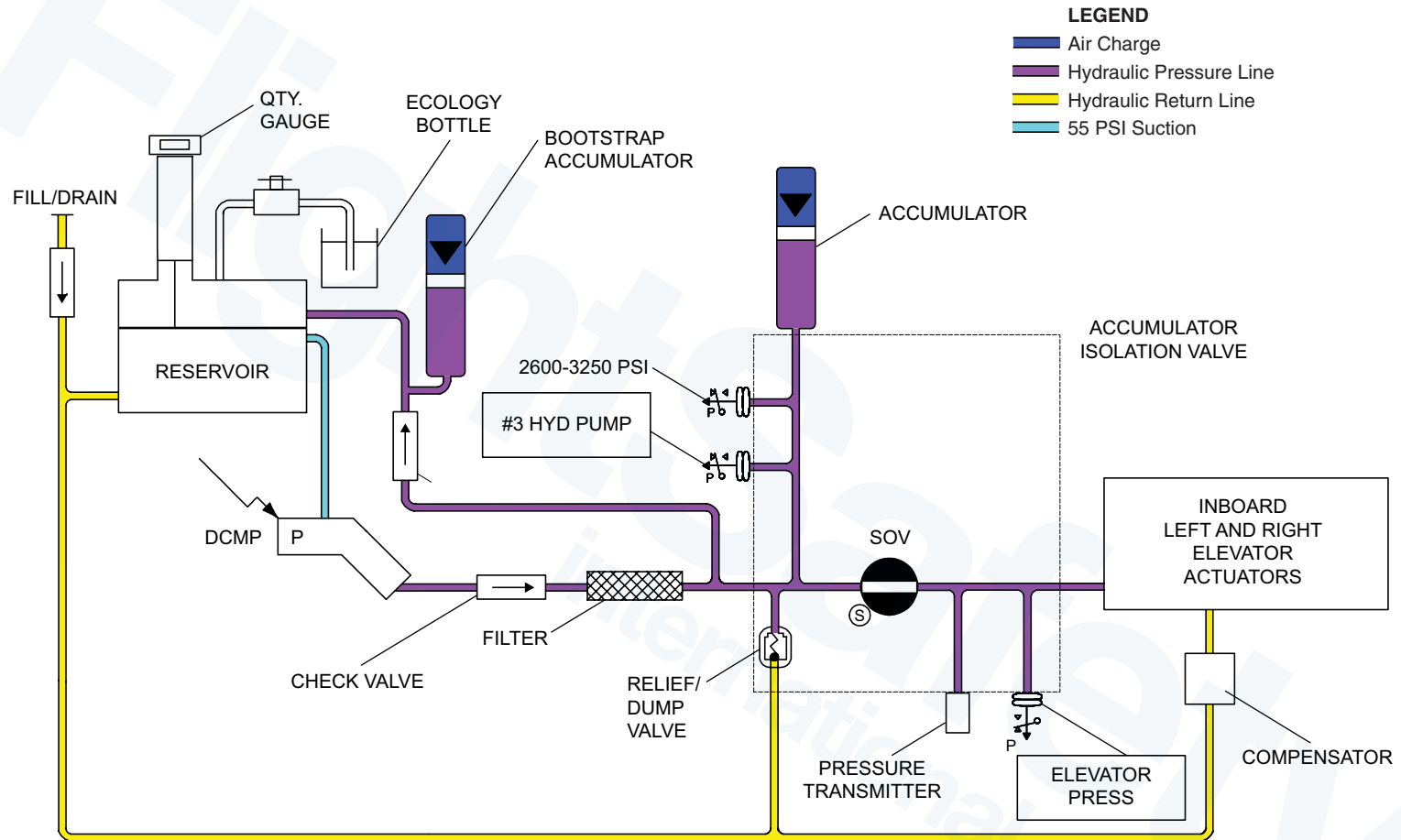
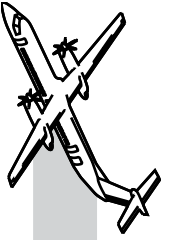
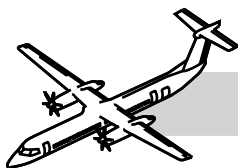


Figure 29-54. No.3 Hydraulic System - Schematic





OPERATION

NOTES

Normal System Operation

Refer to Figure 29-53. Quantity Indicator in No.3 Hydraulic System.

DCMP Control Logic

The DCMP, which is powered by the STBY battery bus, runs intermittently to keep the system accumulator charged.

The DCMP contactor will be energized (DCMP on) when:

- No.1 and/or No.2 engine oil pressure is high and
- Park brake is released and
- Accumulator pressure is less than 2600 psi and
- WOW.

ISOL Valve Control Logic

With WOFFW and a dual engine failure, or the loss of either hydraulic system, the isolation valve is de-energized OPEN and pressure is applied to the inboard elevator PCUs.

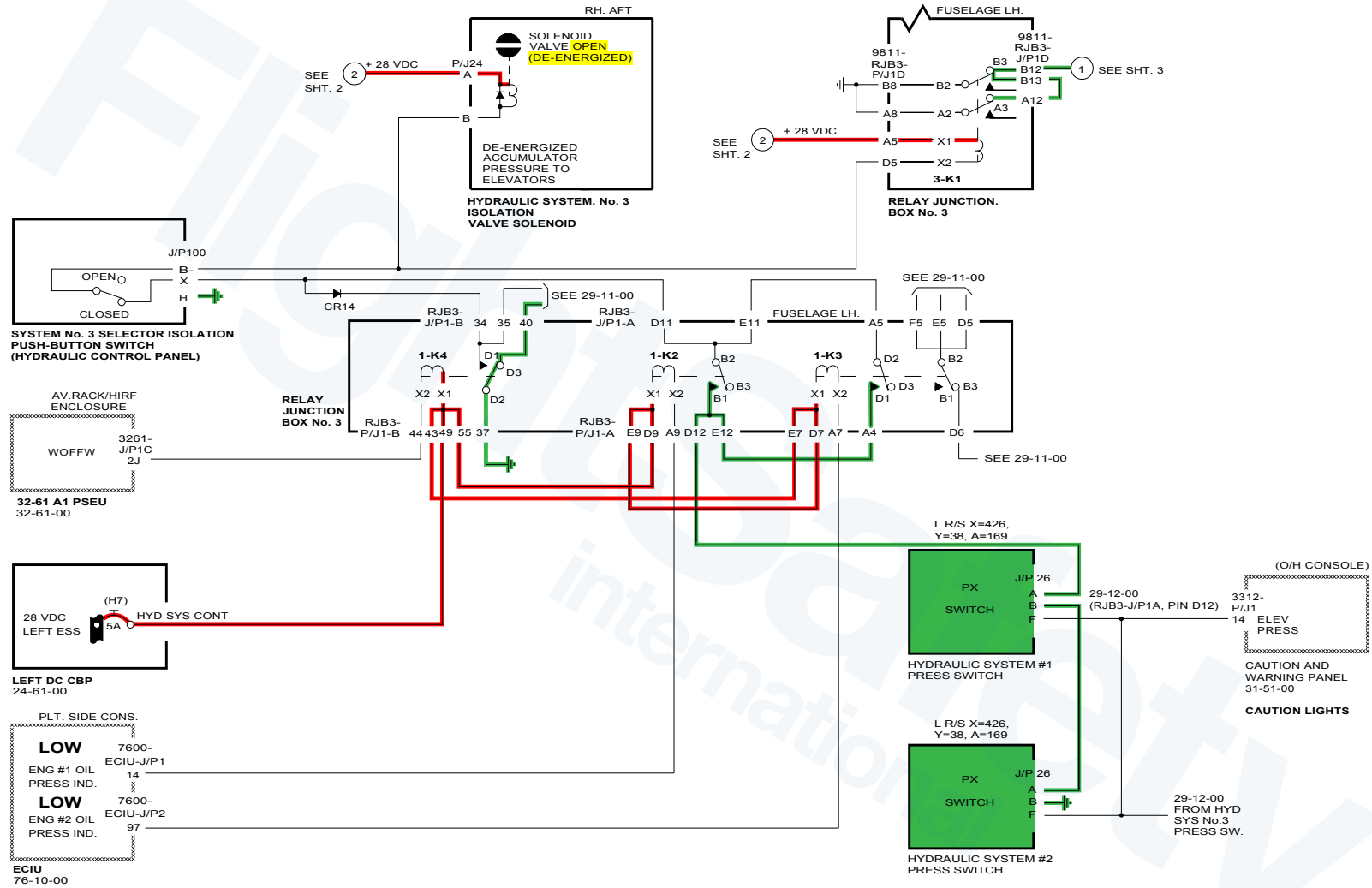
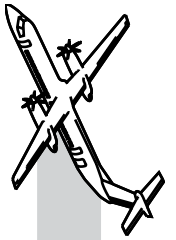
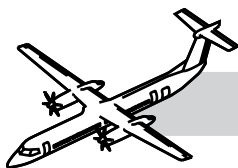


Figure 29-55. No.3 Hydraulic System Isolation Valve (Open) Automatically - Due to a Loss of Both Engines In Flight





No.3 System Isolation Valve Operation (Open) In Flight, Engines Not Running

NOTES

Refer to Figure 29-55. No.3 Hydraulic System Isolation Valve (Open) Automatically - Due to a Loss of Both Engines In Flight.

Condition: Due to a Loss of Both Engines In Flight.

The isolation valve will de-energize (Open) when the ground used to energize the 1-K2 and the 1-K3 relays is no longer provided, due to loss of oil pressure from both engines.

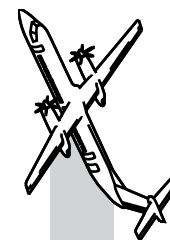
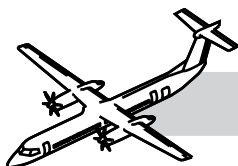


Figure 29-56. No.3 Hydraulic System Isolation Valve (Open) Automatically - Due to No.1 Hydraulic System Failure In Flight



No.3 System Isolation Valve Operation (Open) In Flight, Loss of a One Hydraulic System

NOTES

Refer to Figure 29-56. No.3 Hydraulic System Isolation Valve (Open) Automatically - Due to No.1 Hydraulic System Failure In Flight.

Condition: Due to No.1 Hydraulic System Failure In Flight.

The isolation valve will de-energize (Open) when the ground is no longer provided by either hydraulic pressure switch.

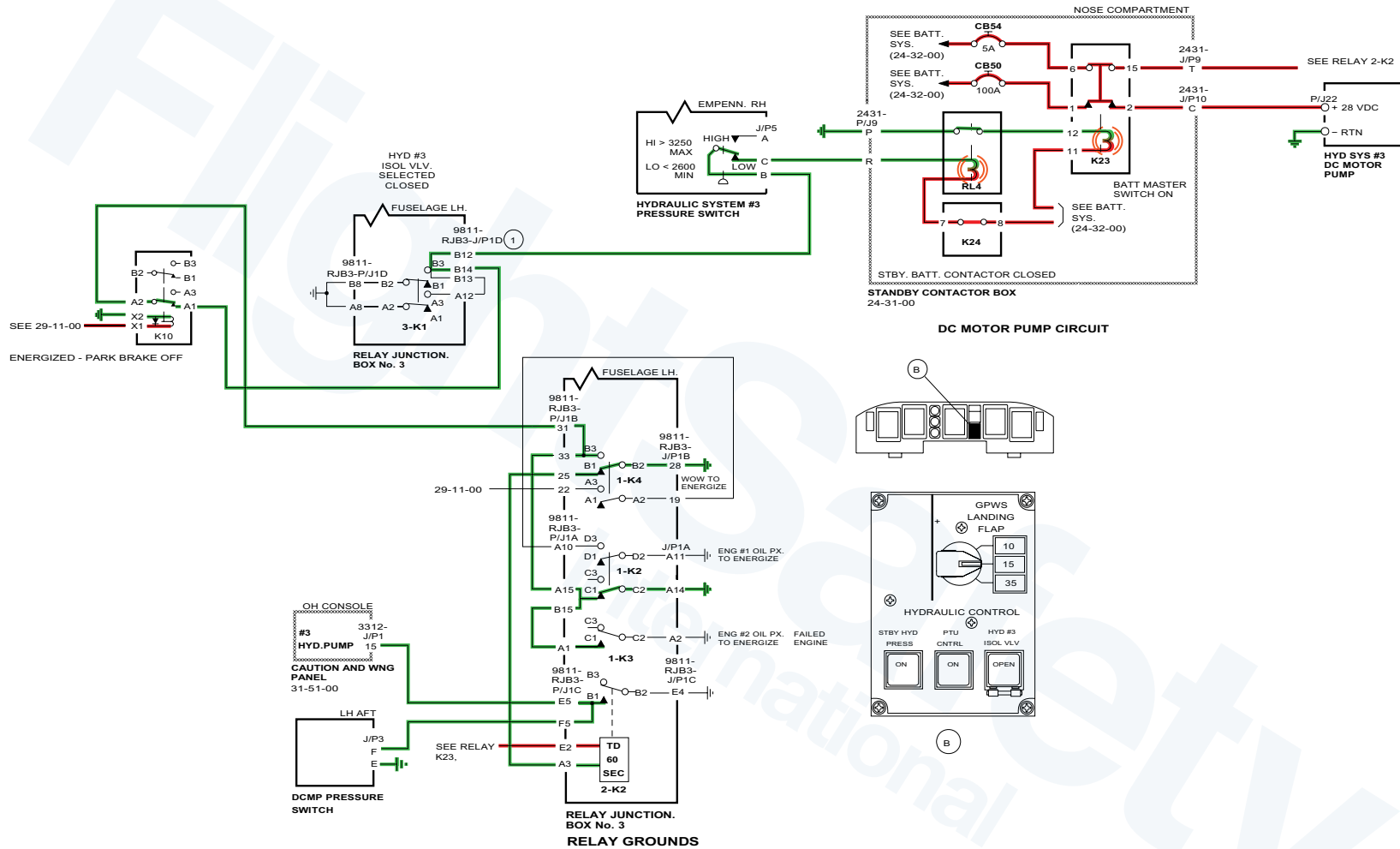
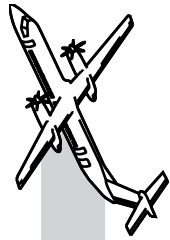
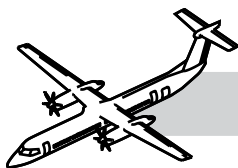


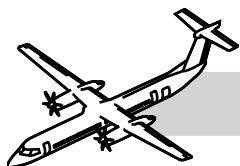
Figure 29-57. DCMP Operation - Park Brake Off - WOW - Single Engine Operating



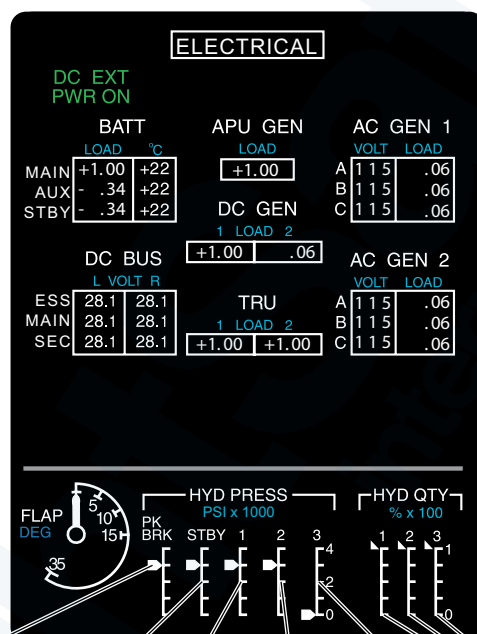
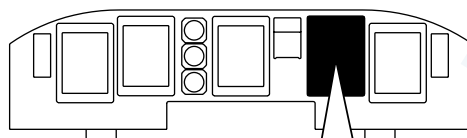
**DCMP Operation - Park Brake Off -
WOW - Single Engine Operating****NOTES**

Refer to Figure 29-57. DCMP Operation - Park Brake Off - WOW - Single Engine Operating.

Hydraulic system No.3 Pressure switch is low and relay K10 is energized by the Park Brake selected OFF and No.1 engine operating. (Oil Pressure).



MAIN INSTRUMENT PANEL



Parking Brake Hydraulic Pressure Indication

#1 System Stand-by Pressure Indication

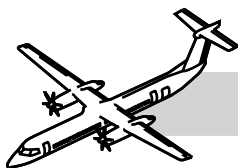
Main System #1 Pressure Indication

Main System #2 Pressure Indication

Main System #3 Pressure Indication

System #1, #2 And #3 Hyd Quantity

Figure 29-58. Hydraulic System Indications (Normal Mode)



29-30-00 INDICATIONS

NOTES

INTRODUCTION

Hydraulic system indications are shown on the:

- MFDs
- CAWP
- HYDRAULIC CONTROL panel
- FPP.

The indicating system shows:

- Hydraulic pressures
- Hydraulic quantities
- System cautions
- System operating status.

GENERAL

Refer to:

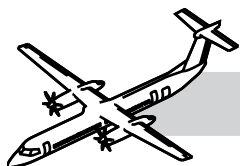
- Figure 29-58. Hydraulic System Indications (Normal Mode).
- Figure 29-59. Hydraulic System Indications Reversion Mode (Composite Mode).

Quantity and pressure transducers give inputs to the MFD for the three hydraulic system indications:

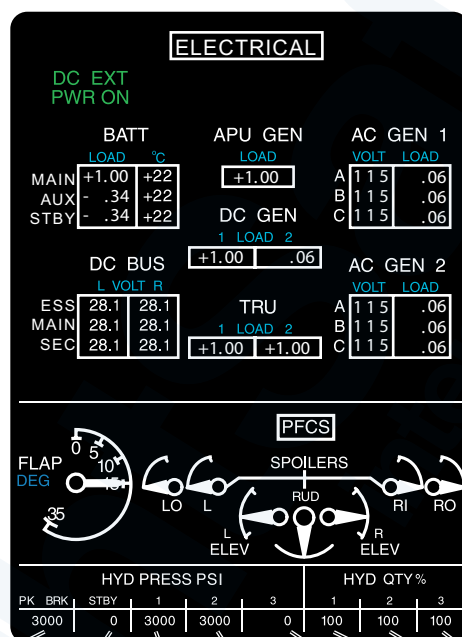
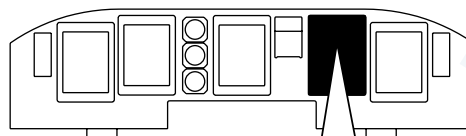
- No.1, No.2 and No.3 main hydraulic system pressure
- No.1, No.2 and No.3 hydraulic systems quantity.

MFD 2 shows data in analog format during normal configuration.

Either MFD can show the data in digital format on a composite page.



MAIN INSTRUMENT PANEL



Parking Brake Hydraulic Pressure Digital Indication

#1 System Stand-by Pressure Indication

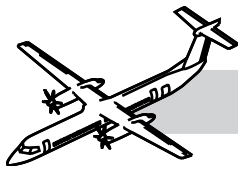
Main System #1 Pressure Indication

Main System #2 Pressure Indication

Main System #3 Pressure Indication

#1, #2 and #3 Hydraulic Fluid Quantity Digital Indications

Figure 29-59. Hydraulic System Indications Reversion Mode (Composite Mode)



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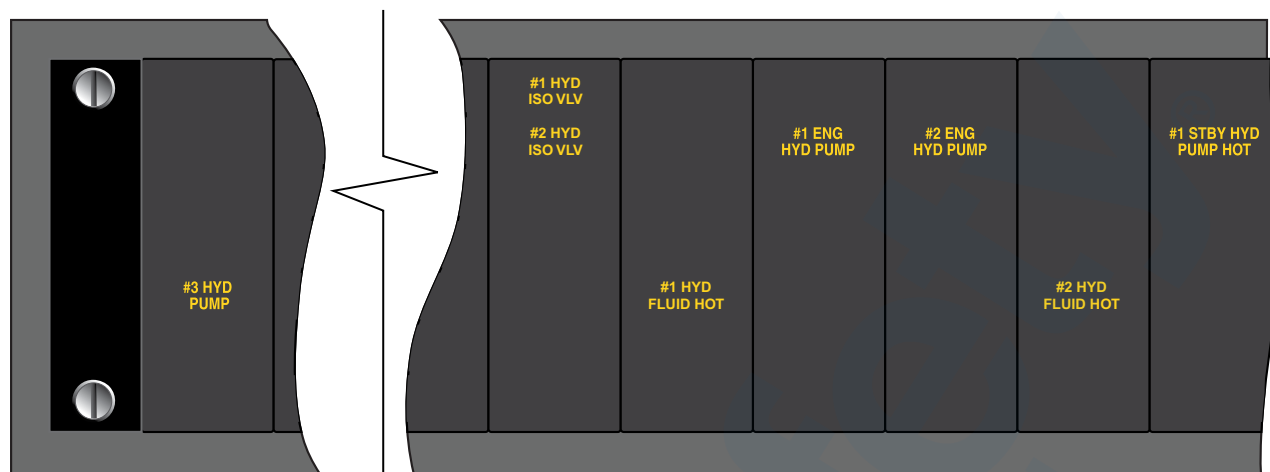
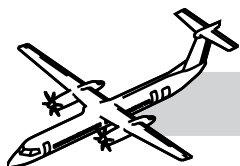


Figure 29-60. Hydraulic System Caution Indications

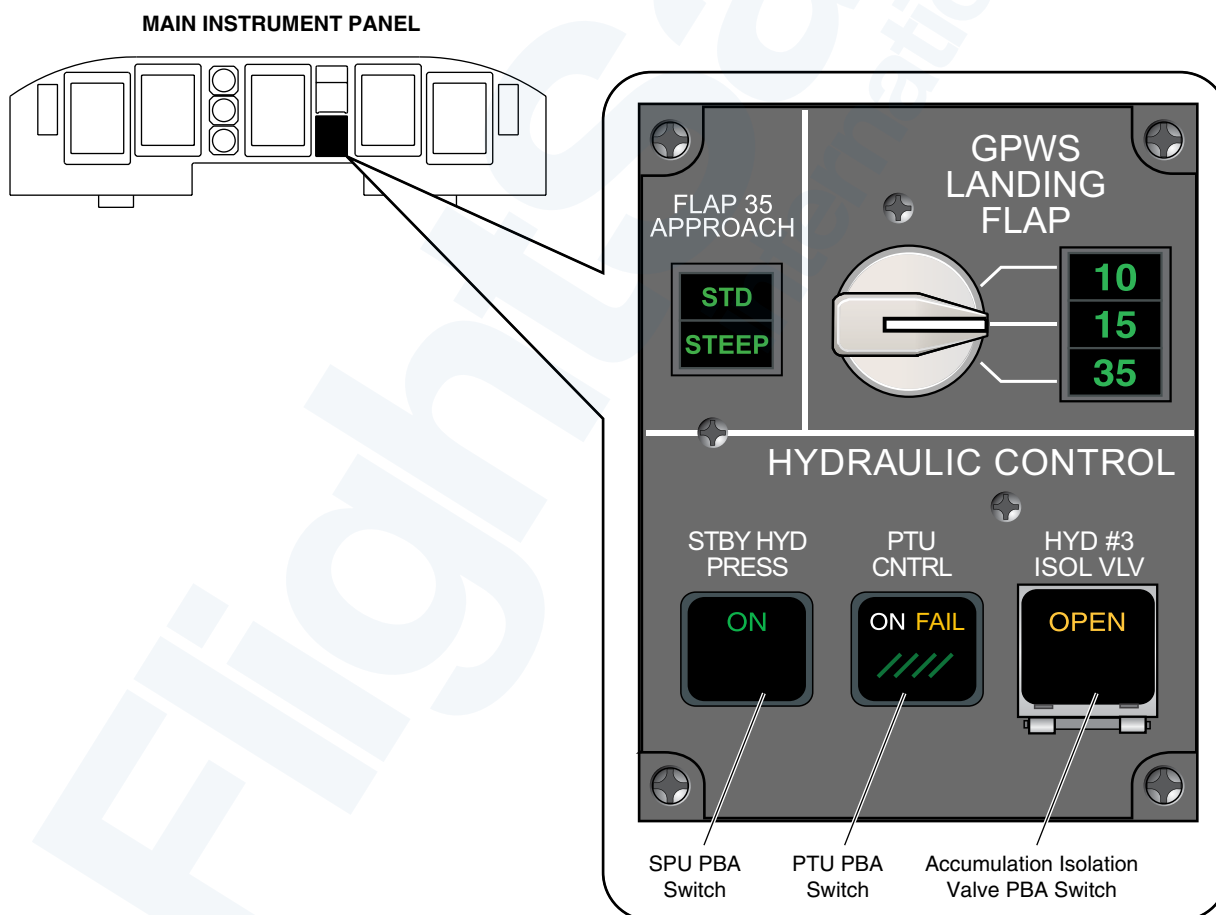
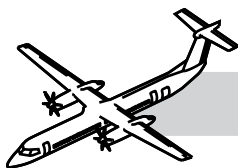


Figure 29-61. Hydraulic Control Panel



INDICATIONS

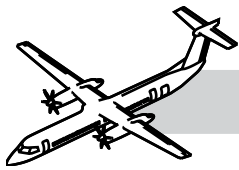
NOTES

Refer to:

- Figure 29-60. Hydraulic System Caution Indications.
- Figure 29-61. Hydraulic Control Panel.

The CAWP shows hydraulic system caution lights. Each hydraulic system includes pressure and temperature switches that illuminate caution lights on this panel.

HYDRAULIC CONTROL Panel switchlights show the operating status for the SPU, PTU and No.3 hydraulic system isolation valve. The SPU switch shows ON with switch selection. HYD #3 ISOL VLV shows OPEN with switch selection. The PTU shows ON with switch selection; the crosshatch when producing pressure and the fail light if the PTU fails to produce 900 psi within five seconds of being commanded on.



Refer to Figure 29-62. Indicating, Fire Protection Panel (FPP).

The FPP shows the advisory indications for the hydraulic FSOVs. During normal operation, an FSOV OPEN green light will illuminate. When the FSOV is closed, a FSOV CLOSED white light will illuminate.

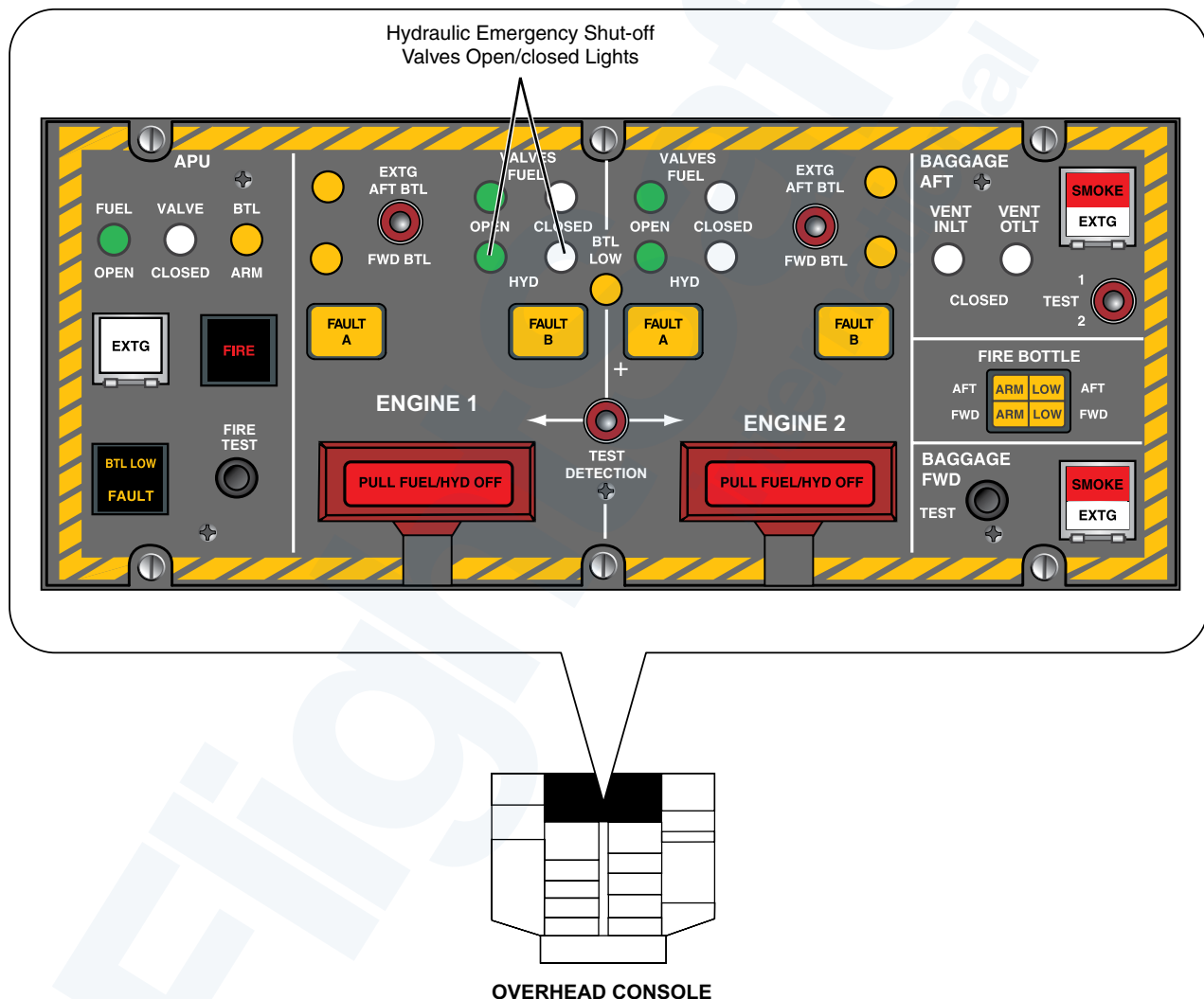
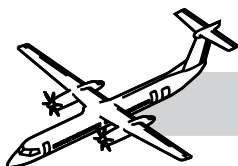


Figure 29-62. Indicating, Fire Protection Panel (FPP)



COMPONENT DESCRIPTION

Hydraulic Pressure Transducers

Refer to Figure 29-63. Hydraulic Pressure Transducers.

There are two pressure transducers installed in the No.1 hydraulic system. A system pressure transducer and a SPU pressure transducer send data to the FDPS.

There is one pressure transducer installed in the No.2 hydraulic system that sends system pressure data to the FDPS.

There is one pressure transducer installed in the No.3 hydraulic system that sends system pressure data to the FDPS.

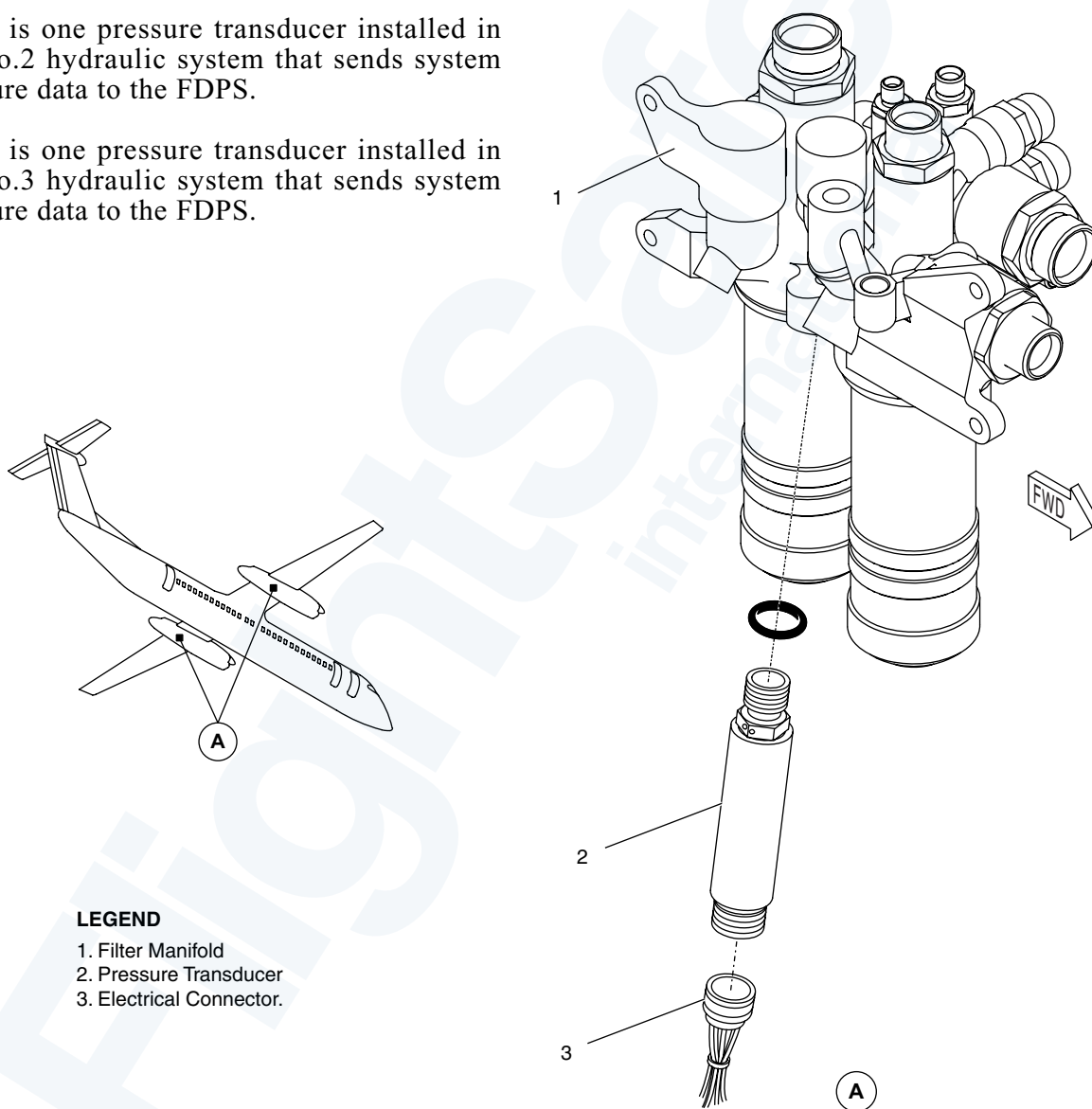
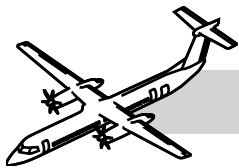


Figure 29-63. Hydraulic Pressure Transducers



Temperature Switches

Refer to Figure 29-64. Temperature Switches.

Two switches are installed on No.1 and No.2 reservoirs.

At 107°C, a HYD FLUID HOT caution light illuminates on the CAWP.

At 135°C, the firewall shut-off valve closes.

If the pump windings are hot, a thermal switch in the SPU turns ON the #1 STBY HYD PUMP HOT caution light.

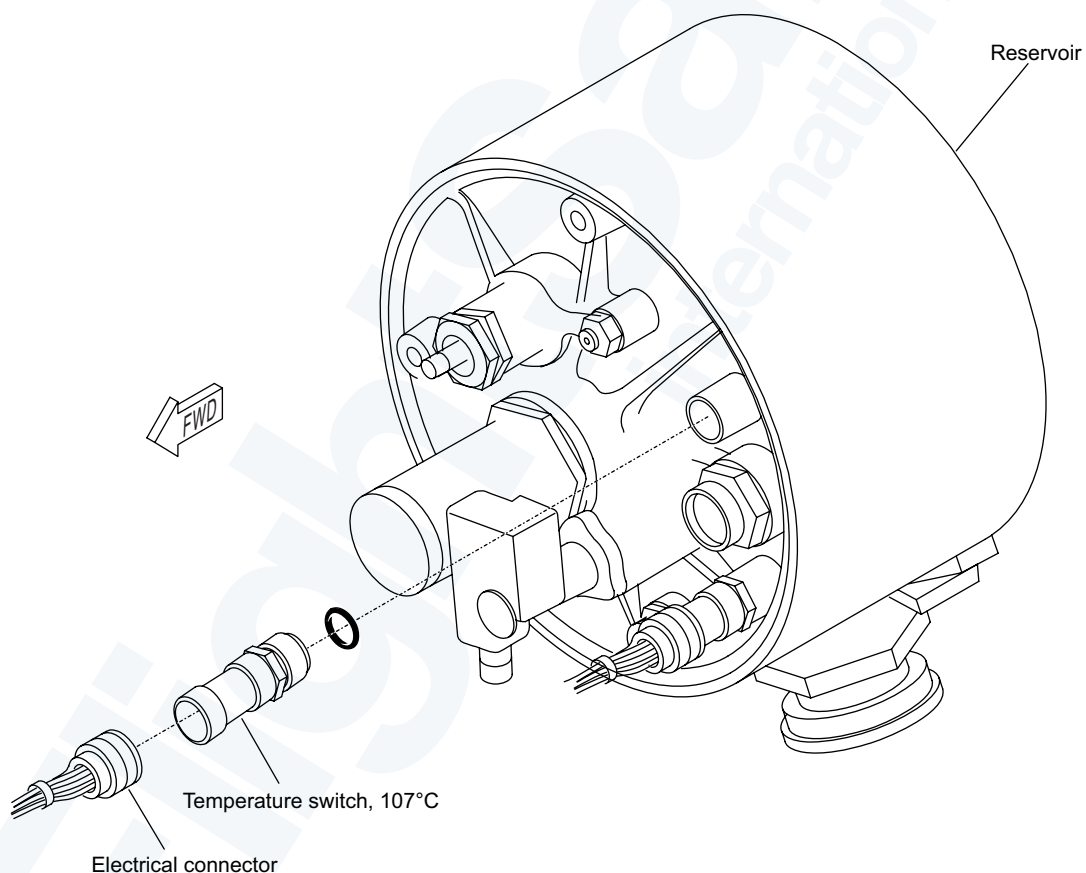
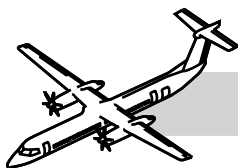


Figure 29-64. Temperature Switches



Hydraulic System Pressure Switch

Refer to Figure 29-65. Hydraulic System Pressure Switch.

If the EDP is not producing pressure, the #1 or #2 ENG HYD PUMP caution light comes ON.

Two pressure switches are installed on the inlet and outlet ports of the No.1 and No.2 hydraulic system isolation valves.

The inlet switch monitors system pressure. This is used for #3 ISOL valve open logic and ELEV PRESS caution light control logic.

The outlet switch brings on the #1 or #2 HYD ISO VLV caution light if it senses pressure less than 900 psi.

The PTU switchlight annunciator indication comes ON if the PTU outlet pressure switch senses pressure.

LEGEND

1. Electrical Connector
2. Isolation Valve (No.1 & 2)
3. Port, Pressure Switch - A
4. Pressure Switch - A
5. Port, Pressure Switch - B
6. Pressure Switch - B

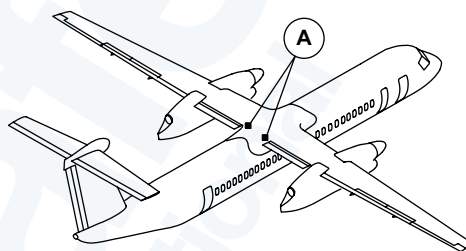
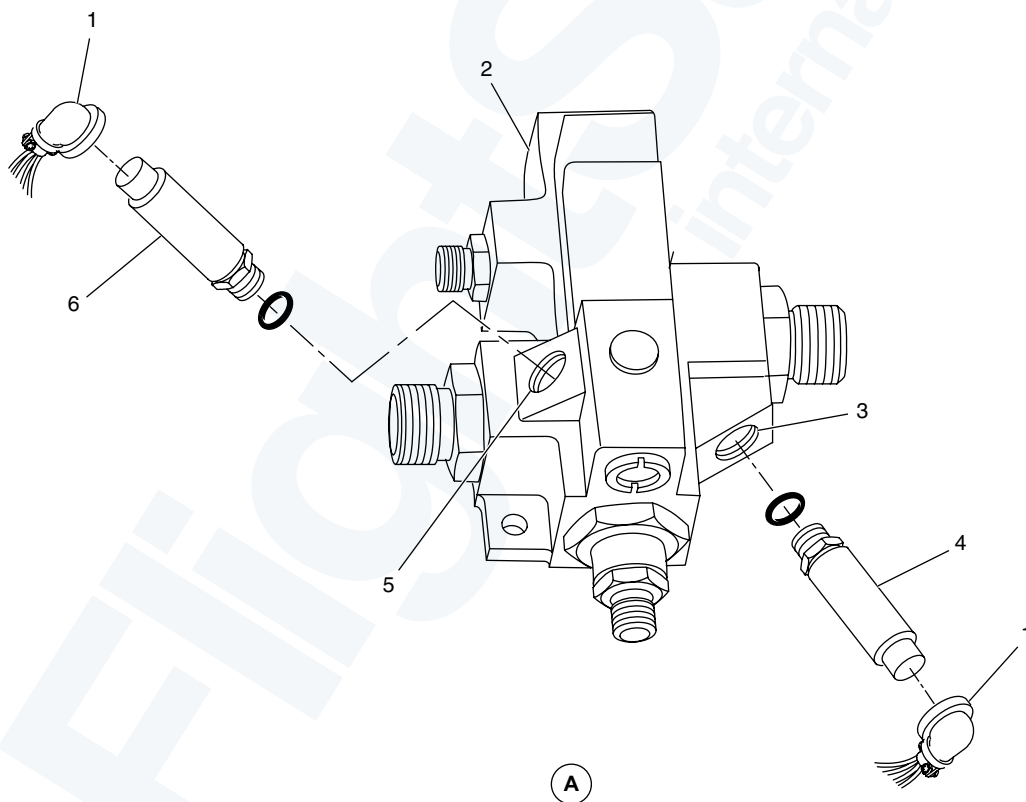
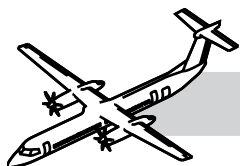


Figure 29-65. Hydraulic System Pressure Switch



No.3 System Hydraulic Warning Pressure Switch

Refer to Figure 29-66. No.3 Hydraulic System Pressure Switch.

The No.3 hydraulic system has a pressure switch that is installed downstream of the accumulator isolation valve.

If the No.1 and No.2 isolation valves inlet pressure switches along with the No.3 system pressure switch all show high pressure, then the ELEV PRESS caution light comes ON on the CAWP.

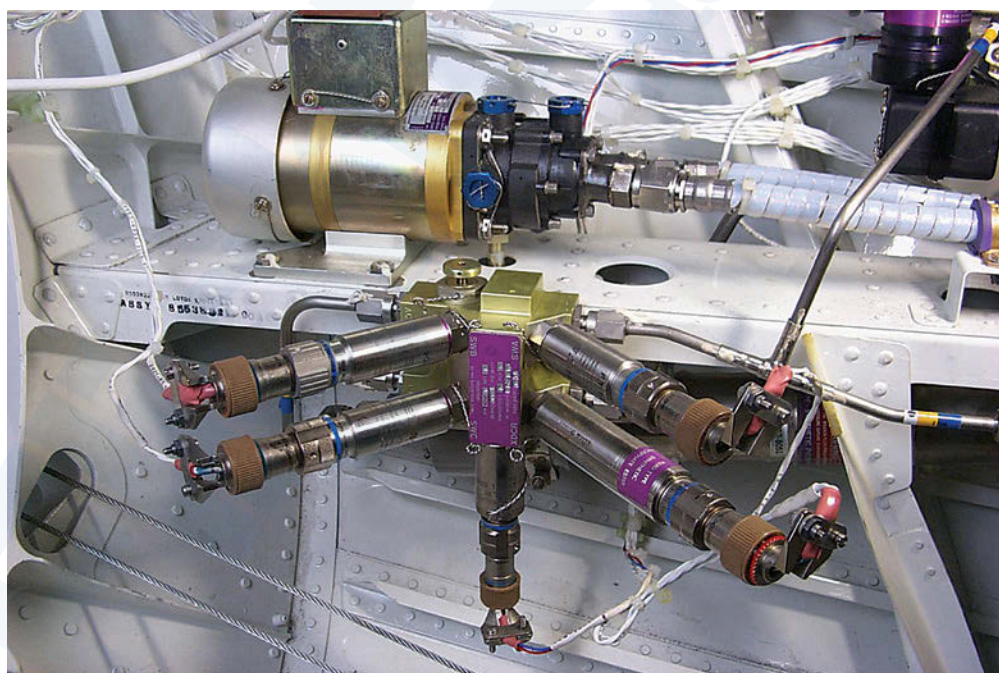
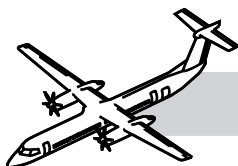


Figure 29-66. No.3 Hydraulic System Pressure Switch



29-00-00 APPENDIX

GENERAL

Maintenance Practices

If you spill hydraulic fluid on the deicer boots or in any case, immediately clean it up.

Pressurization of the Hydraulic Systems

Refer to Figure 29-67. Ground Hydraulic Power Unit (GHPU).

The aircraft hydraulic systems can be pressurized using:

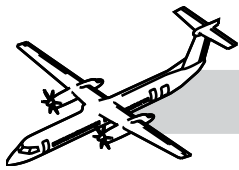
- Ground Hydraulic Power Unit (GHPU)
- External electrical power
- Engine power.

NOTE

The Ground Hydraulic Power Unit (GHPU) does not pressurize the case drain line.



Figure 29-67. Ground Hydraulic Power Unit (GHPU)



Pressurization of the Aircraft Hydraulic Systems with the Ground Hydraulic Power Unit (GHPU)

Refer to:

- Figure 29-68. Ground Hydraulic Power Unit (GHPU).
- Figure 29-69. Hydraulic Access Door.
- Figure 29-70. Servicing No.1 and No.2 Hydraulic Systems.

The GHPU operates in two modes:

1. CLOSED LOOP:

- GHPU pressurizes aircraft hydraulic systems and circulate hydraulic fluid from and back to aircraft hydraulic systems reservoirs. Aircraft hydraulic fluid will not drain into the GHPU reservoir.

2. OPEN LOOP:

- GHPU pressurizes aircraft hydraulic systems and circulates hydraulic fluid from aircraft hydraulic systems reservoirs to GHPU reservoir. Aircraft hydraulic fluid will drain into the GHPU reservoir.

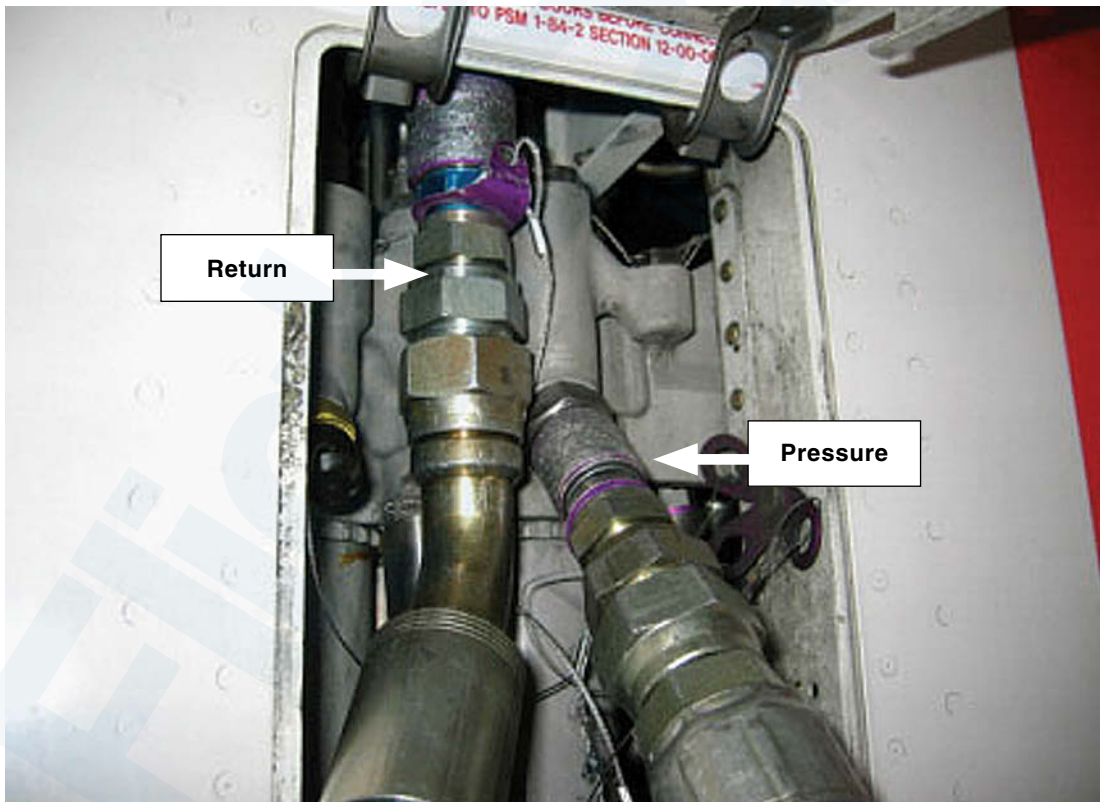


Figure 29-68. Ground Hydraulic Power Unit (GHPU)

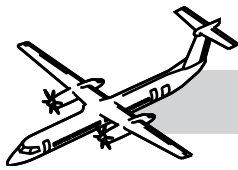


Figure 29-69. Hydraulic Access Door

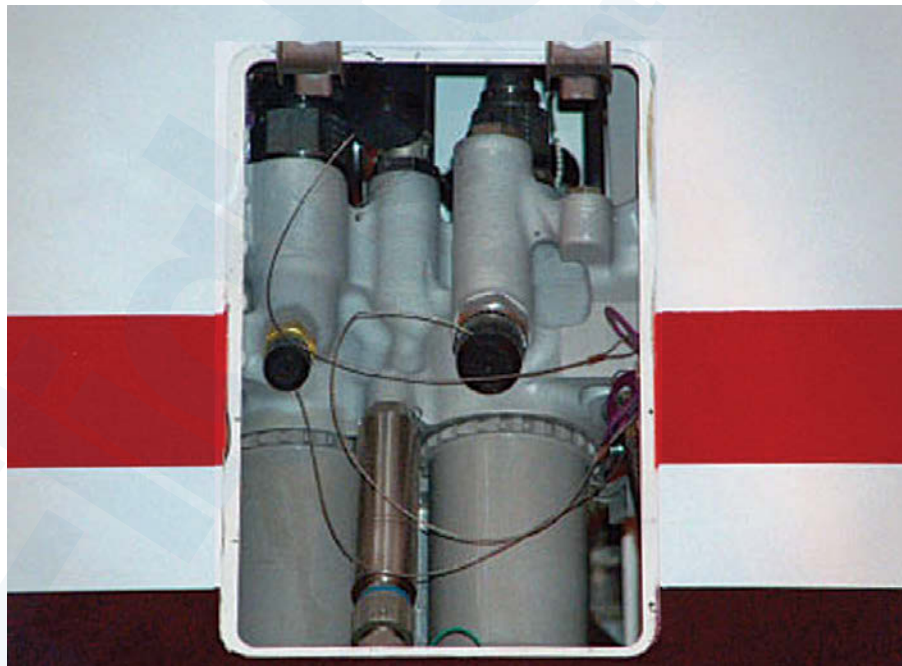
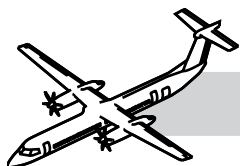


Figure 29-70. Servicing No.1 and No.2 Hydraulic Systems

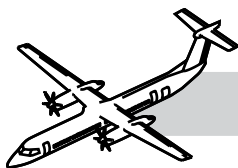


K-4008

Figure 29-71. Hydraulic Sample Bottle



Figure 29-72. No.3 Bleed Relief Valve



29-00-00 SAMPLING OF THE HYDRAULIC SYSTEMS

LIMITATIONS

Refer to:

- Figure 29-71. Hydraulic Sample Bottle.
- Figure 29-72. No.3 Bleed Relief Valve.

New hydraulic fluid is a clear light purple colour, free of haze or solid particles. Dark brown color fluid may be the result of colour leaching from the Engine Driven Pump EDP, seals, hoses. It may also indicate exposure to excessive heat. Loss of fluid colour does not mean loss of fluid properties. Investigate the causes of darkening and check other factors such as viscosity, TAN and solid particle contents. If tests and investigations are satisfactory, do not replace the hydraulic fluid because of the fluid dark colour.

Take a hydraulic fluid sample as follows:

- Pressurize the Hydraulic system to be sampled.
- On the reservoir of the hydraulic system, operate the relief push button.
- Drain 0.25 to 0.50 L of hydraulic fluid into the ecology bottle of the system.
- Remove one ecology bottle from its attachment position on the aircraft.
- Put the clean sample bottle below the inlet fitting to the ecology bottle.
- Operate the relief push button.
- Fill the sample bottle until it is almost full.
- Put a correct label on the sample bottle to identify it.
- Send the hydraulic fluid sampling bottle to a lab for analysis.
- Top up the hydraulic system.
- Get a lab analysis report.

CAUTION

When the hydraulic fluid sample fails to meet the Standard Specifications in the lab report, corrective actions are given to return aircraft to service.

Sampling bottles should be of clear glass and 250 to 500 ML (CC) capacity. Preclean the bottle with distilled water and dry. Do not “RINSE” the bottle with hydraulic sample fluid.

HYDRAULIC LINES

Inspection of Hydraulic Lines

Make sure that there are none of the defects that follow:

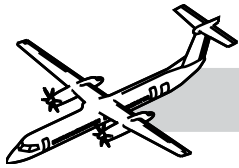
- Cuts
- Fraying
- Kinks
- Wear
- Dents.

NOTE

No damage (wear or dents) is permitted in the bend area of hydraulic tubes.

If there are dents or wear, measure the defect with the vernier caliper or optical micrometer.

Make sure the dents or wear are in the recommended limits for the tube material.



HYDRAULIC LINE REPAIR

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

There are two approaches to repair damage tubing (permanent and temporary repair).

To minimize aircraft downtime and expedite return to service.

Repair the damaged tube with a serviceable piece using permaswage sleeves or with flex hose.

The procedure also includes information regarding the Quick Fix Kit for the field repair of the damaged tubes.

Temporary Repair

Install a flexible hose as a temporary repair, this will allow 5000 hours of operation provided an inspection is done after every 1000 hours to ensure the serviceability and condition of the flex hose.

29-10-00 FLUSHING OF THE HYDRAULIC SYSTEM(S)

Reasons to flush the hydraulic system(s):

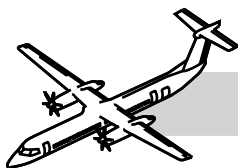
- Particle contamination,
- Chemical contamination,
- Fluid properties degradation,
- Failure of the Engine Driven Pump (EDP), Standby Power Unit (SPU) or Power Transfer Unit (PTU)

Repeated extensions of the Differential Pressure Indicators (DPI's).

Perform General Visual Inspection of the Hydraulic System Filter Differential Pressure Indicators (DPI's).

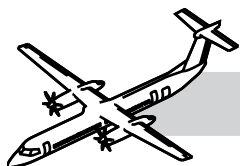
NOTE

Material found by the analysis of a fluid sample by Materials Lab will determine the additional task requirements.



29-00-00 SPECIAL TOOL & TEST EQUIPMENT

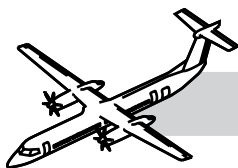
- GSB2440006 DC Rectifier (50/60 Hz, 28.50 V dc).
- GSB2440003 Electrical AC power cart.
- GSB29000XX Ground Hydraulic Power Unit.
- GSB29000YY Ground Hydraulic Power Unit.
- GSB2900001 Hydraulic Power Unit.
- GSB2900002 Hydraulic Coupling Kit.
- 11-8073-1010 DC Ground Power Unit
- MS21914 - Bootstrap pressure port cap.
- H102 AVTRON or equivalent portable hydraulic test stand.
- 18-4208-0010- or equivalent booster and manifold/regulator.
- B0302164 or MS33514 or equivalent, commercially available dapter and fitting between the GHPU hose and the heat exchanger line for the flushing of the No. 1 or No. 2 hydraulic system heat exchanger line.
- GSB2900001 GHPU
- Commercially Available Various adaptors hoses and fittings of size, length and type to suit container to capture fluid while sampling 575 VAC, 3 Phase, 60 Hz or equivalent Standard Electric Power.
- GSB1216010 Nitrogen cart.
- GSB1210006 Servicing Unit
- GSB1210005 Reservoir Servicing Unit - Single Container 5 gal (19 L) and 150 psi (1034 kPa), Phosphate Ester Compatible.
- GSB1000020 Pin, NLG Door Ground Lock
- GSB1000028 Pin, MLG Door Ground Lock
- GSB2400001, Multimeter-Hand Held
- ACS28008 Hydraulic Pipe Swage Repair Kit
- GSB1210003 Reservoir Servicing Quick-Disconnect Coupling
- GSB1210004 Hydraulic Fluid Dispenser (SAE AS 1241)
- GSB1210007 Portable Hydraulic Service Unit (SAE AS 1241)
- GSB1000004 Wheel Chocks - MLG and NLG
- Commercially available Glenair TG70 Strap Wrench or Equivalent.
- Commercially available McMaster Carr Replacement Strap P/N: 5448A33 (101-170 lbf in torque) or Equivalent
- GSB2700008-32 Target - De-Actuator (Copper).
- GSB2700008-16D Target - Actuator (Steel).
- GSB3411011 Test Set - Air Data.



29-00-00 MAINTENANCE PRACTICES

Refer to the Bombardier AMM PSM 1-84-2 for details on these maintenance procedures:

- AMM 29-00-00-863-801 Pressurization of the No. 1 and No. 2 Hydraulic Systems
- AMM 29-00-00-863-802 Removal of the Hydraulic Pressure from the No. 1 and No. 2 Hydraulic Systems
- AMM 29-00-00-863-803 Pressurization of the No. 3 Hydraulic System
- AMM 29-00-00-863-804 Removal of the Hydraulic Pressure from the No. 3 Hydraulic System
- AMM 12-10-29-210-801 Fluid Quantity Check of the No. 1 and No. 2 Hydraulic System Reservoirs
- AMM 12-10-29-611-802 Servicing of the No. 3 Hydraulic System Reservoir
- AMM 29-00-00-220-801 Sampling of the No. 1, No. 2 and No. 3 Hydraulic Systems
- AMM 29-10-00-280-802 Detailed Visual Inspection of the No. 1, No. 2 and No. 3 Hydraulic Systems
- AMM 29-11-00-210-802 General Visual Inspection of the Hydraulic System No. 1 and No. 2 Filter Differential Pressure Indicators (Delta P) (MRB 291100-201)
- AMM 29-11-26-000-801 Removal of the No. 1 and No. 2 Hydraulic System Ecology Bottles
- AMM 29-11-26-400-801 Installation of the No. 1 and No. 2 Hydraulic System Ecology Bottles
- AMM 29-11-61-000-801 Removal of the Hydraulic Power Transfer Unit
- AMM 29-11-61-400-801 Installation of the Hydraulic Power Transfer Unit
- AMM 29-10-00-870-801 Bleeding of the No. 1 Hydraulic System
- AMM 29-10-00-870-802 Bleeding of the No. 2 Hydraulic System
- AMM 12-00-16-490-801 Connection of the Ground Hydraulic Power
- AMM 12-00-16-090-801 Removal of the Ground Hydraulic Power
- FIM 29-11-00-810-801 #1 HYD ISO VLV (Caution) – Fault Isolation
- FIM 29-11-00-810-802 HYD VALVE CLOSED (Advisory) – Fault Indication
- FIM 29-11-00-810-803 HYDRAULIC PRESSURE LOSS (Status) – Fault Isolation
- FIM 29-11-00-810-804 #2 HYD ISO VLV (Caution) – Fault Isolation
- AMM 29-11-00-710-802 Operational Check of the Hydraulic System 1 Engine Driven Pump (EDP) Pressure Switch (CMR#291100-103)
- AMM 29-11-06-710-801 Operational Test of the Firewall Shut-off Hydraulic Valve
- AMM 29-11-36-710-801 Operational Test of the Standby Power Unit
- AMM 29-11-61-710-801 Operational Test of the Power Transfer Unit
- AMM 29-12-01-710-801 Operational Test of the No.3 Hydraulic System after replacement of the DC Motor Driven Pump (DCMP)



- AMM 29-21-06-000-802 Removal of the Secondary MLG Selector Valve Filter Fitting
- AMM 29-21-06-400-802 Installation of the Secondary MLG Selector Valve Filter Fitting
- AMM 29-21-16-000-801 Removal of the In-Line Filter Screen
- AMM 29-21-16-400-801 Installation of the In-Line Filter Screen
- AMM 29-21-00-611-801 Servicing of the Landing Gear Alternate Extension Hydraulic Reservoir (MRB #292000-201)
- AMM 12-10-29-611-803 Servicing of the Alternate Landing-Gear Extension Hydraulic Reservoir
- AMM 29-21-00-617-801 Flushing of the Alternate Landing Gear Extension Hydraulic System
- AMM 29-21-00-870-801 Bleeding of the Alternate Landing Gear Extension Hydraulic System
- AMM 29-21-00-710-802 Operational Check of the Landing Gear Alternate Extension Hydraulic System Hand Pump (CMR#292000-103)
- FIM 29-30-00-810-801 ELEVATOR PRESS (Caution) – Fault Isolation
- FIM 29-30-00-810-802 #1 ENG HYD PUMP (Caution) – Fault Isolation
- FIM 29-30-00-810-803 #1 HYD FLUID HOT (Caution) – Fault Isolation
- FIM 29-30-00-810-804 #1 STBY HYD PUMP HOT (Caution) – Fault Isolation
- FIM 29-30-00-810-805 #2 ENG HYD PUMP (Caution) – Fault Isolation
- FIM 29-30-00-810-806 #2 HYD FLUID HOT (Caution) – Fault Isolation